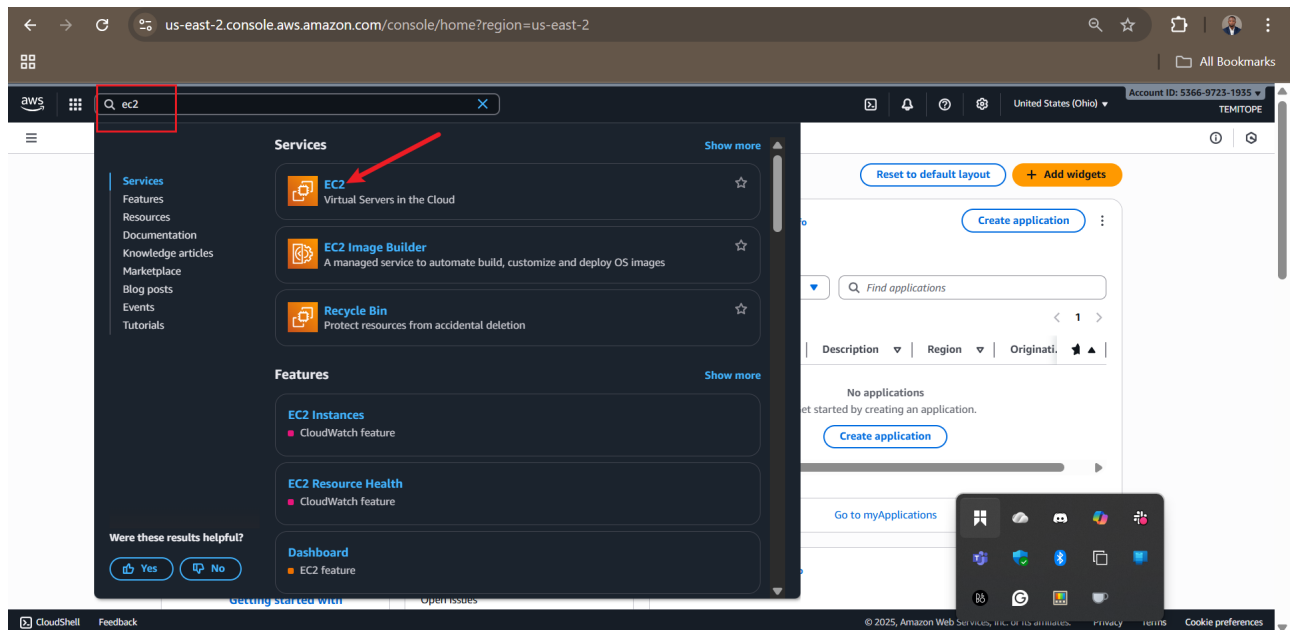


Configuring Auto Scaling with Application Load Balancer using Launch Template

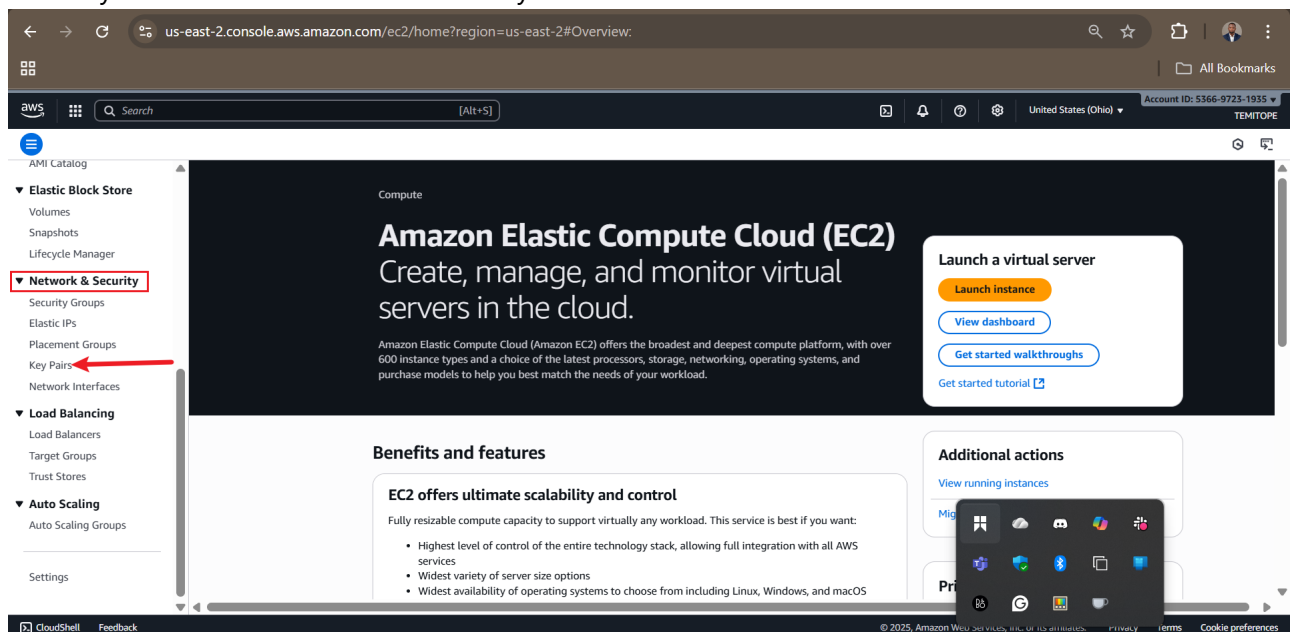
This project demonstrates the step-by-step process of configuring an Auto Scaling group (ASG) integrated with an Application Load Balancer (ALB) using a Launch Template in AWS EC2. The steps are organized sequentially based on the numbered screenshots provided.

Step-by-Step Guide

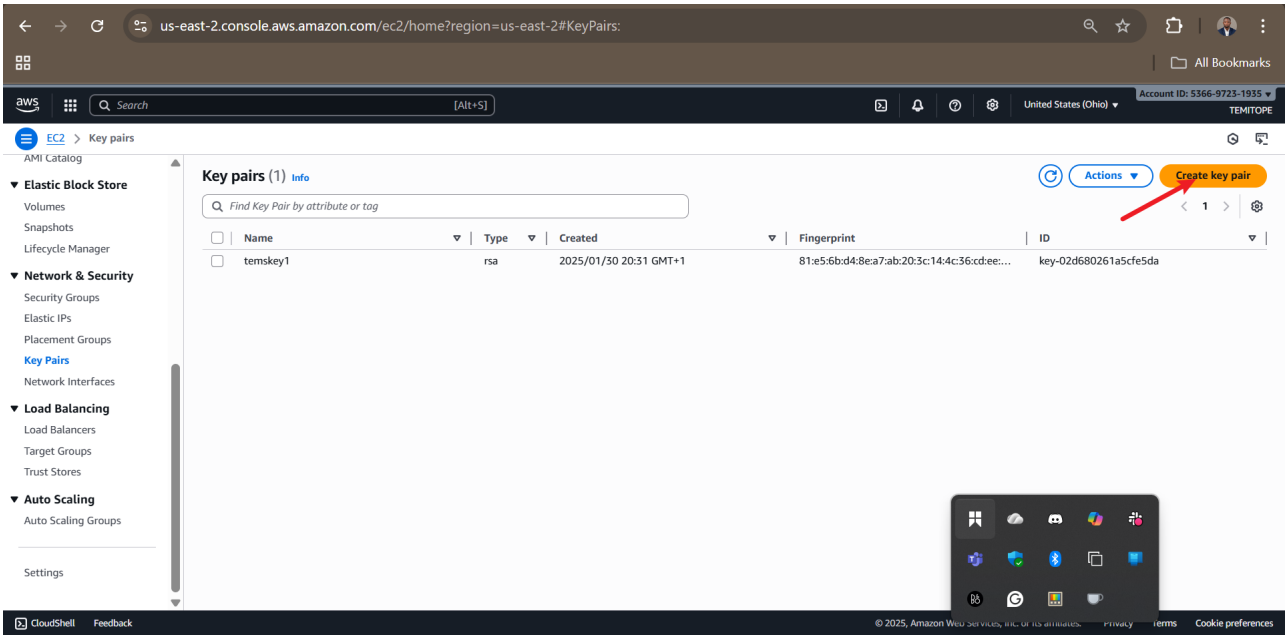
1. Search for EC2 and click on it.



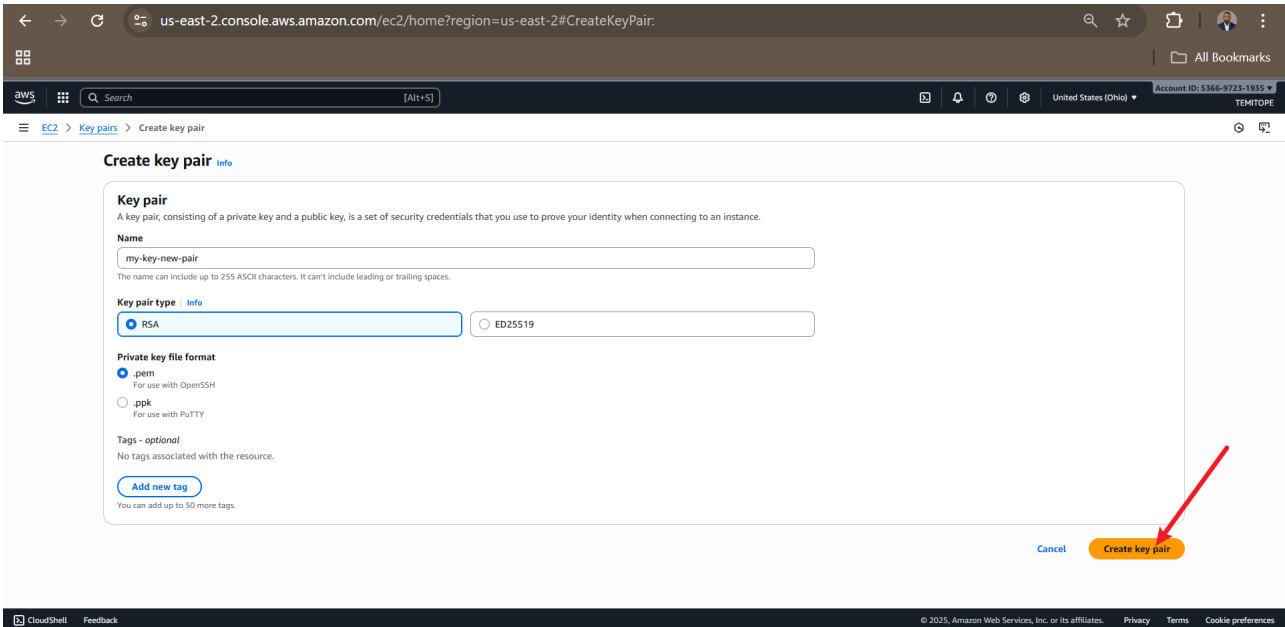
2. Find Key Pairs under Network and Security on the left sidebar and click on it.



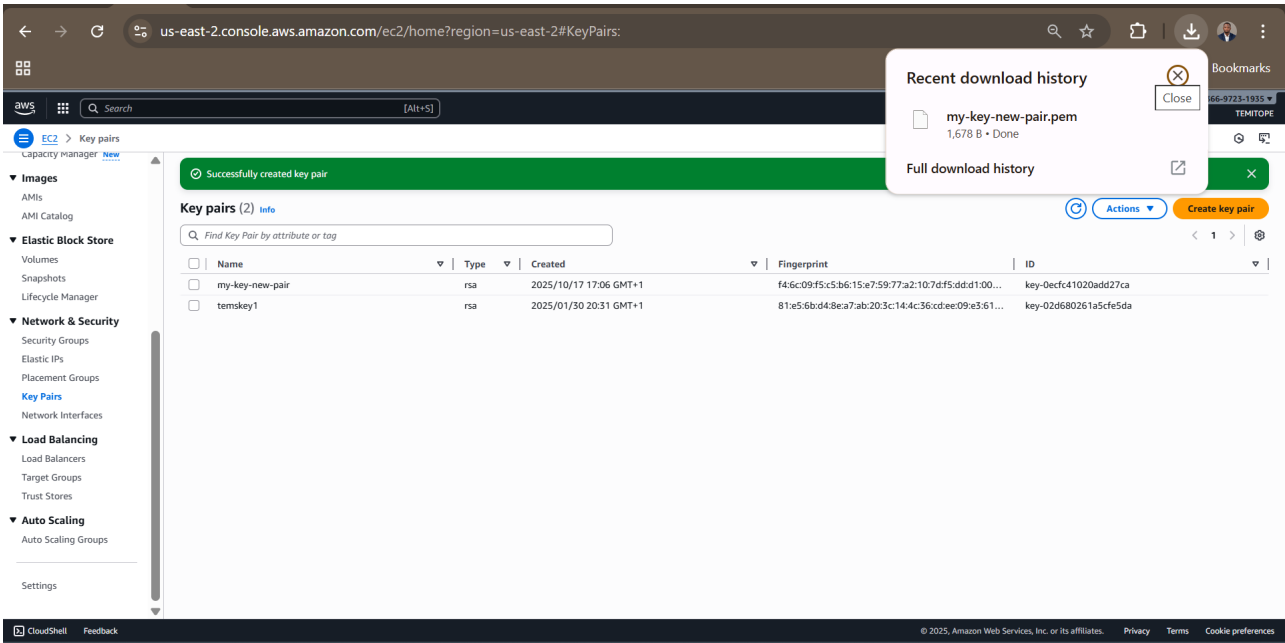
3. On the keypair page click create keypair.



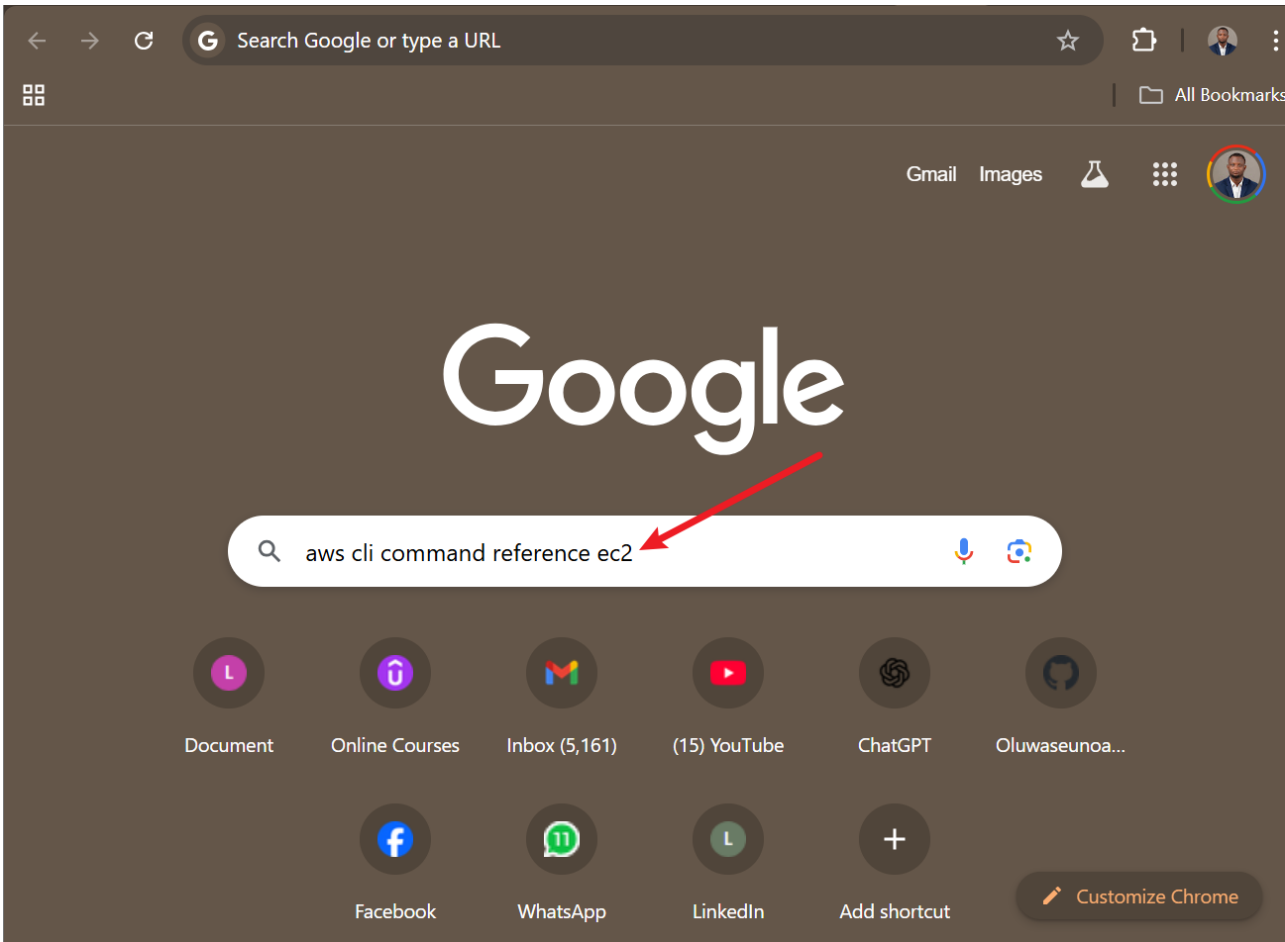
4. Name key pair select type and format and click create key pair.



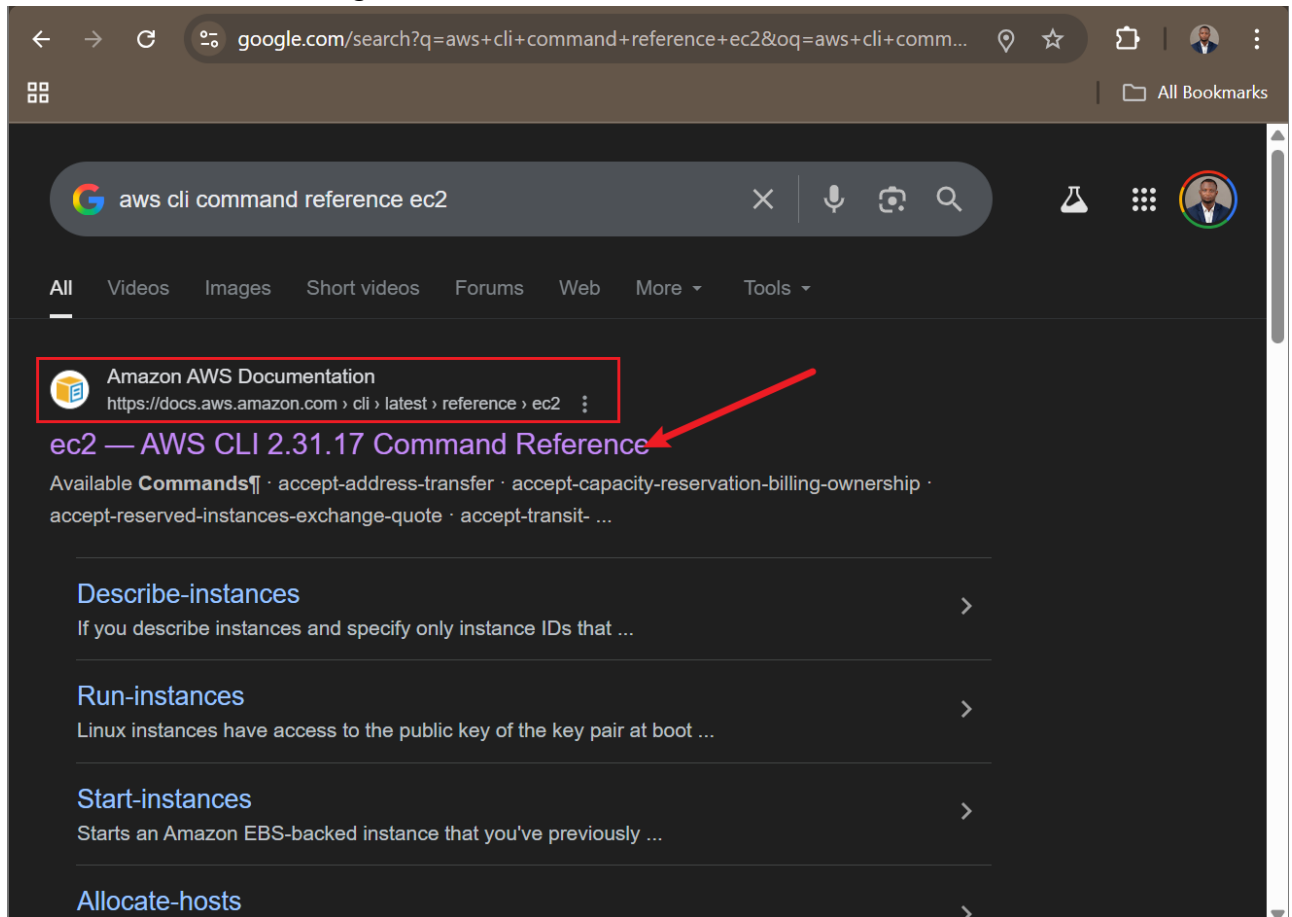
5. Keypair created and downloaded.



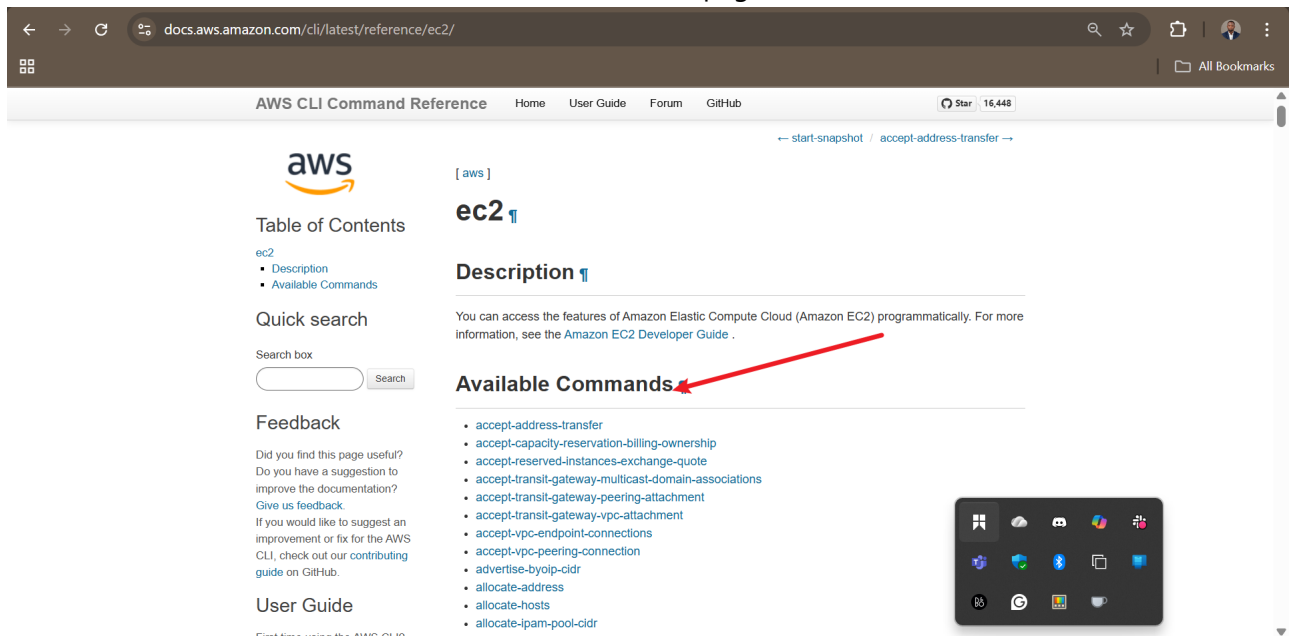
6. Search for AWS CLI command reference EC2.



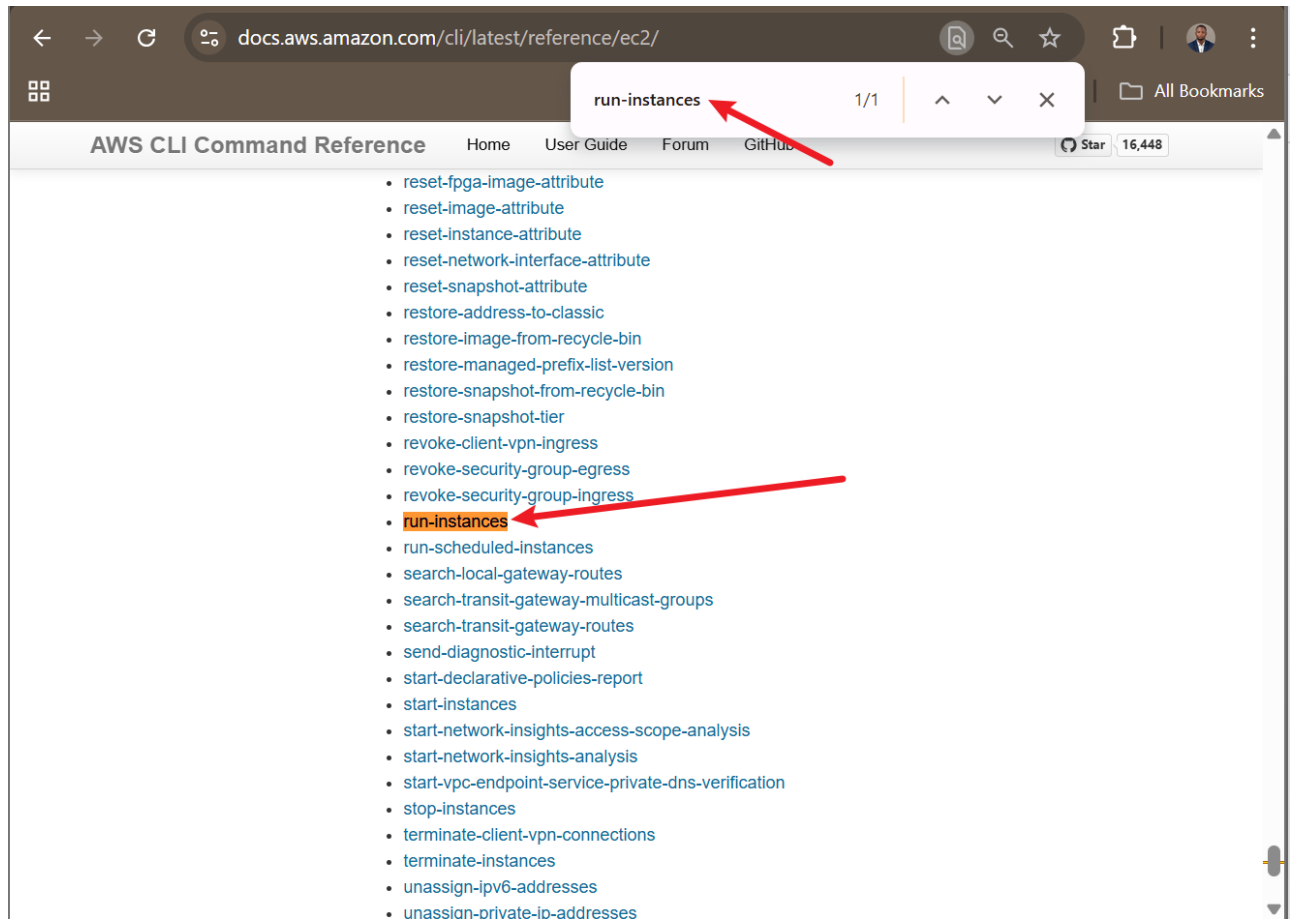
7. Click on the first link showing CLI latest reference EC2.



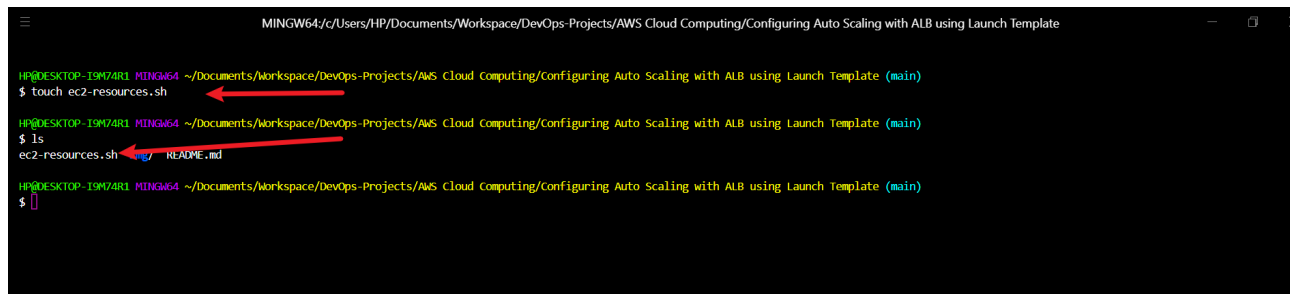
8. Now on AWS CLI commands official documentation webpage.



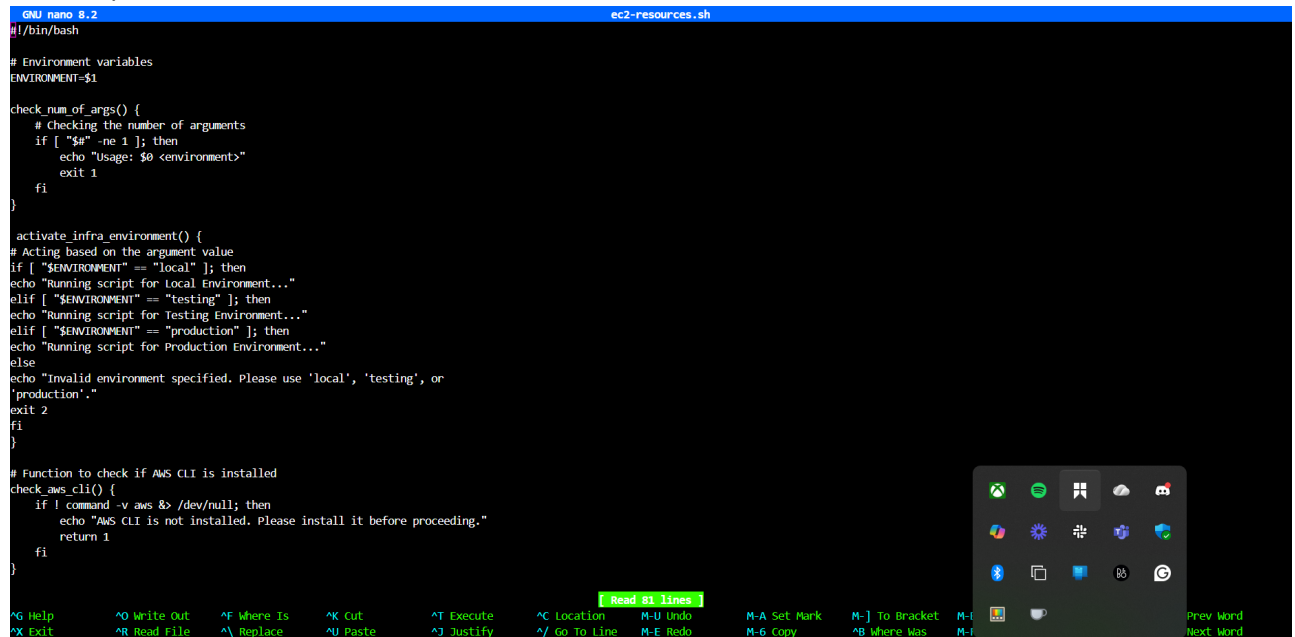
9. Search for run-instances with Ctrl-F and click on it when found.



10. Create ec2-resources.sh.



11. Add script to automate EC2 creation.



```

GNU nano 8.2 ec2-resources.sh
#!/bin/bash

# Environment variables
ENVIRONMENT=$1

check_num_of_args() {
    # Checking the number of arguments
    if [ "$#" -ne 1 ]; then
        echo "Usage: $0 <environments>"
        exit 1
    fi
}

activate_infra_environment() {
    # Acting based on the argument value
    if [ "$ENVIRONMENT" == "local" ]; then
        echo "Running script for Local Environment..."
    elif [ "$ENVIRONMENT" == "testing" ]; then
        echo "Running script for Testing Environment..."
    elif [ "$ENVIRONMENT" == "production" ]; then
        echo "Running script for Production Environment..."
    else
        echo "Invalid environment specified. Please use 'local', 'testing', or 'production'."
        exit 2
    fi
}

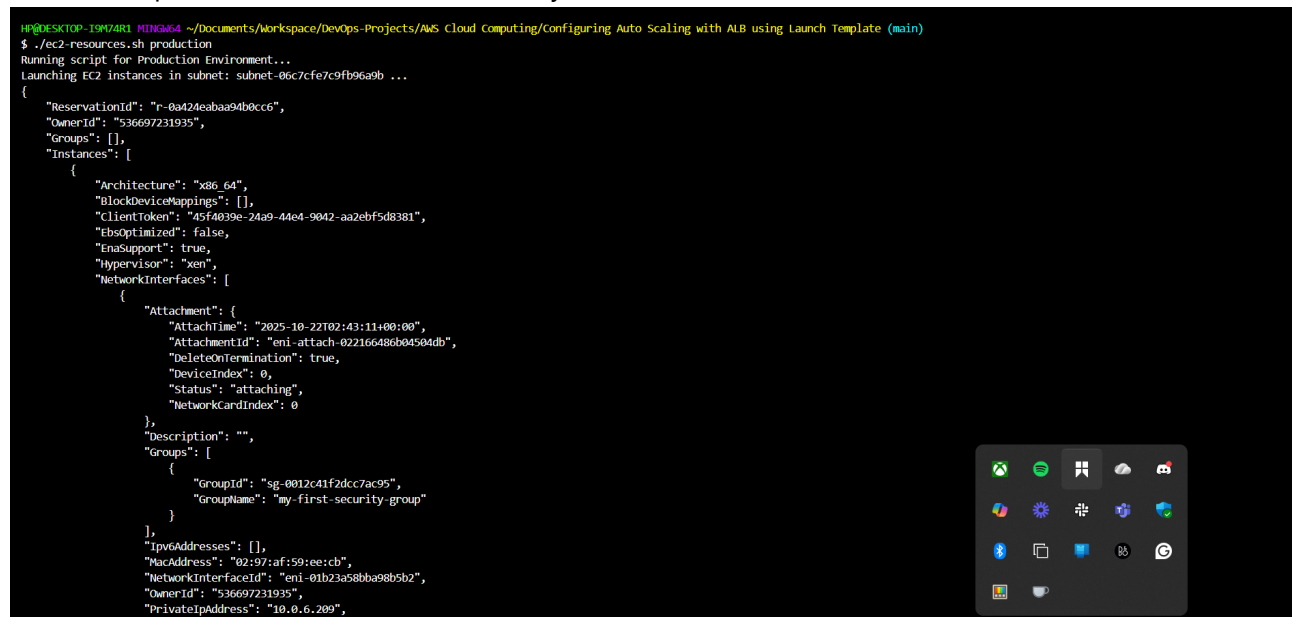
# Function to check if AWS CLI is installed
check_aws_cli() {
    if ! command -v aws && /dev/null; then
        echo "AWS CLI is not installed. Please install it before proceeding."
        return 1
    fi
    return 0
}

```

12. Note that create_ec2_instances() will create two instances in a public subnet.

! [Note that create_ec2_instances() will create two instances in a public subnet] (./img/12.note_that_create_ec2_instances()_will_create_two_instances_in_a_public_subnet.png)

13. Execute script and EC2 resources successfully created.

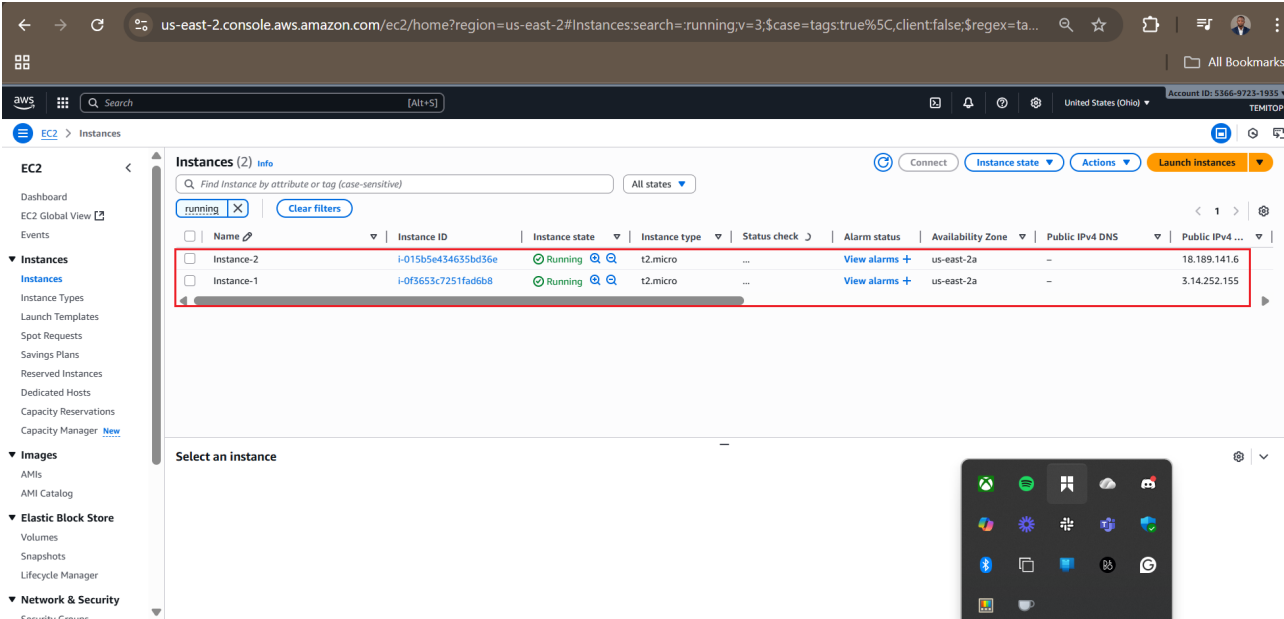


```

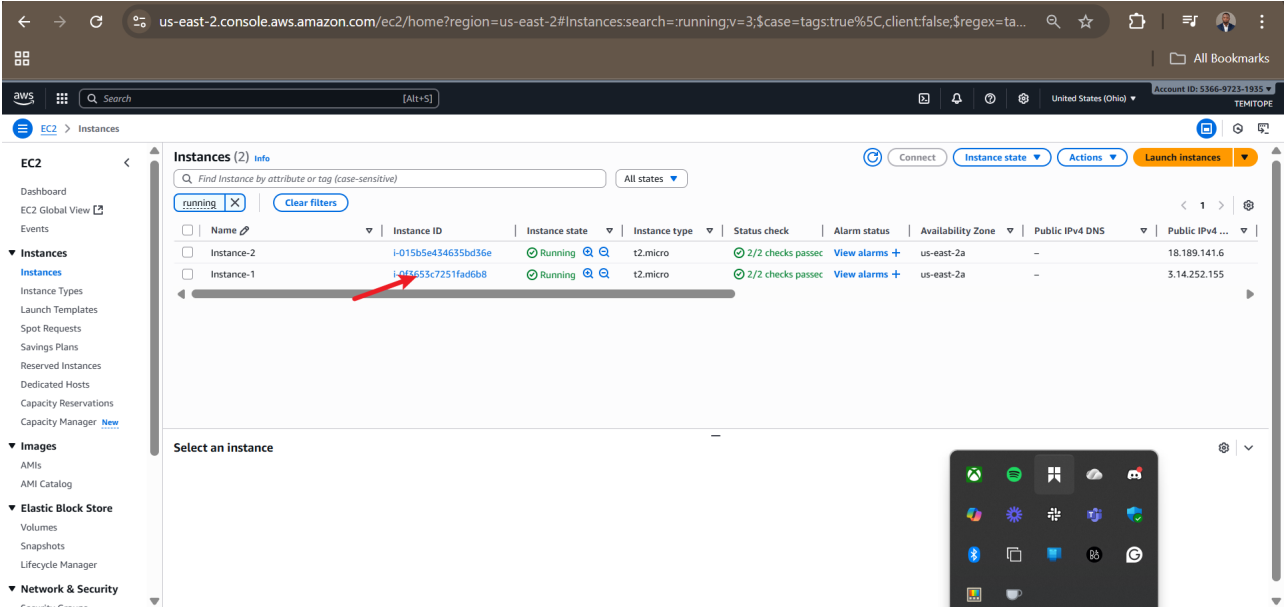
HP@DESKTOP-19M74R1 MINGW64 ~\Documents\workspace\DevOps-Projects\AWS Cloud Computing\configuring Auto Scaling with ALB using Launch Template (main)
$ ./ec2-resources.sh production
Running script for Production Environment...
Launching EC2 instances in subnet: subnet-06c7cfe7c9fb96a9b ...
{
  "ReservationId": "r-0a424eabaa94b0cc6",
  "OwnerId": "536697231935",
  "Groups": [],
  "Instances": [
    {
      "Architecture": "x86_64",
      "BlockDeviceMappings": [],
      "ClientToken": "45f4039e-24a9-44e4-9042-aa2ebf5d8381",
      "EbsOptimized": false,
      "EnaSupport": true,
      "Hypervisor": "xen",
      "NetworkInterfaces": [
        {
          "Attachment": {
            "AttachTime": "2025-10-22T02:43:11+00:00",
            "AttachmentId": "eni-attach-022166486b04504db",
            "DeleteOnTermination": true,
            "DeviceIndex": 0,
            "Status": "attaching",
            "NetworkCardIndex": 0
          },
          "Description": "",
          "Groups": [
            {
              "GroupId": "sg-0012c41f2d0c7ac95",
              "GroupName": "my-first-security-group"
            }
          ],
          "Ipv6Addresses": [],
          "MacAddress": "02:97:af:59:ee:cb",
          "NetworkInterfaceId": "eni-01b23a58bba98b5b2",
          "OwnerId": "536697231935",
          "PrivateIpAddress": "10.0.6.209",

```

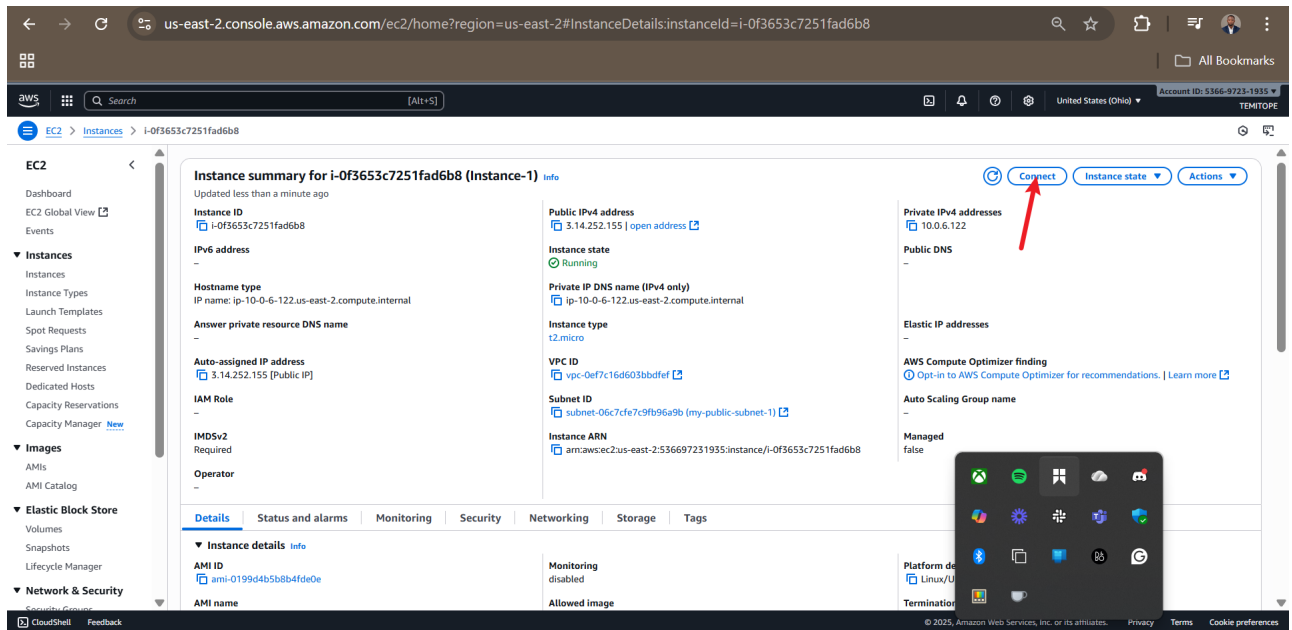
14. Two instances seen created and running on the console.



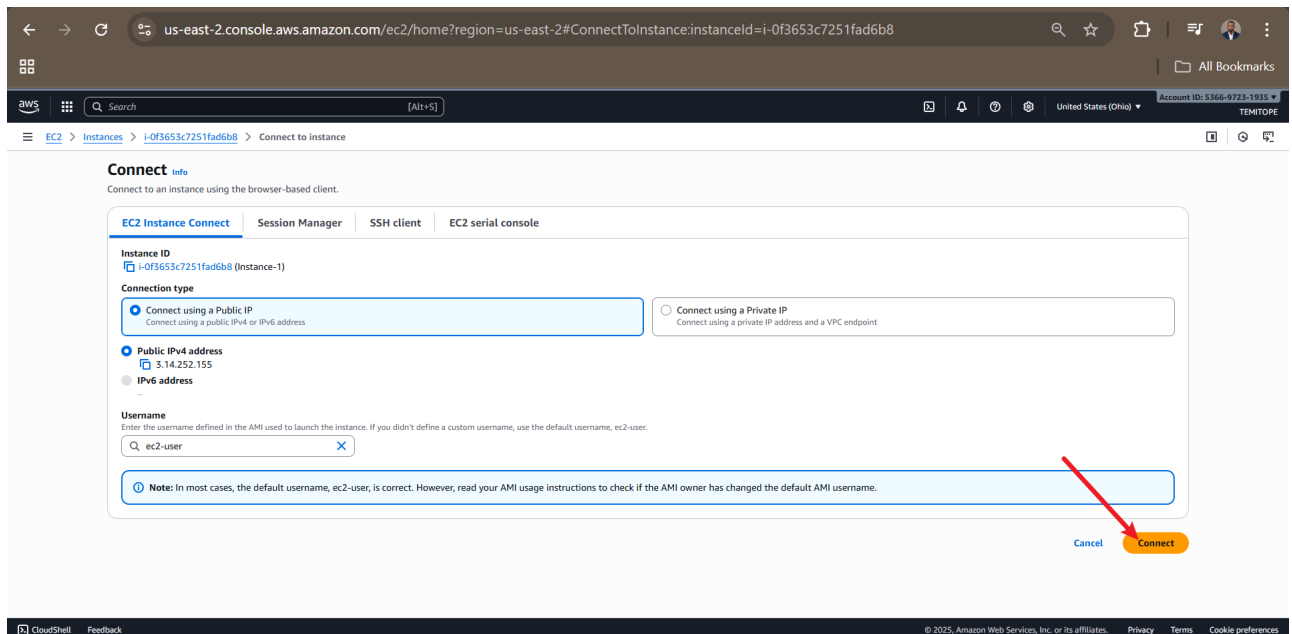
15. Click on instance-1 ID (may have another name in yours).



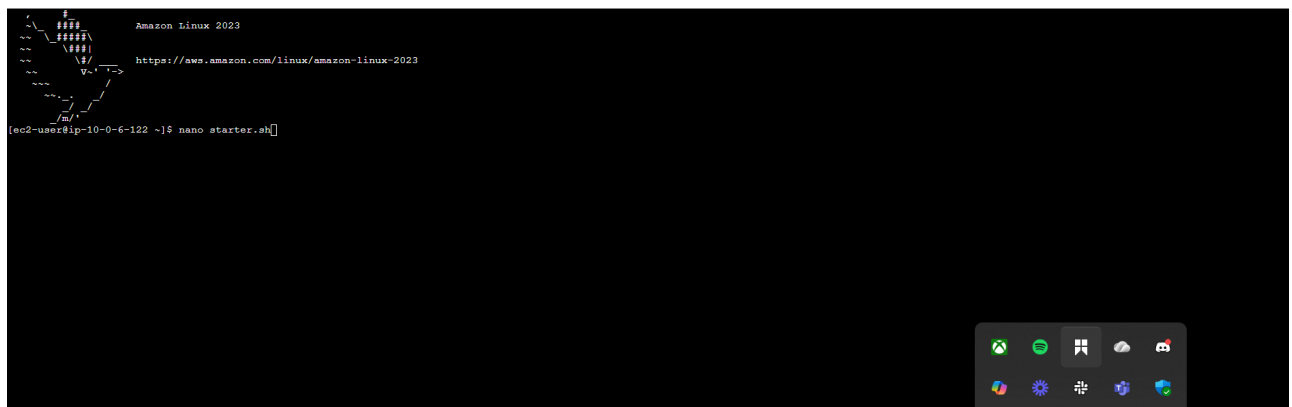
16. On instance-1 dashboard click connect.



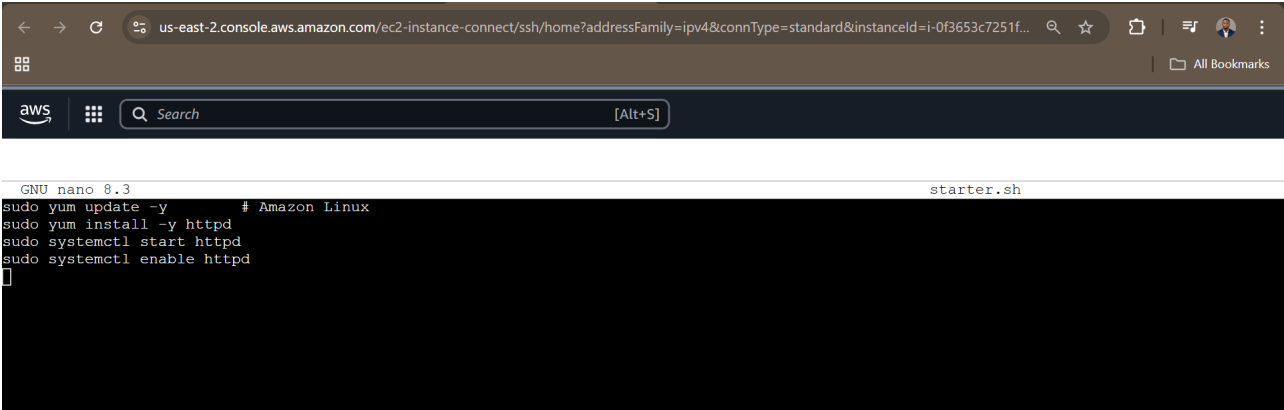
17. Under EC2 instance connect tab click connect.



18. Now connected create starter-sh with nano.



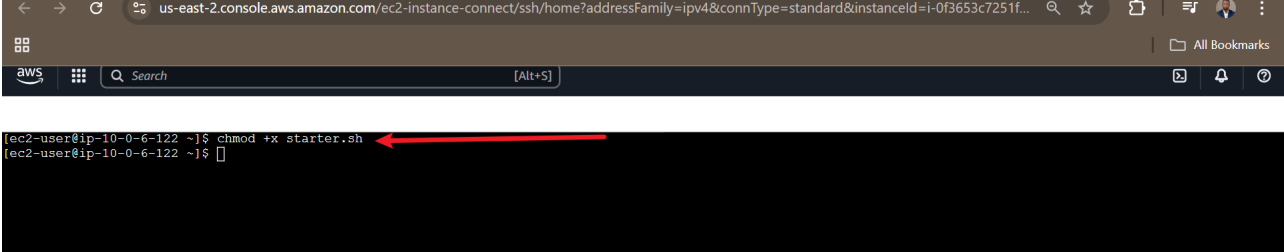
19. Add Apache installation script (I have Amazon Linux instance).



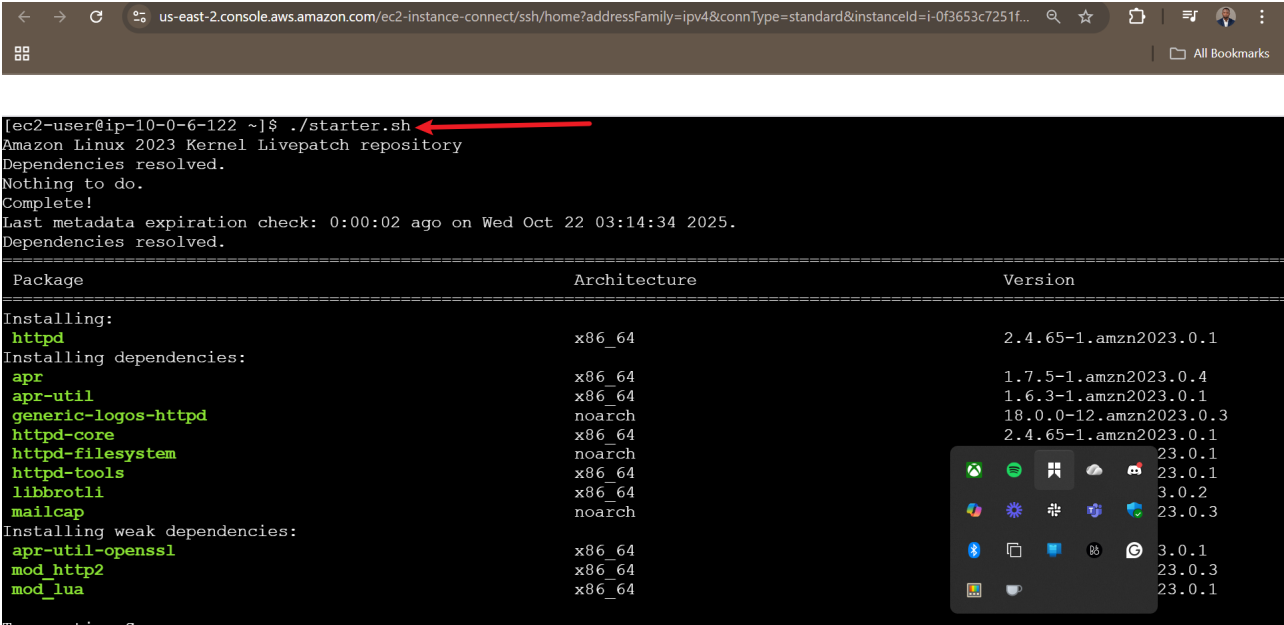
20. Save and exit (Ctrl-X and Y Enter).



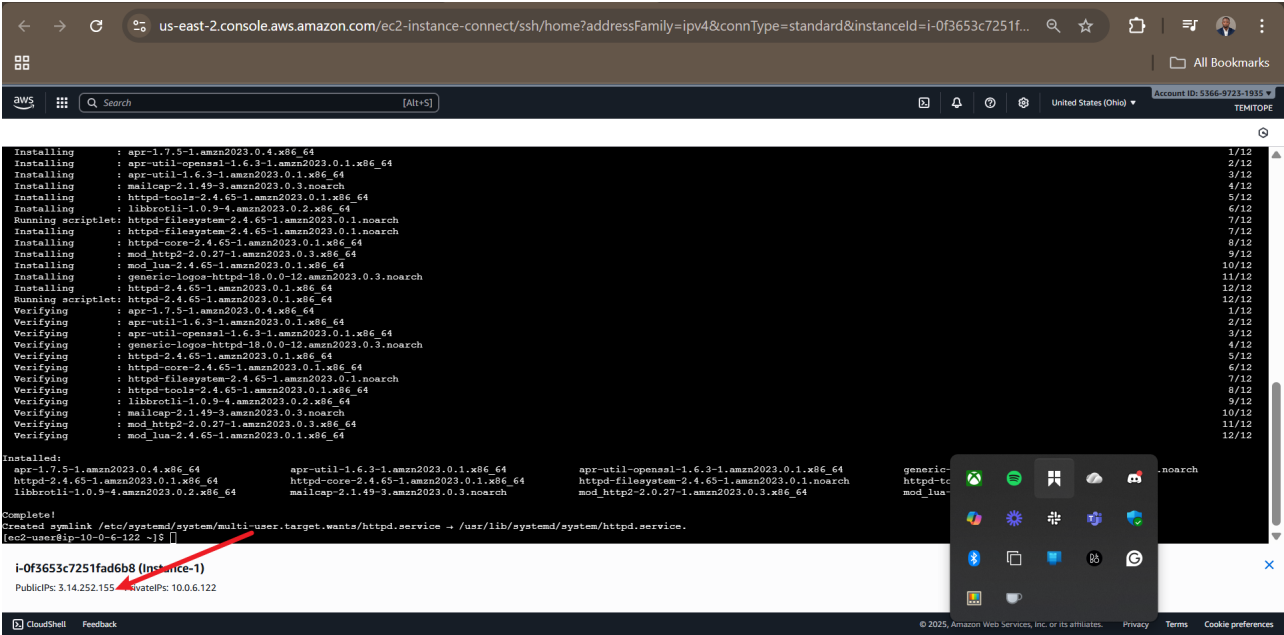
21. Turn starter-sh to executable (chmod +x).



22. Execute starter-sh to install httpd and enable it.



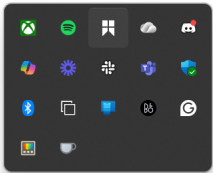
23. Copy the public IP.



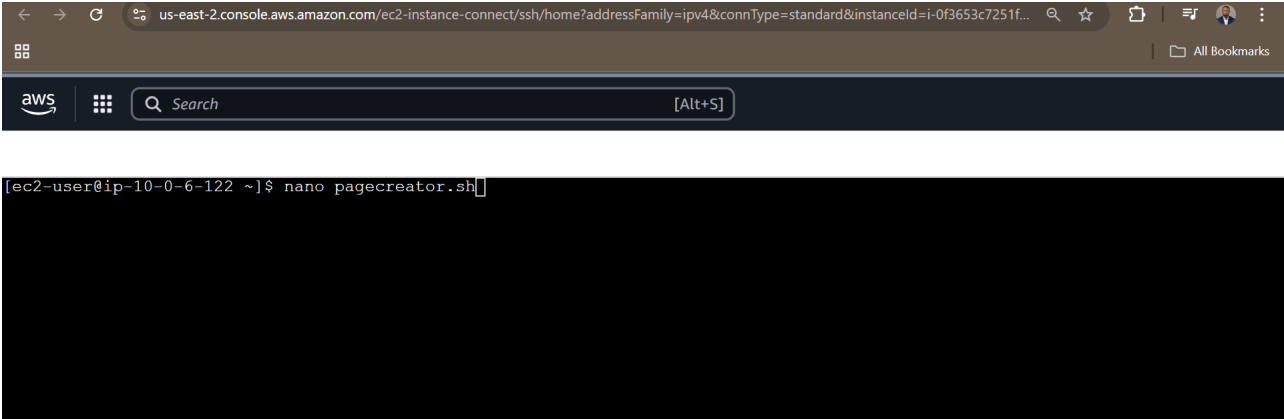
24. Paste it on a browser and visit it (via http) and the default page will be displayed.



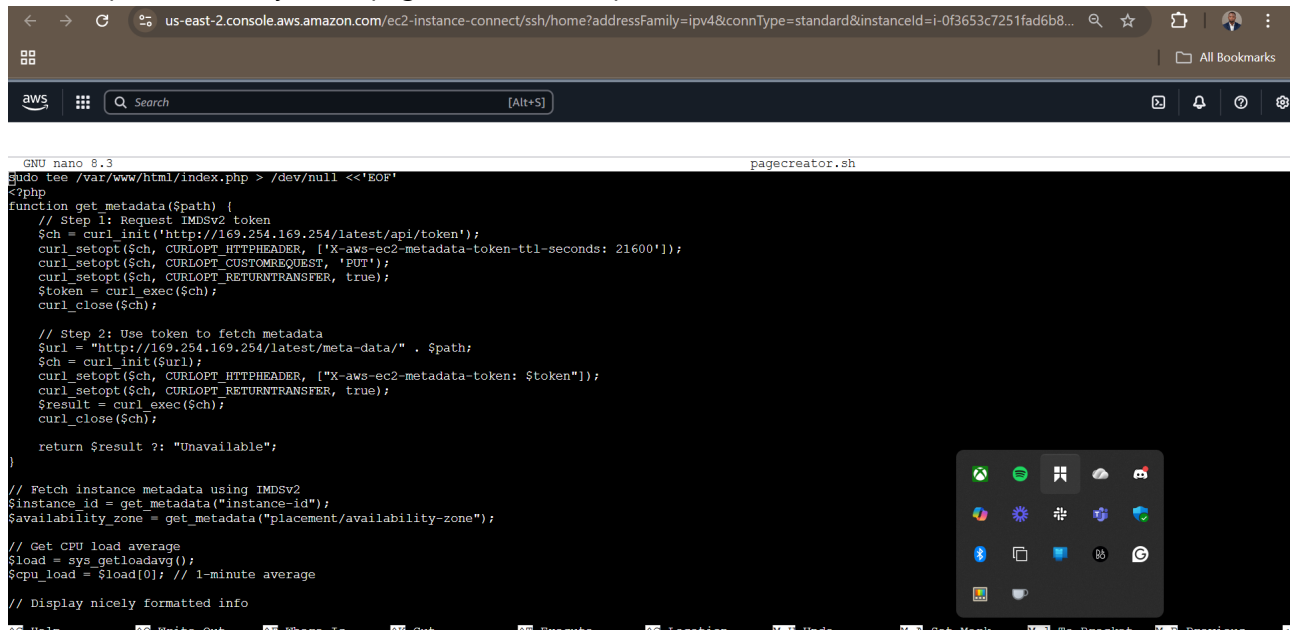
It works!



25. Back to instance connect CLI create pagecreator-sh (nano).



26. Add script to create dynamic page with PHP and provide data.



```
GNU nano 8.3 pagecreator.sh
sudo tee /var/www/html/index.php > /dev/null <<'EOF'
<?php
function get_metadata($path) {
    // Step 1: Request IMDSv2 token
    $ch = curl_init('http://169.254.169.254/latest/api/token');
    curl_setopt($ch, CURLOPT_HTTPHEADER, ['X-aws-ec2-metadata-token-ttl-seconds: 21600']);
    curl_setopt($ch, CURLOPT_CUSTOMREQUEST, 'PUT');
    curl_setopt($ch, CURLOPT_RETURNTRANSFER, true);
    $token = curl_exec($ch);
    curl_close($ch);

    // Step 2: Use token to fetch metadata
    $url = "http://169.254.169.254/latest/meta-data/" . $path;
    $ch = curl_init($url);
    curl_setopt($ch, CURLOPT_HTTPHEADER, ["X-aws-ec2-metadata-token: $token"]);
    curl_setopt($ch, CURLOPT_RETURNTRANSFER, true);
    $result = curl_exec($ch);
    curl_close($ch);

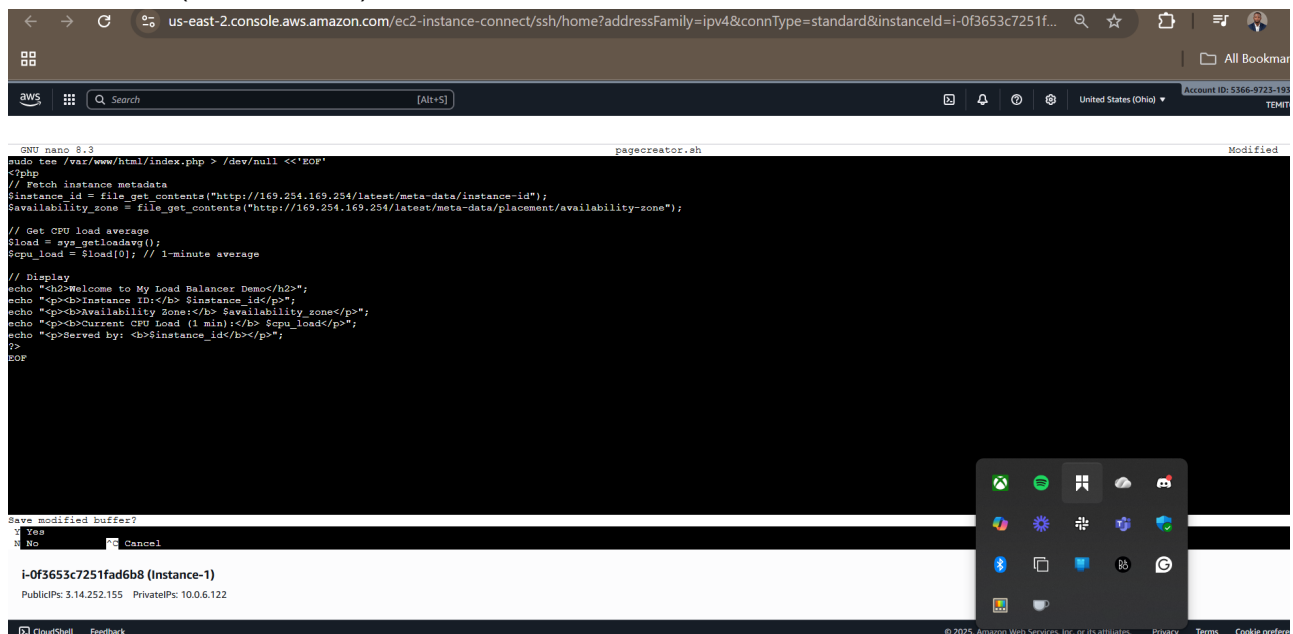
    return $result ?: "Unavailable";
}

// Fetch instance metadata using IMDSv2
$instance_id = get_metadata("instance-id");
$availability_zone = get_metadata("placement/availability-zone");

// Get CPU load average
$load = sys_getloadavg();
$cpu_load = $load[0]; // 1-minute average

// Display nicely formatted info
echo "Instance ID: $instance_id\n";
echo "Availability Zone: $availability_zone\n";
echo "CPU Load (1 min): $cpu_load\n";
EOF
```

27. Save and exit (Ctrl+X Y Enter).



```
GNU nano 8.3 pagecreator.sh Modified
sudo tee /var/www/html/index.php > /dev/null <<'EOF'
<?php
// Fetch instance metadata
$instance_id = file_get_contents("http://169.254.169.254/latest/meta-data/instance-id");
$availability_zone = file_get_contents("http://169.254.169.254/latest/meta-data/placement/availability-zone");

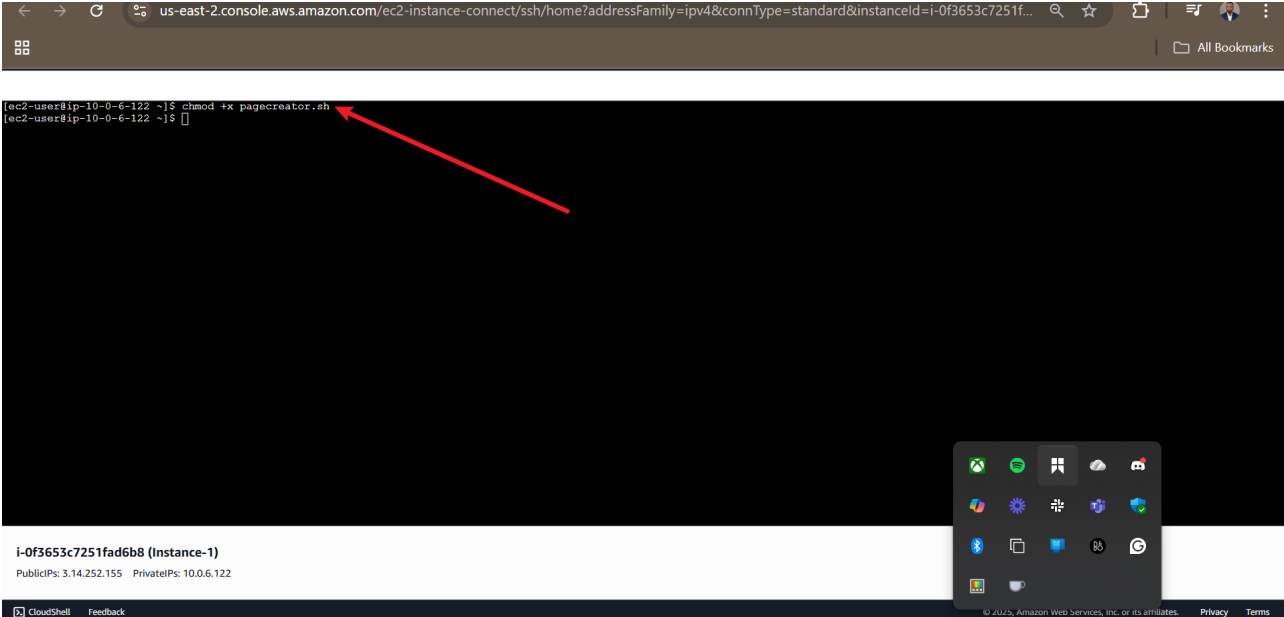
// Get CPU load average
$load = sys_getloadavg();
$cpu_load = $load[0]; // 1-minute average

// Display
echo "<h2>Welcome to My Load Balancer Demo</h2>";
echo "<p><b>Instance ID:</b> $instance_id</p>";
echo "<p><b>Availability Zone:</b> $availability_zone</p>";
echo "<p><b>Current CPU Load (1 min):</b> $cpu_load</p>";
echo "<p>Served by: <b>$instance_id</b></p>";
EOF

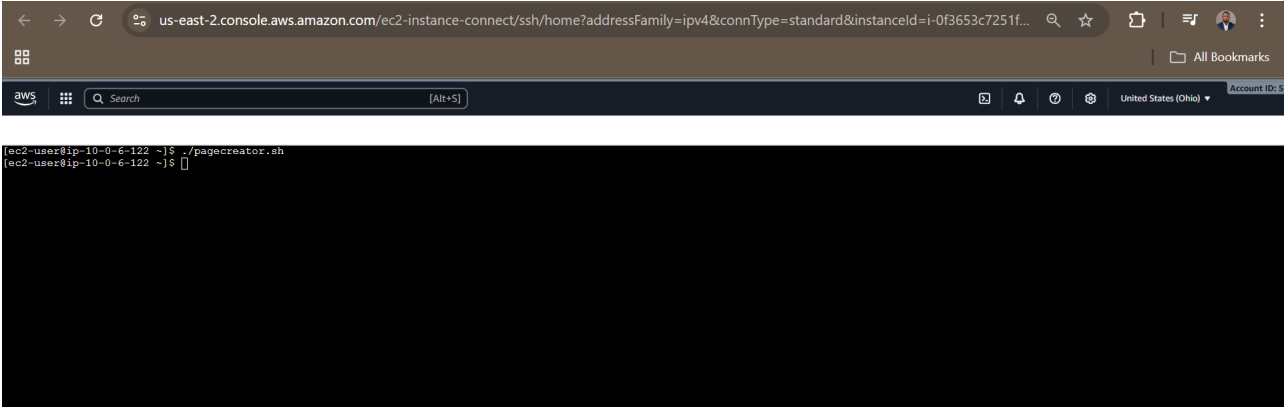
Save modified buffer?
Y Yes
N No
Cancel

i-0f3653c7251fad6b8 (Instance-1)
PublicIPs: 3.14.252.155 PrivateIPs: 10.0.6.122
```

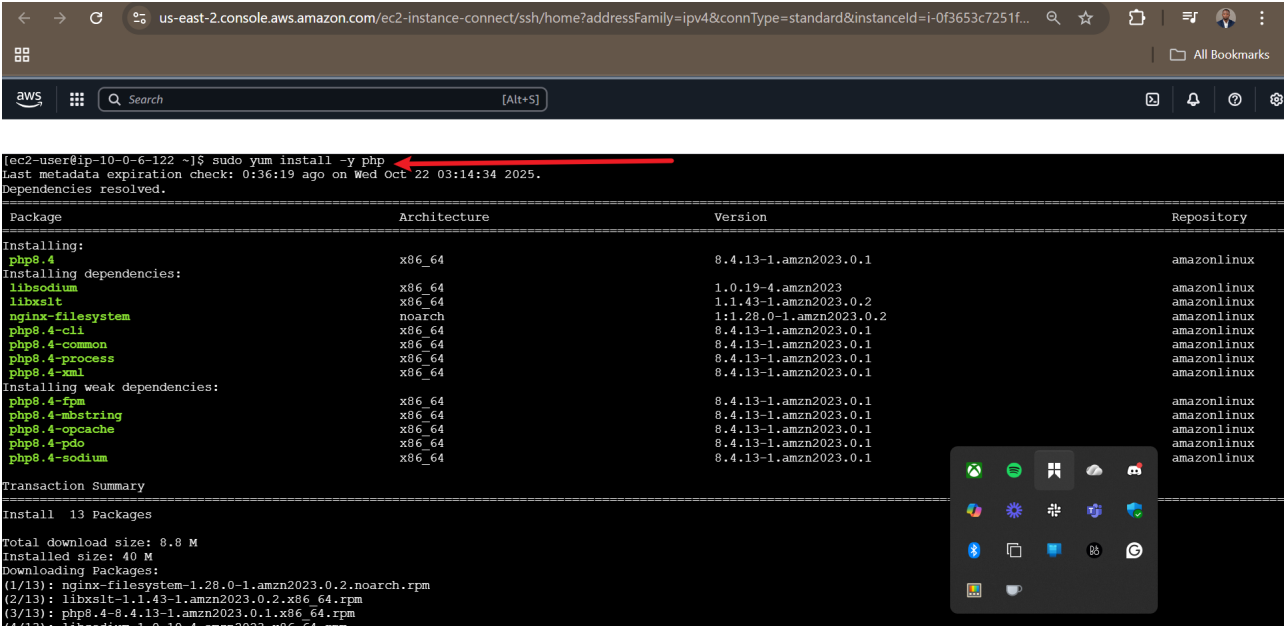
28. Turn pagecreator-sh to executable (chmod +x).



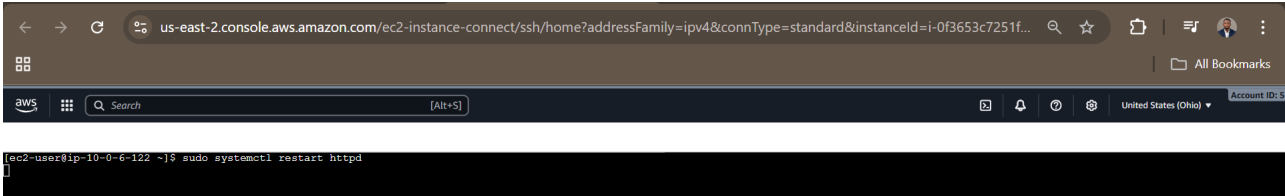
29. Execute page creator-sh to create page.



30. Yum install PHP.



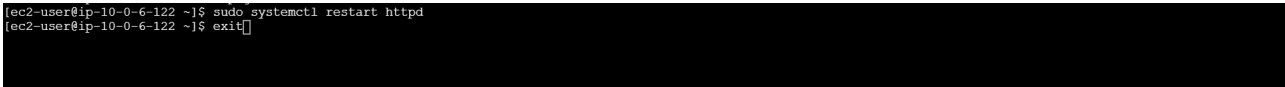
31. Restart Apache with systemctl.



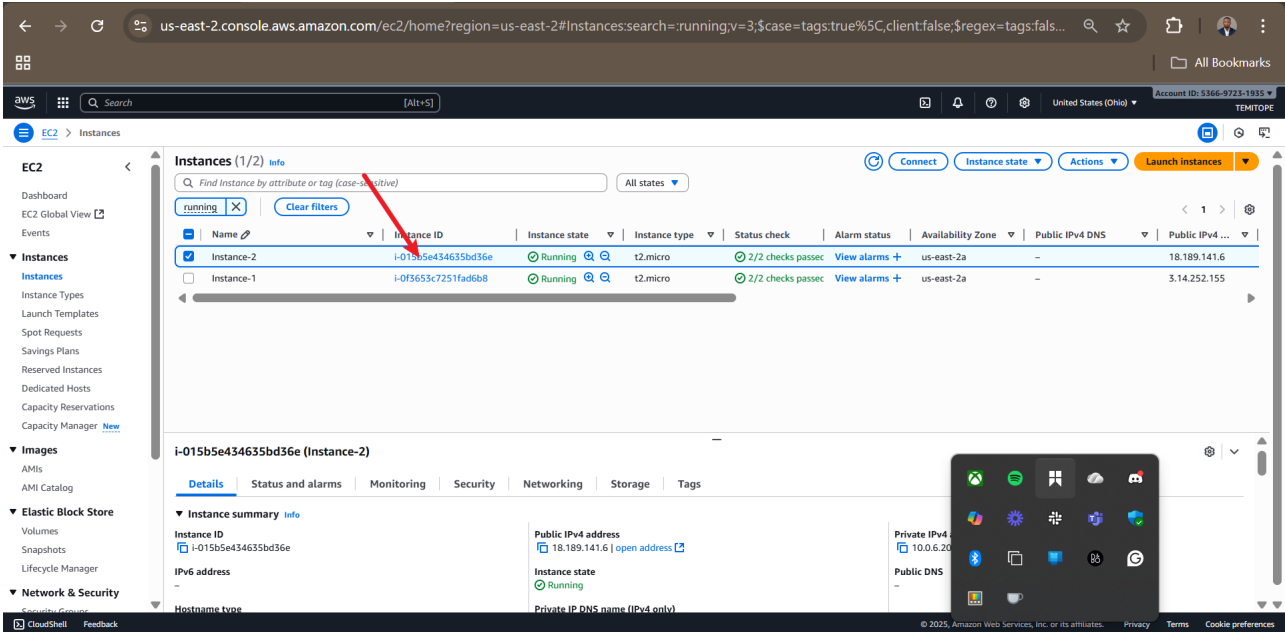
32. Visit the page again website now display instance-1 details.



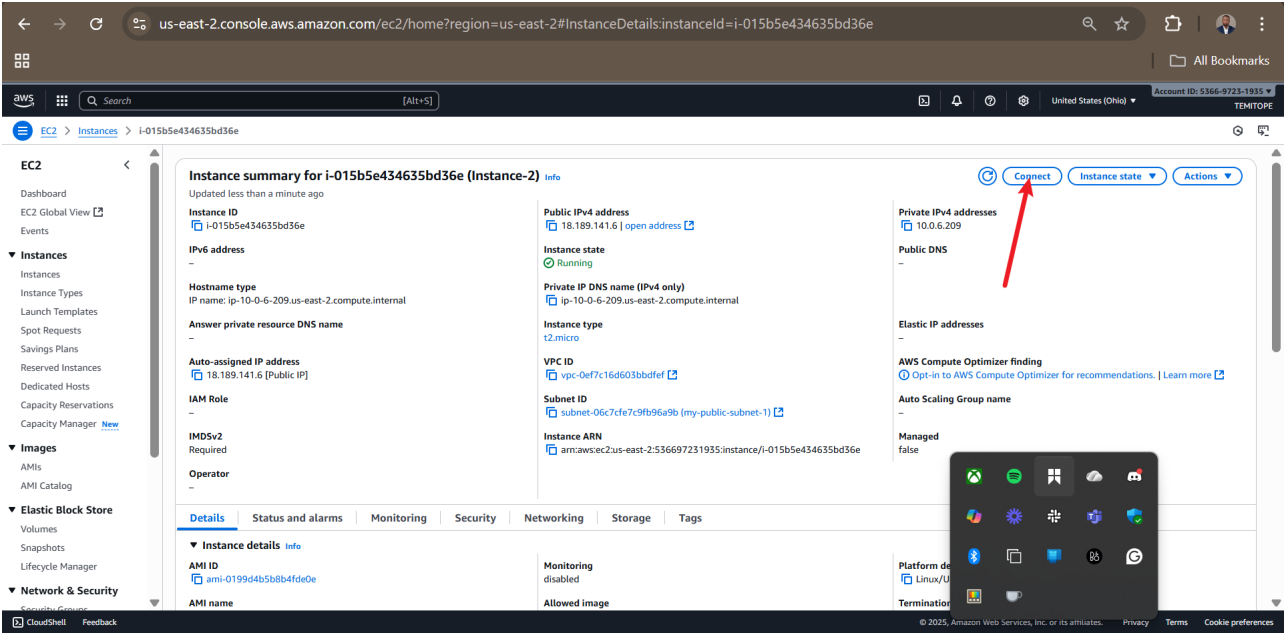
33. Exit CLI.



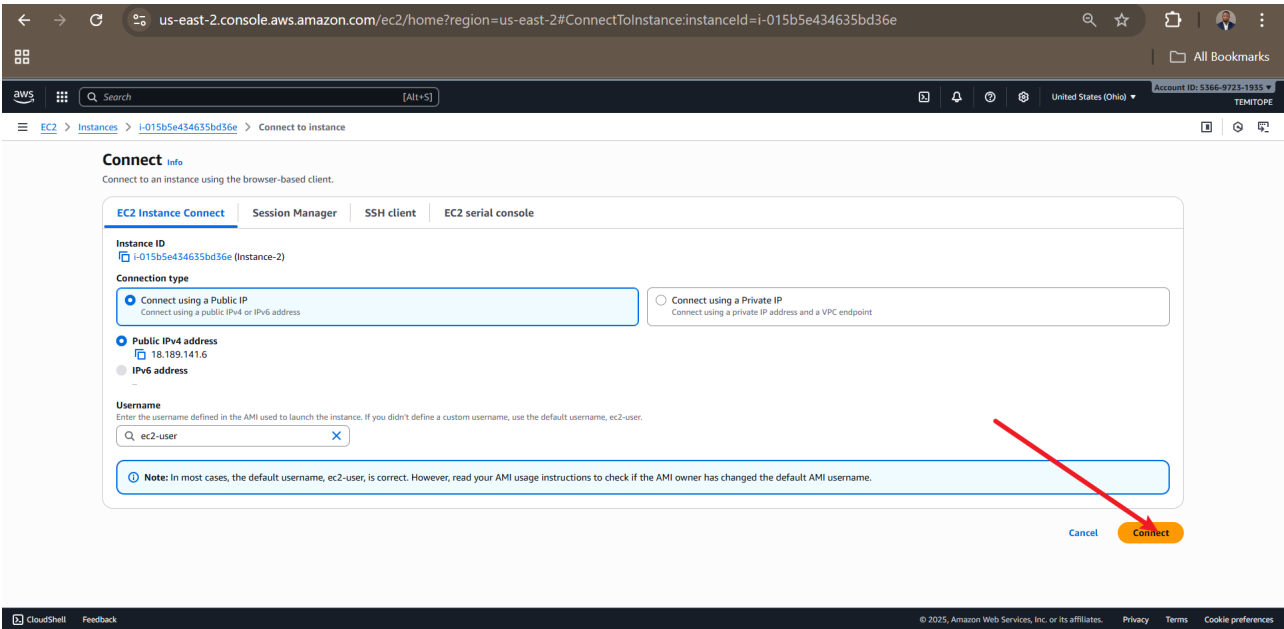
34. Navigate back to instances dashboard and click on instance-2 ID.



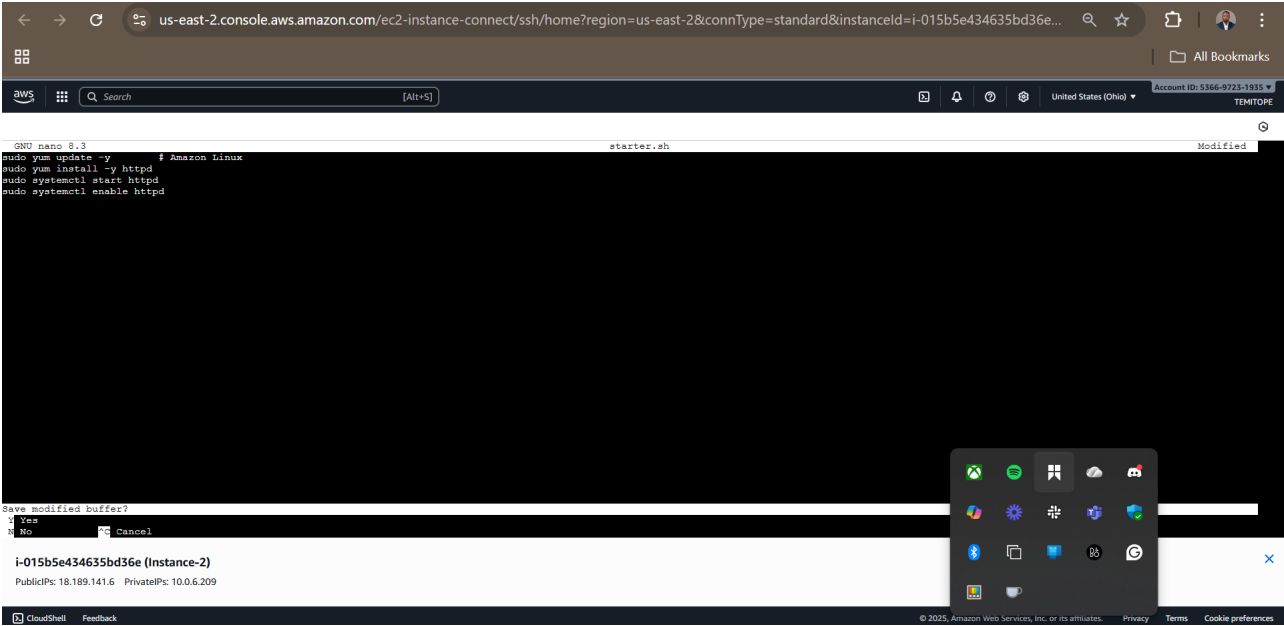
35. On instance-2 dashboard click connect.



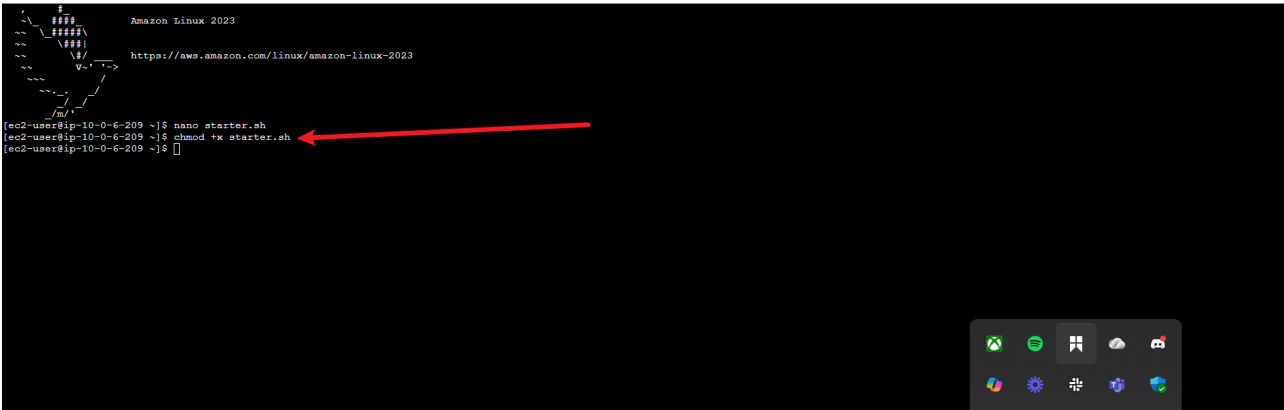
36. Under EC2 instance connect tab click connect.



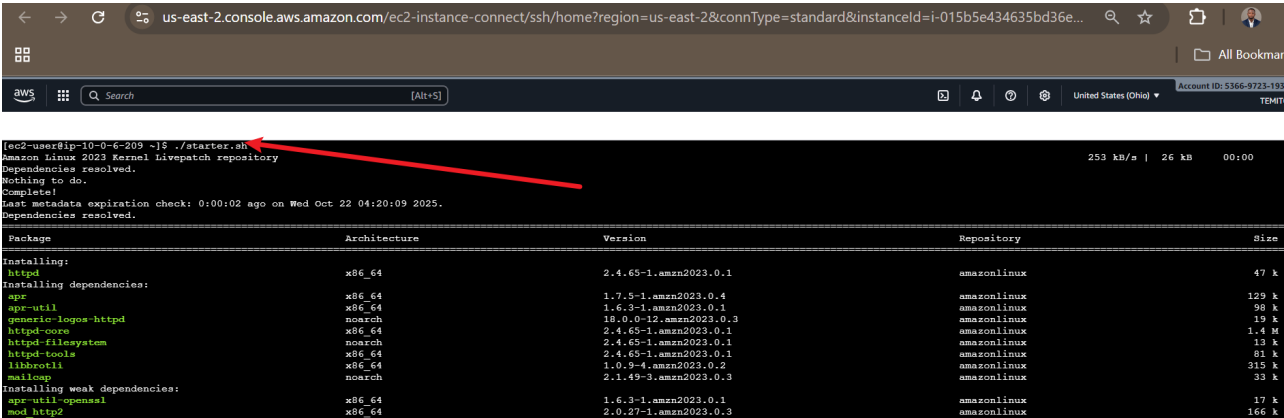
37. Nano starter-sh add Apache installation script save and exit.



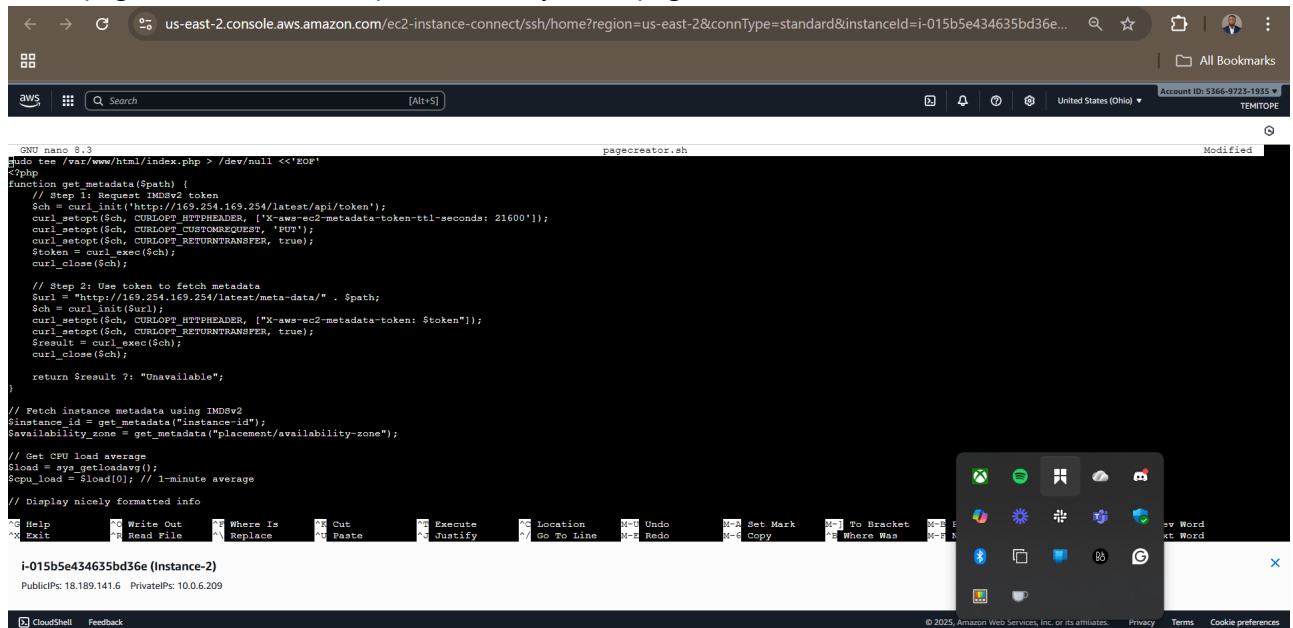
38. Turn starter-sh to executable (chmod +x).



39. Execute starter-sh to install httpd and enable it.



40. Nano pagecreator-sh add script to create dynamic page.



```
GNU nano 8.3 pagecreator.sh
#!/bin/bash
sudo tee /var/www/html/index.php > /dev/null <<'EOF'
<:php
function get_metadata($path) {
    // Step 1: Request IMDSv2 token
    $ch = curl_init('http://169.254.169.254/latest/api/token');
    curl_setopt($ch, CURLOPT_HTTPHEADER, ['X-aws-ec2-metadata-token-ttl-seconds: 21600']);
    curl_setopt($ch, CURLOPT_CUSTOMREQUEST, 'PUT');
    curl_setopt($ch, CURLOPT_RETURNTRANSFER, true);
    $token = curl_exec($ch);
    curl_close($ch);

    // Step 2: Use token to fetch metadata
    $url = "http://169.254.169.254/latest/meta-data/" . $path;
    $ch = curl_init($url);
    curl_setopt($ch, CURLOPT_HTTPHEADER, ["X-aws-ec2-metadata-token: $token"]);
    curl_setopt($ch, CURLOPT_RETURNTRANSFER, true);
    $result = curl_exec($ch);
    curl_close($ch);

    return $result ?: "Unavailable";
}

// Fetch instance metadata using IMDSv2
$instance_id = get_metadata("instance-id");
$availability_zone = get_metadata("placement/availability-zone");

// Get CPU load average
$load = sys_getloadavg();
$cpu_load = $load[0]; // 1-minute average

// Display nicely formatted info
echo "Instance ID: $instance_id\n";
echo "Availability Zone: $availability_zone\n";
echo "CPU Load (1-min): $cpu_load\n";

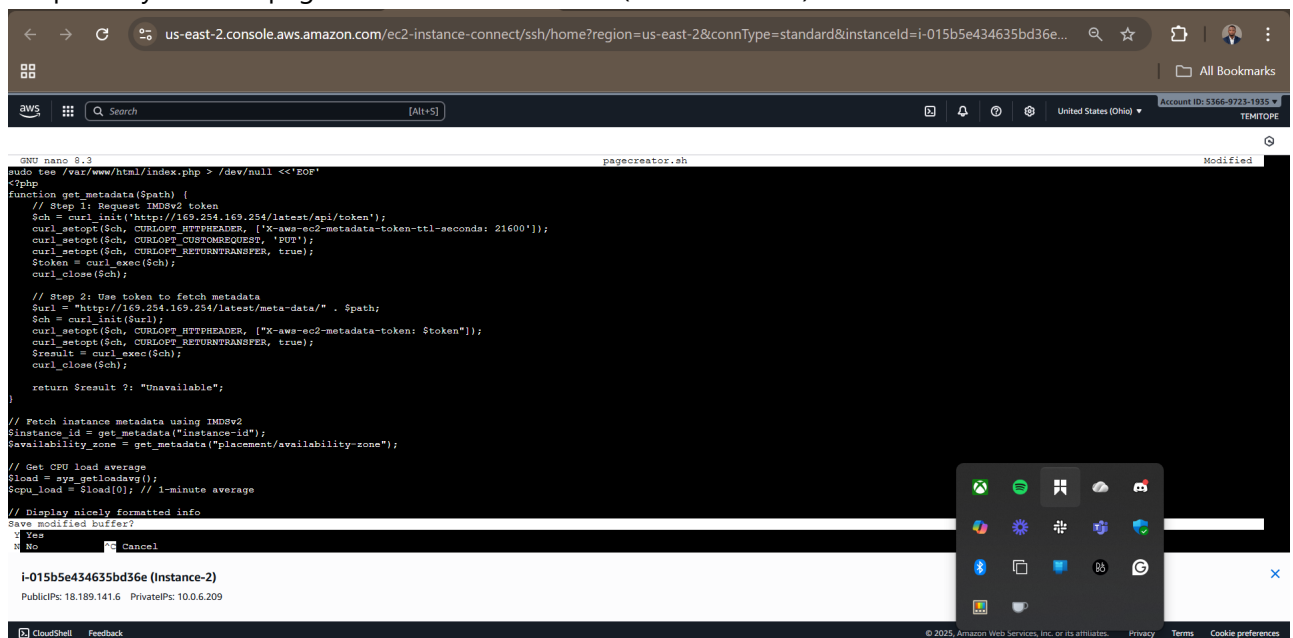
}

Help Write Out Where Is Out Execute Location Go To Line Undo Get Mark To Bracket
Exit Read File Replace Paste Justify Where Was Copy Where Was Word at Word

i-015b5e434635bd36e (Instance-2)
PublicIPs: 18.189.141.6 PrivateIPs: 10.0.6.209

CloudShell Feedback
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```

41. Script ready to serve page and data save and exit (Ctrl+X Y Enter).



```
GNU nano 8.3 pagecreator.sh
#!/bin/bash
sudo tee /var/www/html/index.php > /dev/null <<'EOF'
<:php
function get_metadata($path) {
    // Step 1: Request IMDSv2 token
    $ch = curl_init('http://169.254.169.254/latest/api/token');
    curl_setopt($ch, CURLOPT_HTTPHEADER, ['X-aws-ec2-metadata-token-ttl-seconds: 21600']);
    curl_setopt($ch, CURLOPT_CUSTOMREQUEST, 'PUT');
    curl_setopt($ch, CURLOPT_RETURNTRANSFER, true);
    $token = curl_exec($ch);
    curl_close($ch);

    // Step 2: Use token to fetch metadata
    $url = "http://169.254.169.254/latest/meta-data/" . $path;
    $ch = curl_init($url);
    curl_setopt($ch, CURLOPT_HTTPHEADER, ["X-aws-ec2-metadata-token: $token"]);
    curl_setopt($ch, CURLOPT_RETURNTRANSFER, true);
    $result = curl_exec($ch);
    curl_close($ch);

    return $result ?: "Unavailable";
}

// Fetch instance metadata using IMDSv2
$instance_id = get_metadata("instance-id");
$availability_zone = get_metadata("placement/availability-zone");

// Get CPU load average
$load = sys_getloadavg();
$cpu_load = $load[0]; // 1-minute average

// Display nicely formatted info
echo "Instance ID: $instance_id\n";
echo "Availability Zone: $availability_zone\n";
echo "CPU Load (1-min): $cpu_load\n";

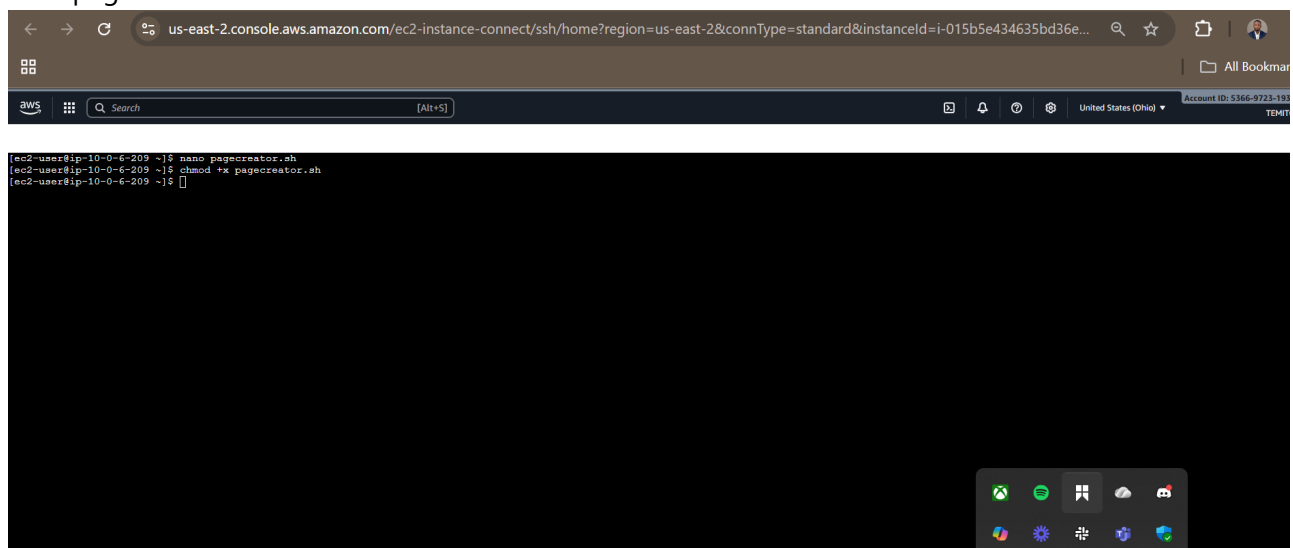
}

Help Write Out Where Is Out Execute Location Go To Line Undo Get Mark To Bracket
Exit Read File Replace Paste Justify Where Was Copy Where Was Word at Word

i-015b5e434635bd36e (Instance-2)
PublicIPs: 18.189.141.6 PrivateIPs: 10.0.6.209

CloudShell Feedback
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```

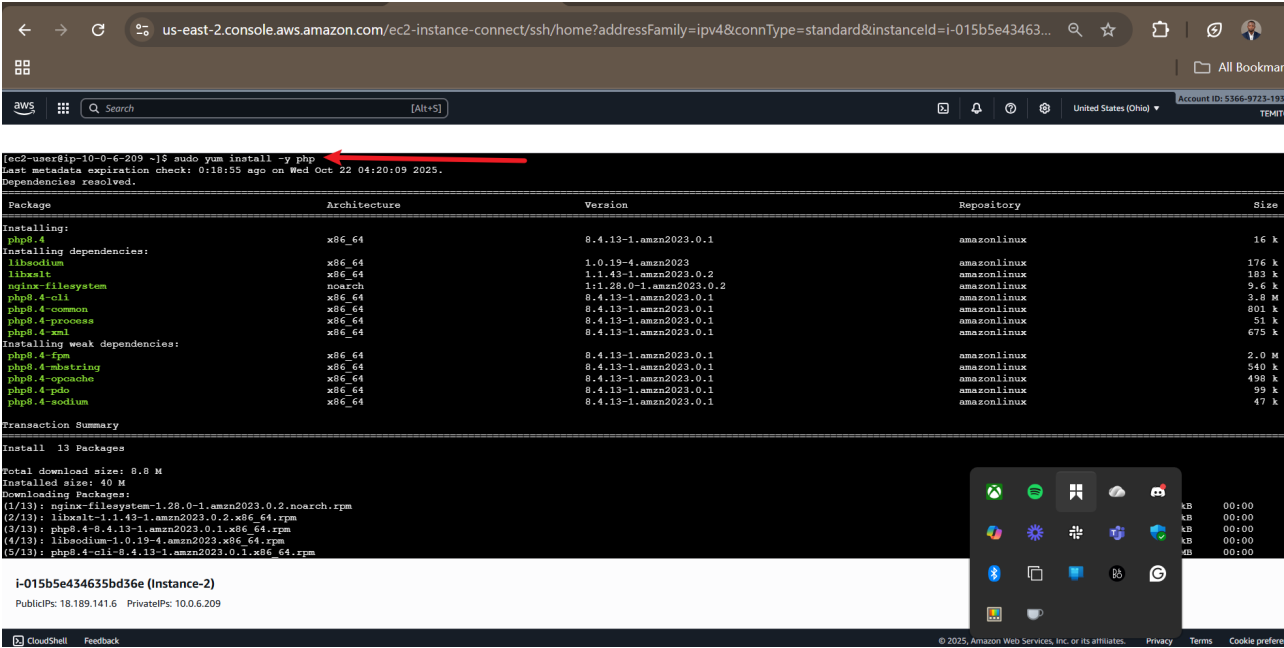
42. Turn page creator-sh to executable and execute it.



```
[ec2-user@ip-10-0-6-209 ~]$ nano pagecreator.sh
[ec2-user@ip-10-0-6-209 ~]$ chmod +x pagecreator.sh
[ec2-user@ip-10-0-6-209 ~]$ ./pagecreator.sh

Instance ID: i-015b5e434635bd36e
Availability Zone: us-east-2a
CPU Load (1-min): 0.05
```

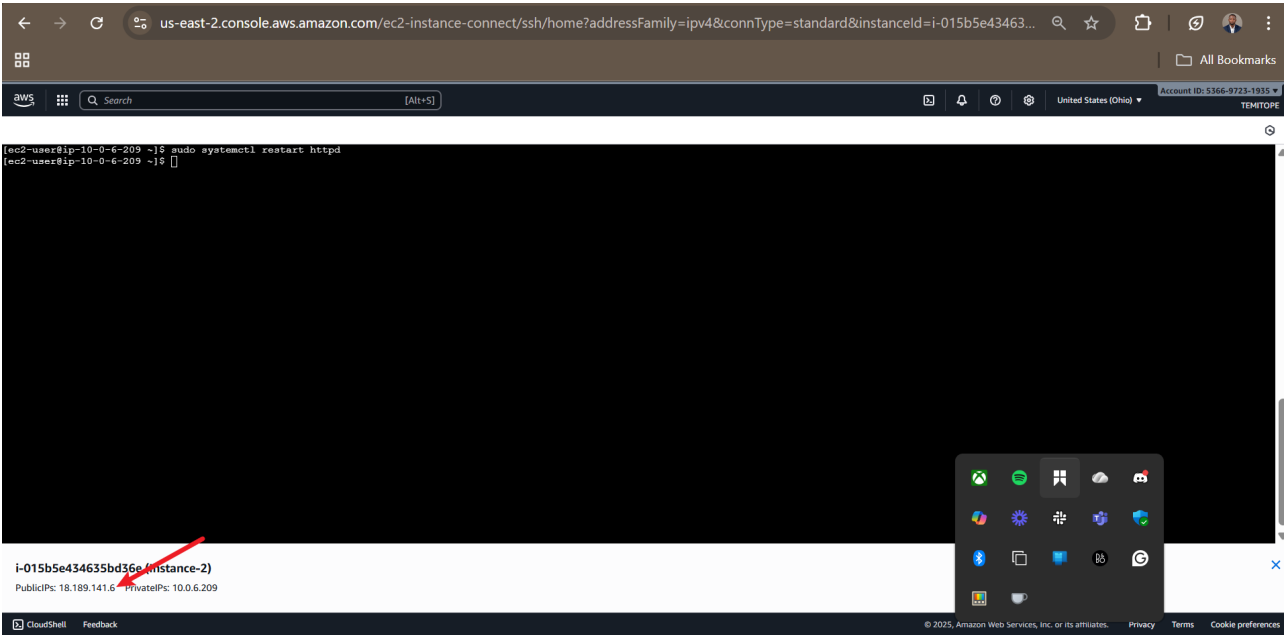
43. Yum install PHP.



44. Restart Apache with systemctl.



45. Copy the public IP to visit it on the browser (http).



46. Instance 2 is showing its details.

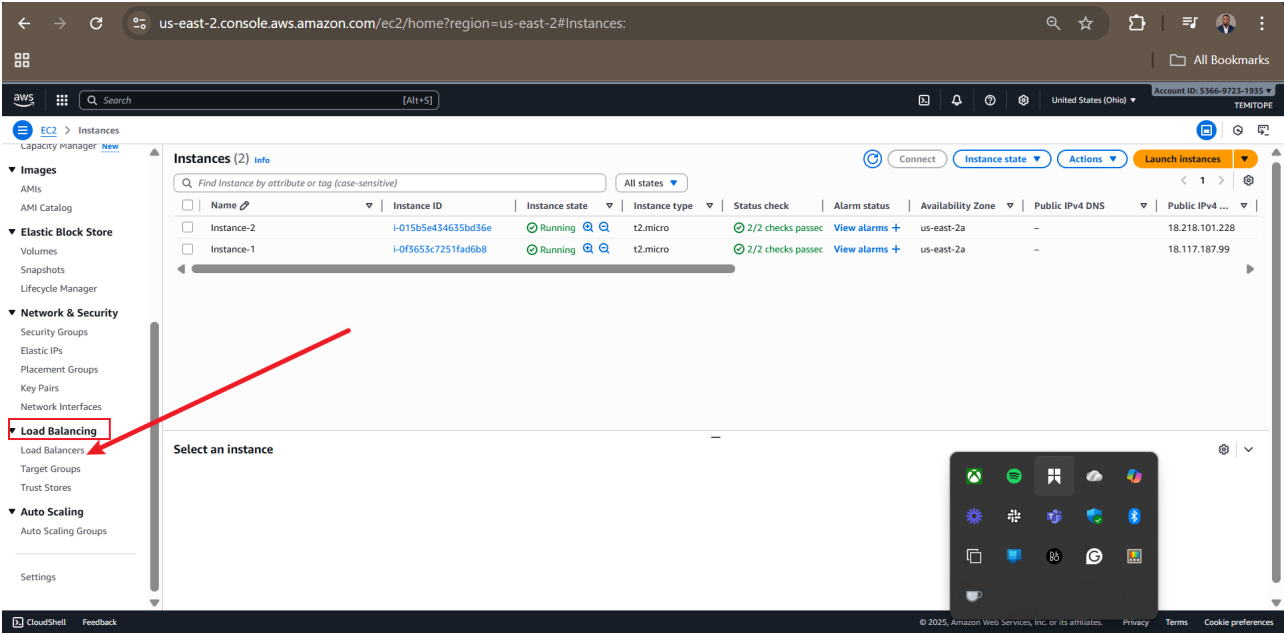
The screenshot shows the AWS Management Console interface. At the top, there's a navigation bar with the AWS logo and a search bar. Below that, the main content area displays the 'Welcome to Instance-2' page. The page has a dark blue header with the title 'Welcome to Instance-2' in white. Below the header, there's a section with instance details: 'Instance ID: i-015b5e434635bd36e', 'Availability Zone: us-east-2a', 'Current CPU Load (1 min): 0%', and 'Served by: i-015b5e434635bd36e'. At the bottom right of the screenshot, a Windows taskbar is visible, showing various application icons and the system clock.

47. To create load balancer navigate to instances dashboard.

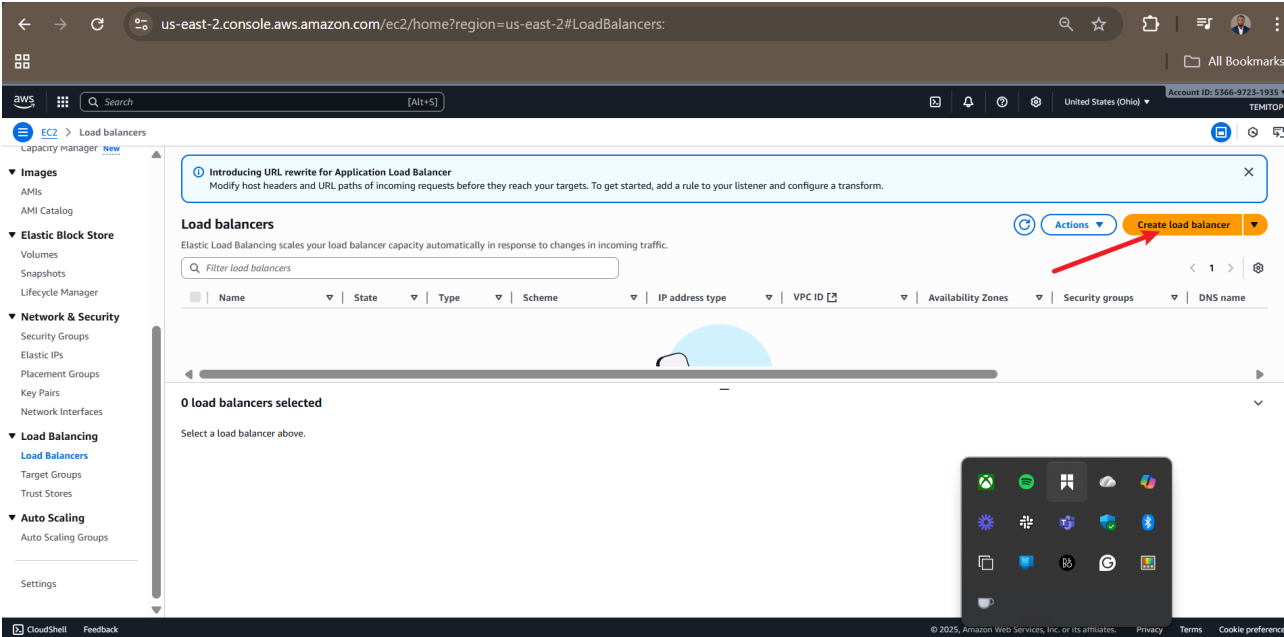
The screenshot shows the AWS Management Console for the 'us-east-2' region. The 'Instances' page is active, displaying a table of two EC2 instances. The left sidebar contains a navigation menu with categories like Dashboard, Events, Instances, Images, Elastic Block Store, and Network & Security. The top navigation bar includes the AWS logo, search bar, and account information. The main content area has a table of instances with columns for Name, Instance ID, Instance state, Instance type, Status check, Alarm status, Availability Zone, Public IPv4 DNS, and Public IPv4 address.

Name	Instance ID	Instance state	Instance type	Status check	Alarm status	Availability Zone	Public IPv4 DNS	Public IPv4 address
Instance-2	i-015b5e434635bd36e	Running	t2.micro	2/2 checks passed	View alarms +	us-east-2a	-	18.218.101.228
Instance-1	i-0f3655c7251fad6b8	Running	t2.micro	2/2 checks passed	View alarms +	us-east-2a	-	18.117.187.99

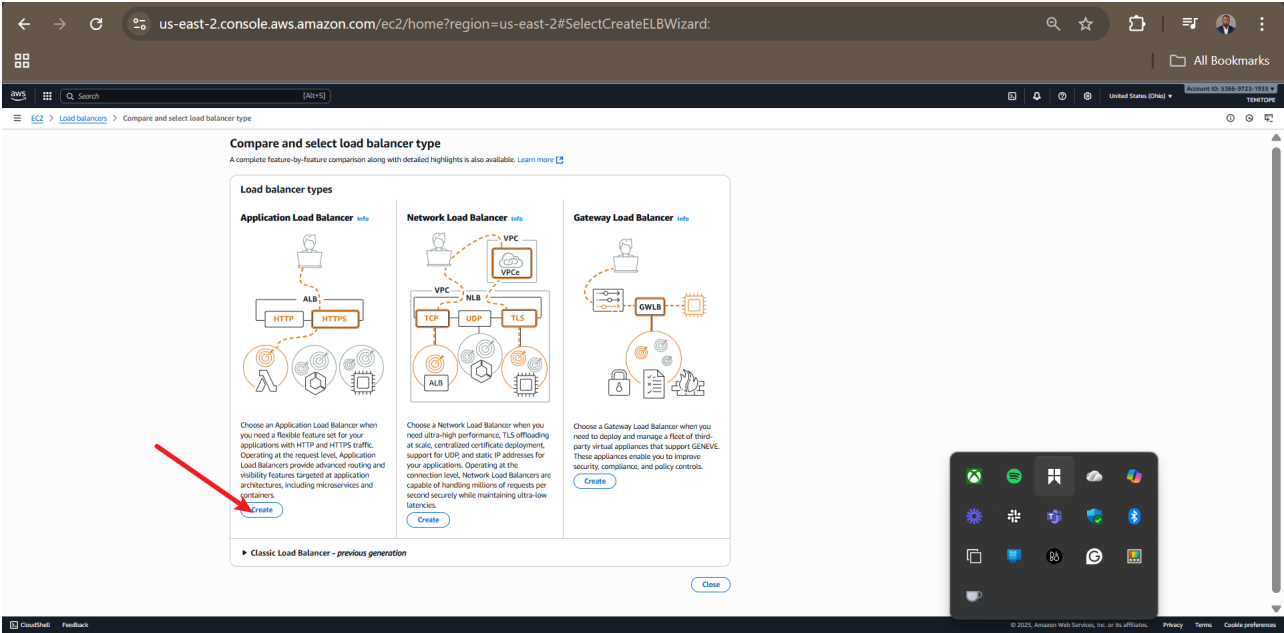
48. On the side menu scroll down to load balancing menu and click on load balancer.



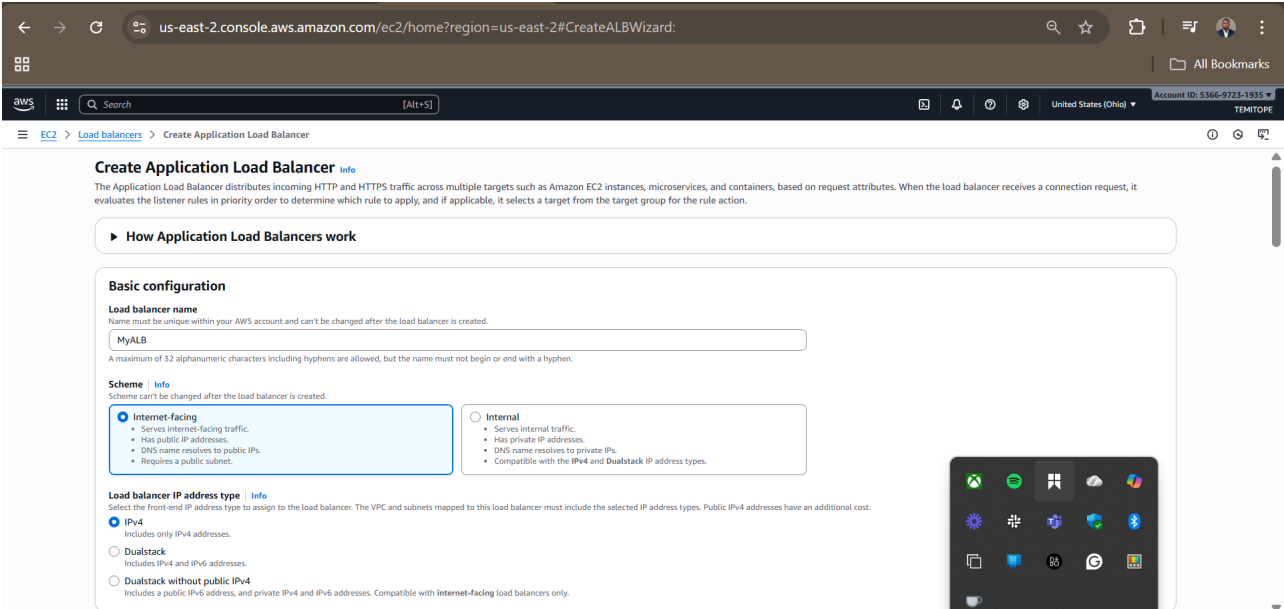
49. On load balancers dashboard click on create load balancer.



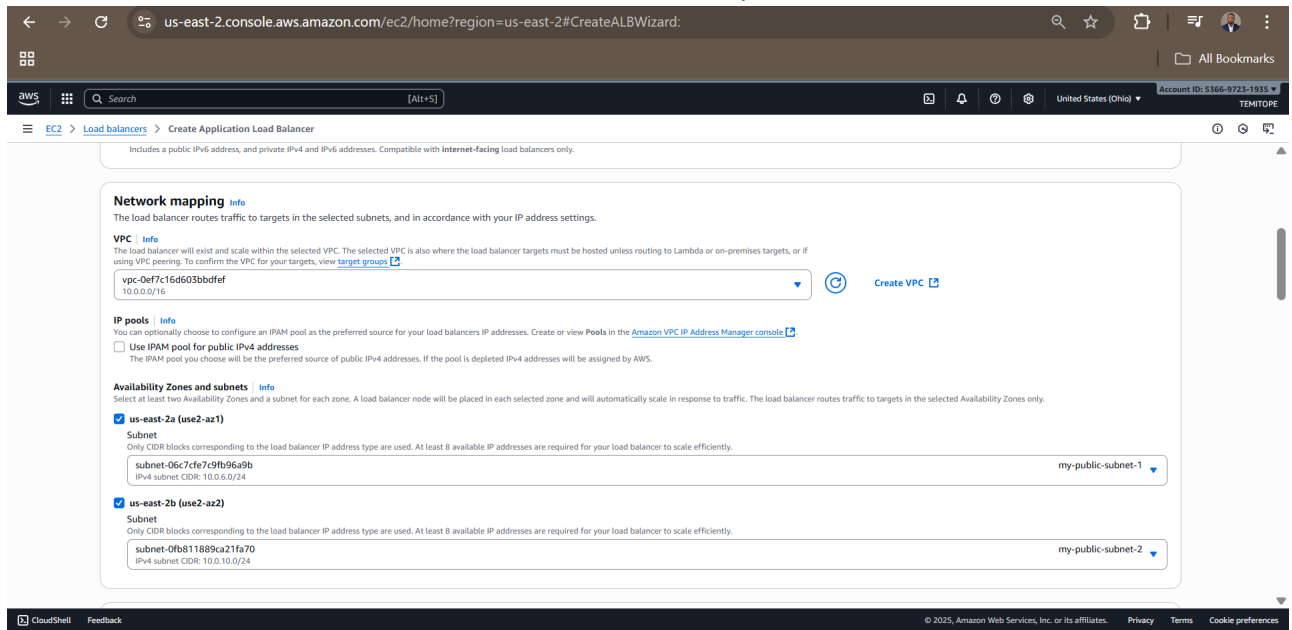
50. Click to create application load balancer.



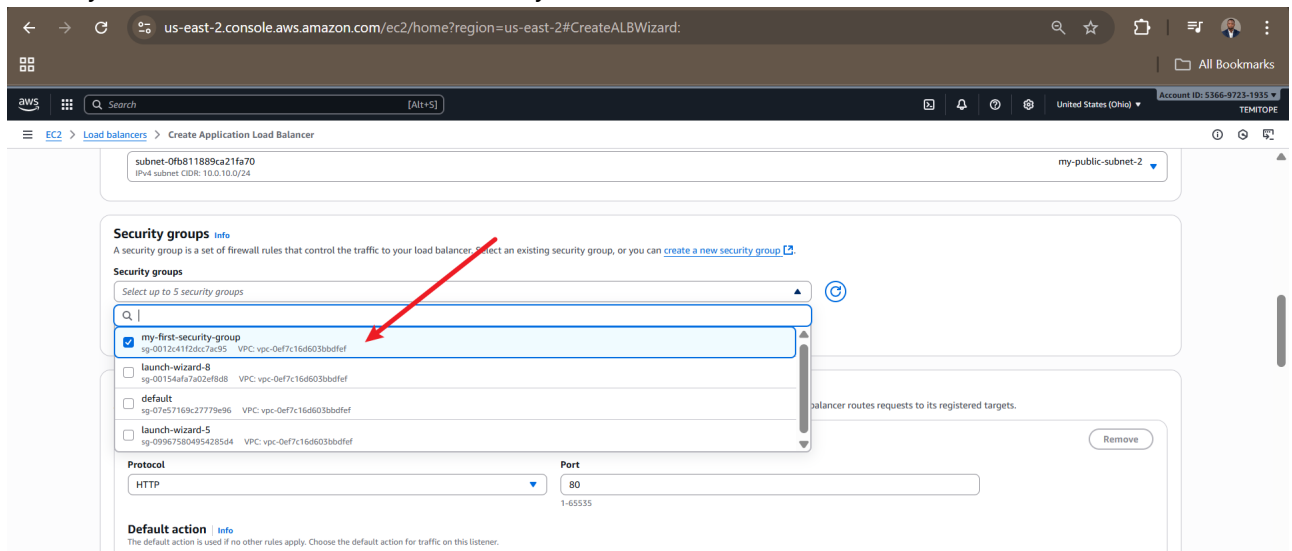
51. Name load balancer select internet facing and IPv4 then scroll down.



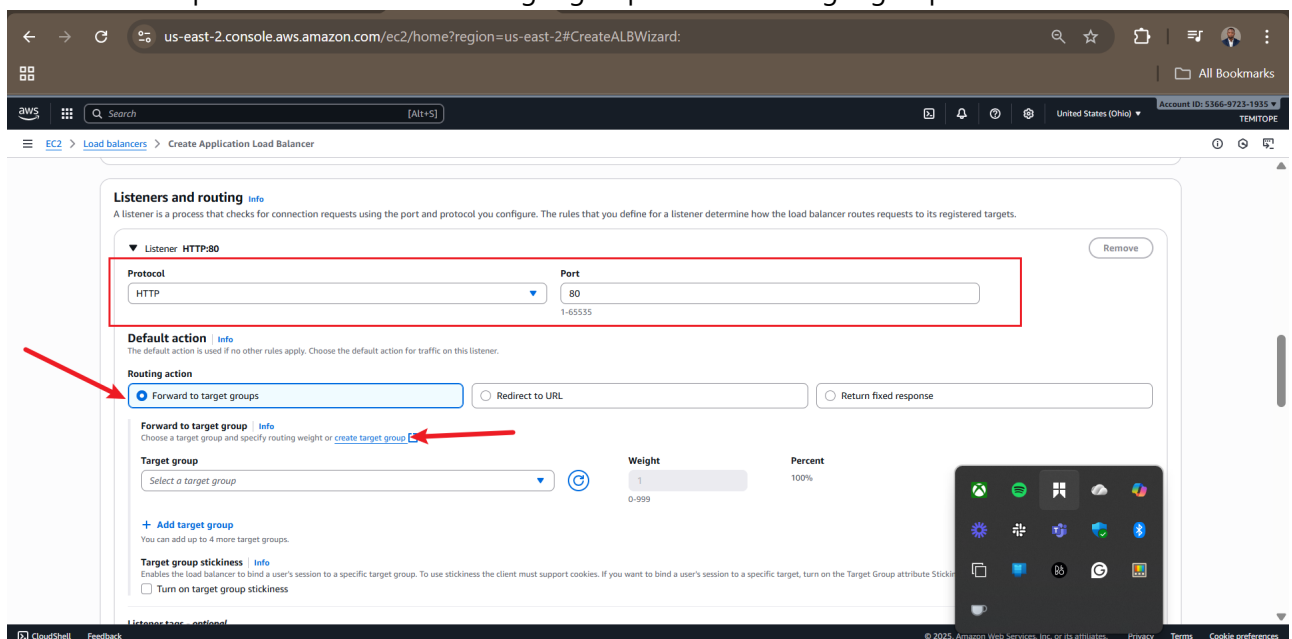
52. Select VPC the servers are check at least 2 AZ and subnets (public subnet) and scroll down.



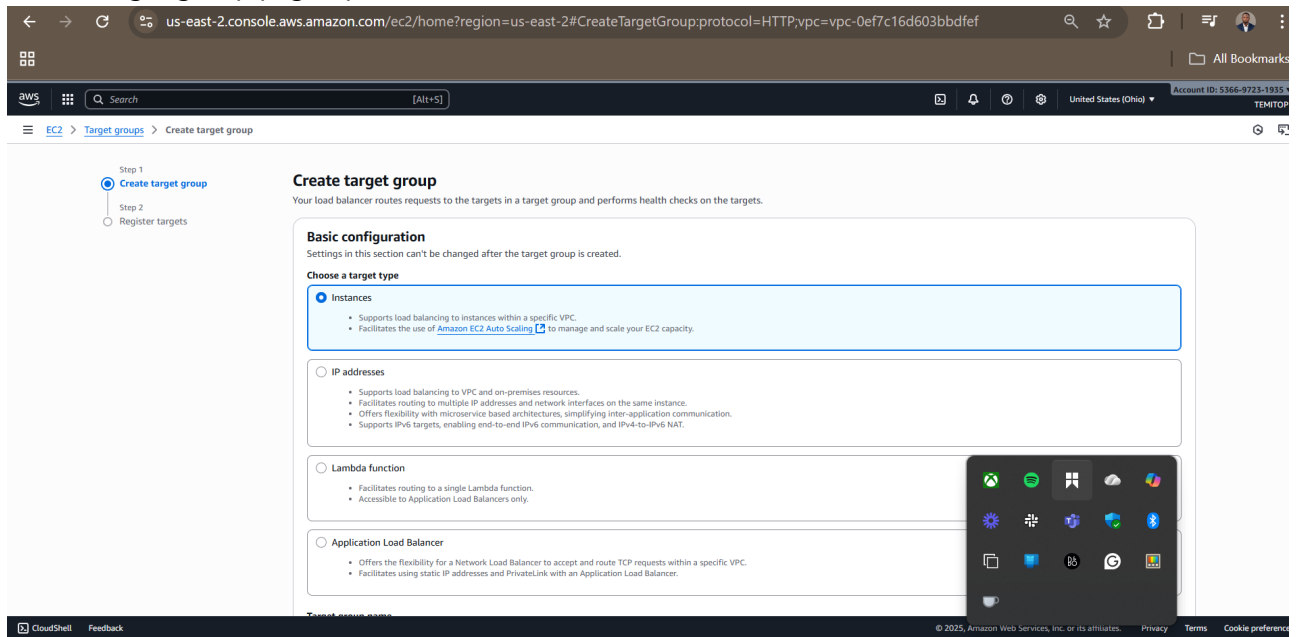
53. Select your SG or create one that allows any HTTP and SSH traffic and scroll down.



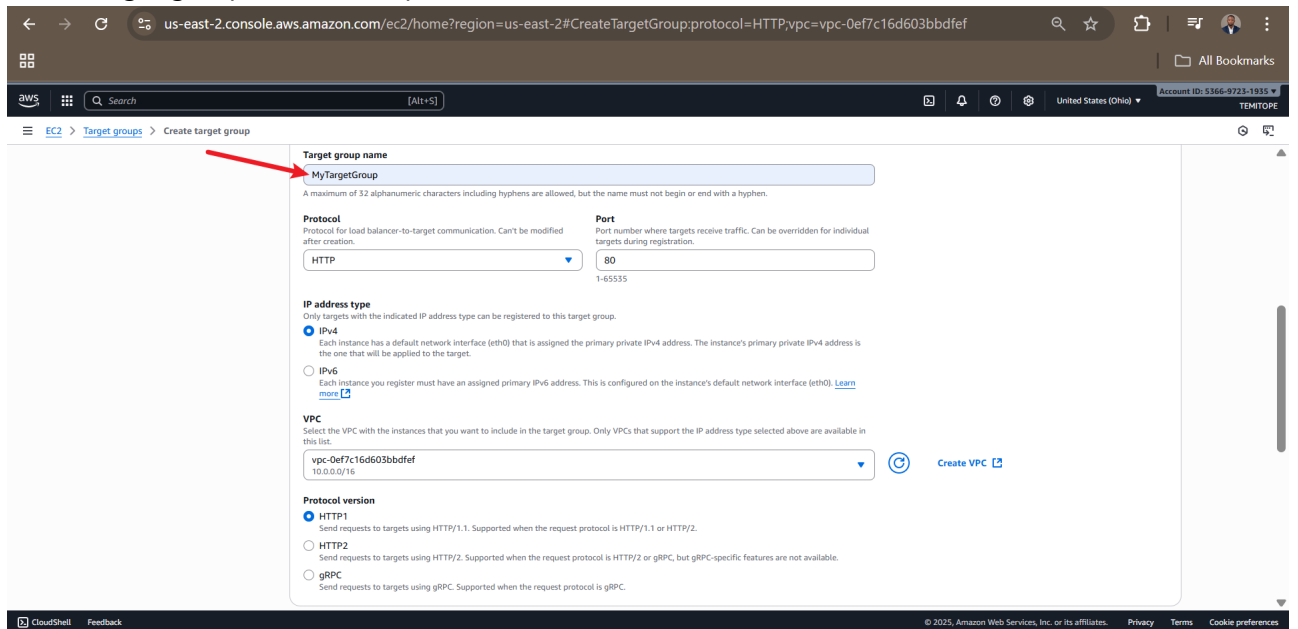
54. Protocol HTTP port 80 click forward to target group and create target group.



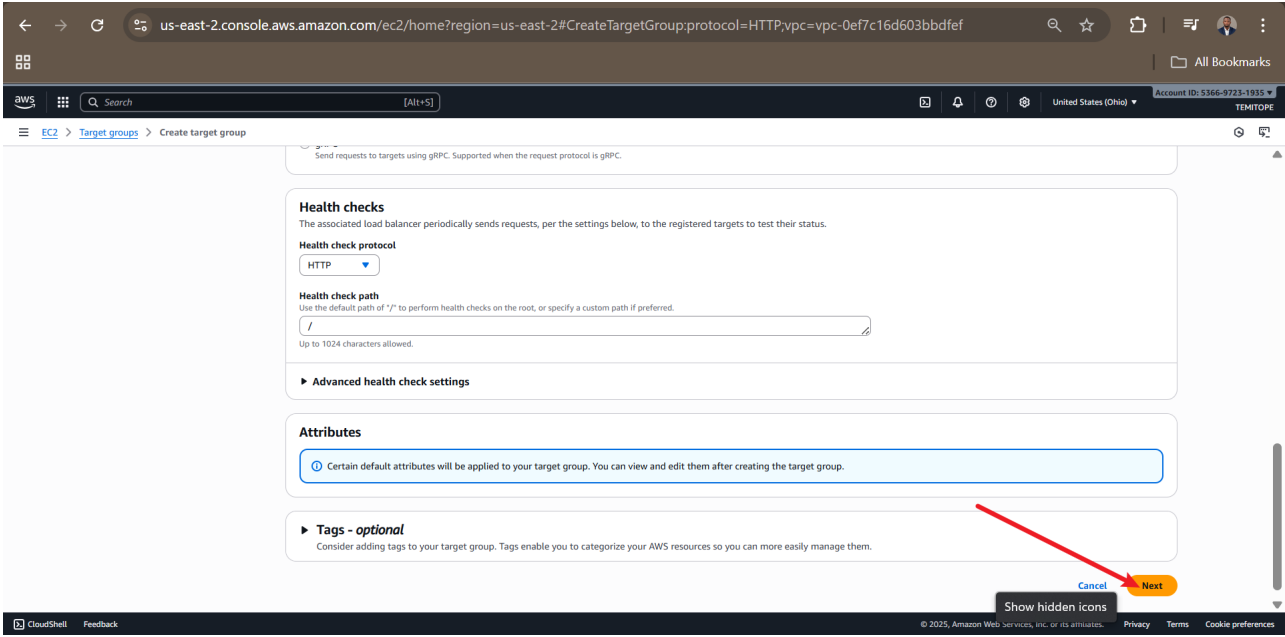
55. Create target group page open select instances and scroll down.



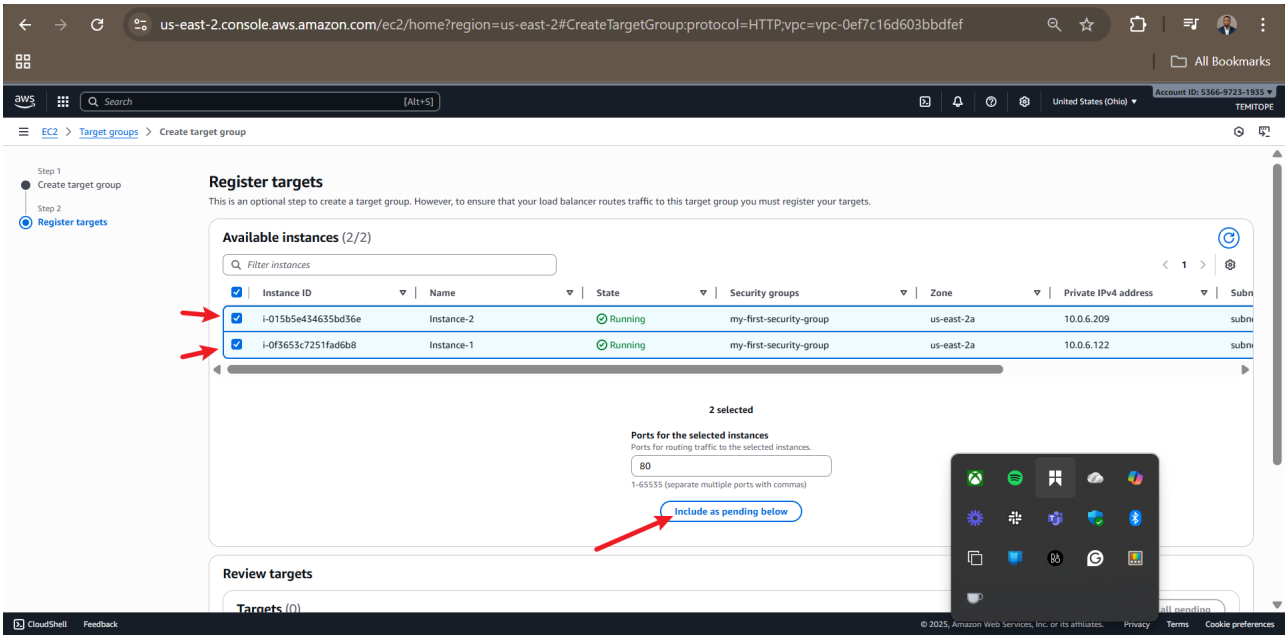
56. Name target group select HTTP port 80 choose the VPC and scroll down.



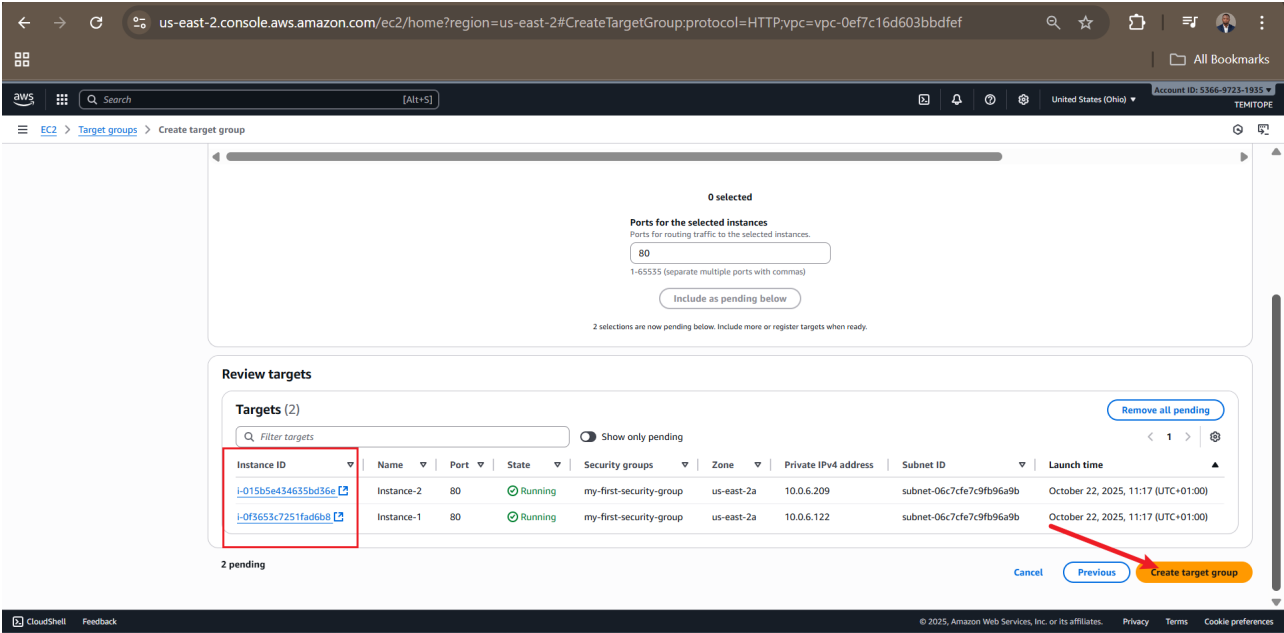
57. Leave everything at default and click next.



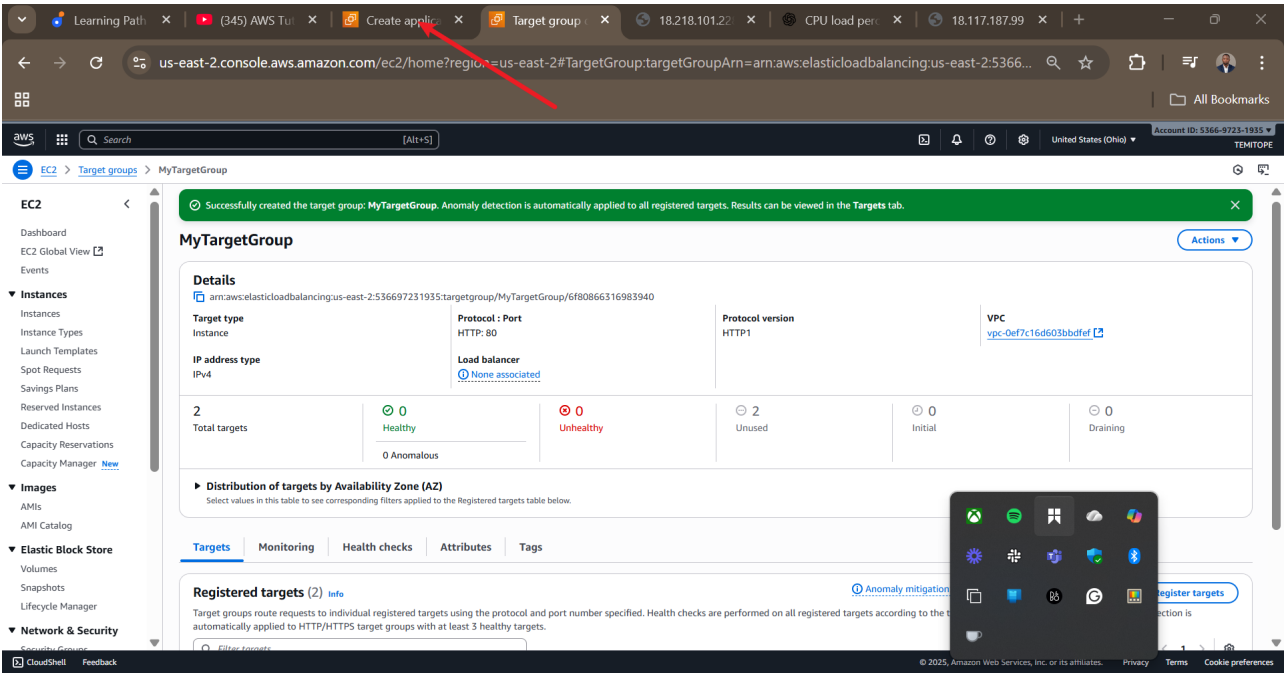
58. Select the servers and click include as pending below.



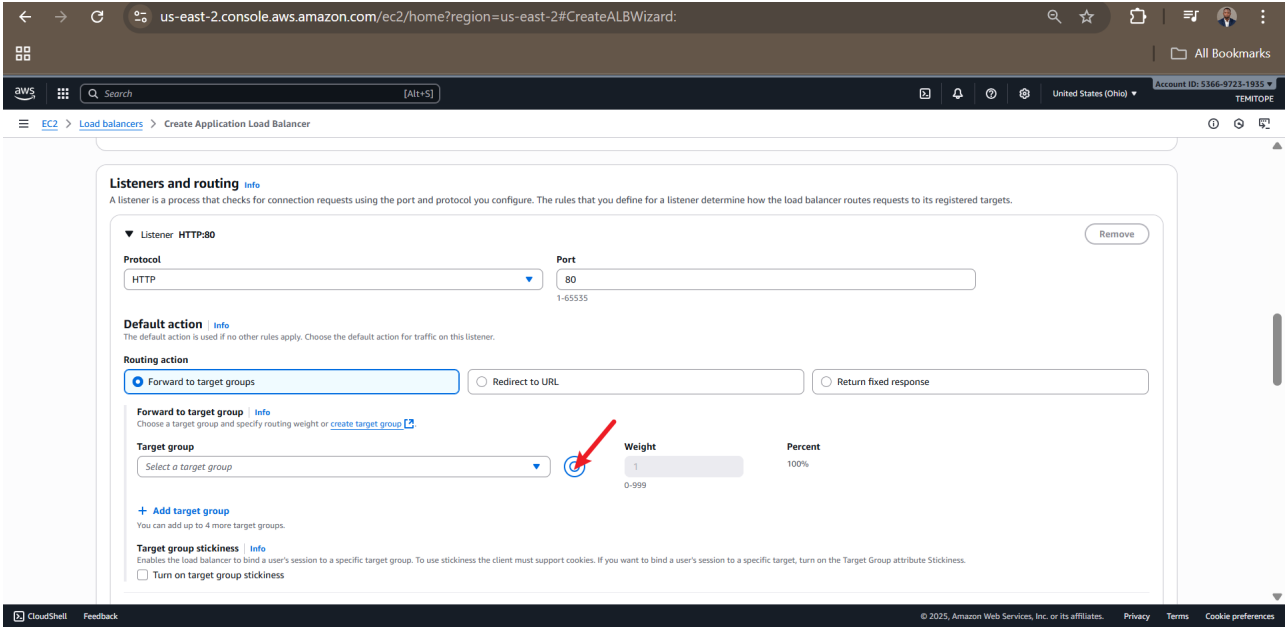
59. Servers are now included click on create target group.



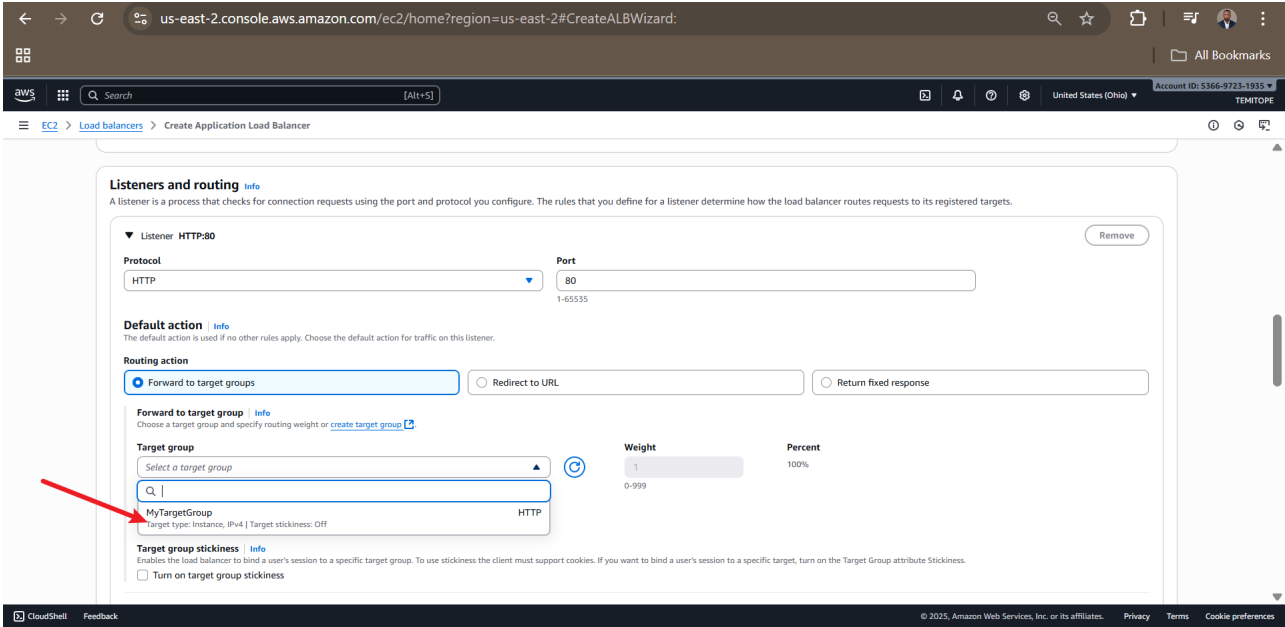
60. Target group successfully created now go back to LB tab to continue.



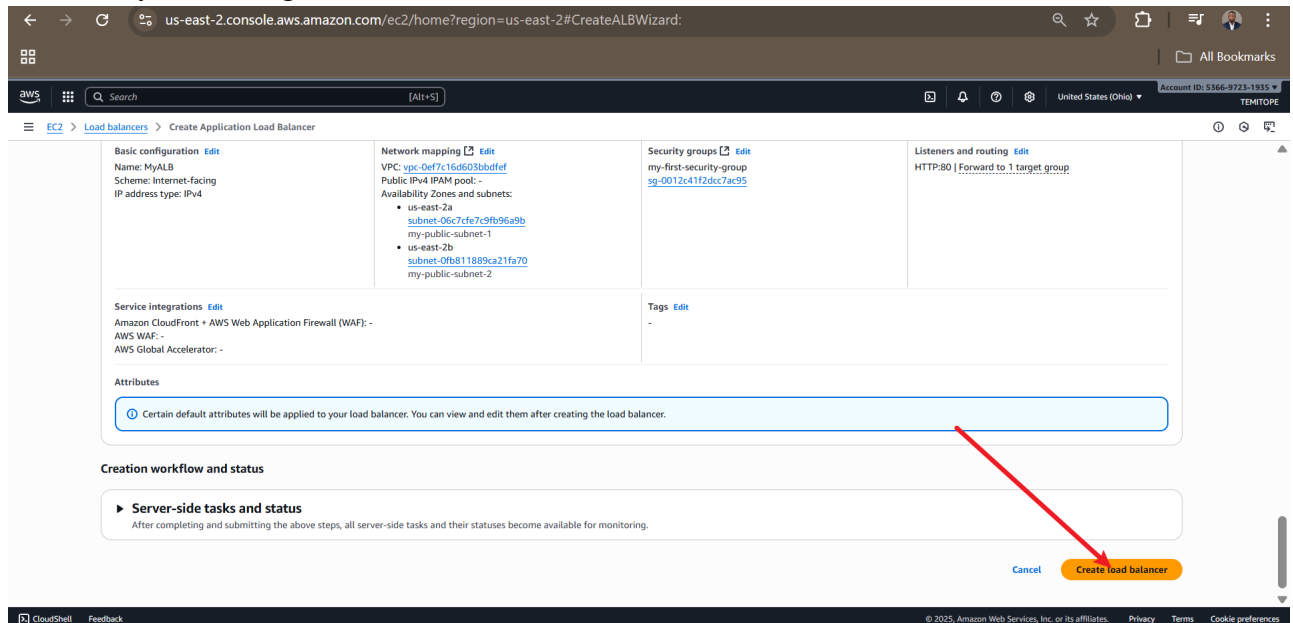
61. Click the refresh button to load the created TG.



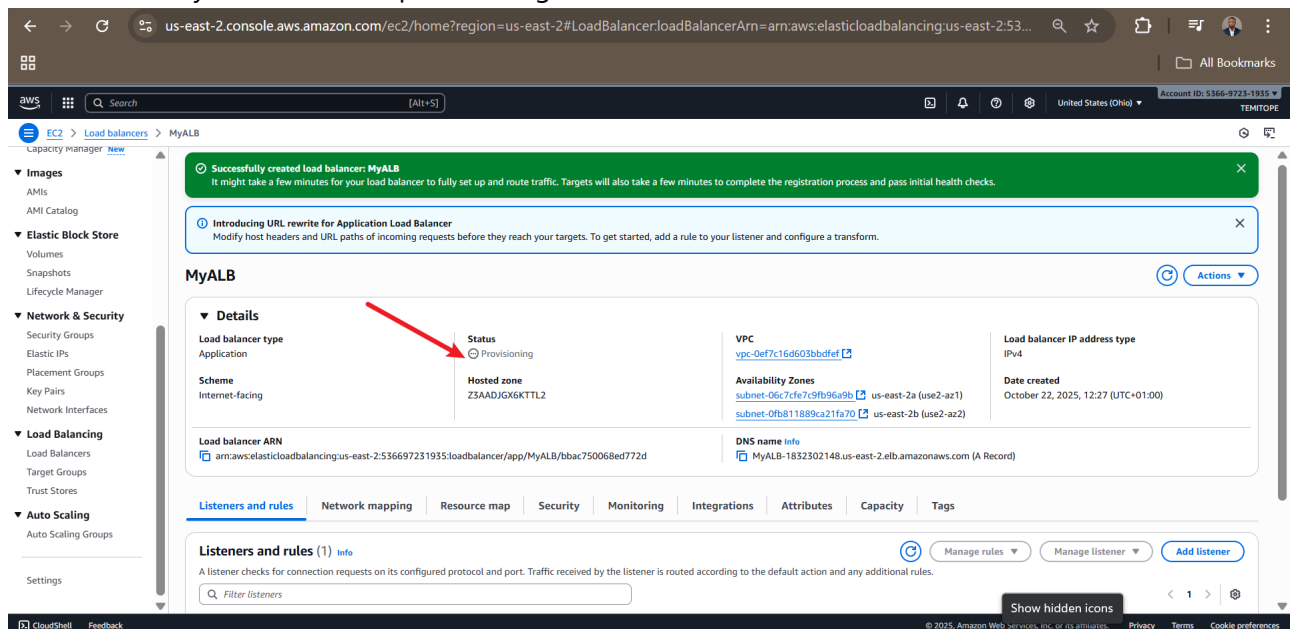
62. Select the newly created target group scroll down.



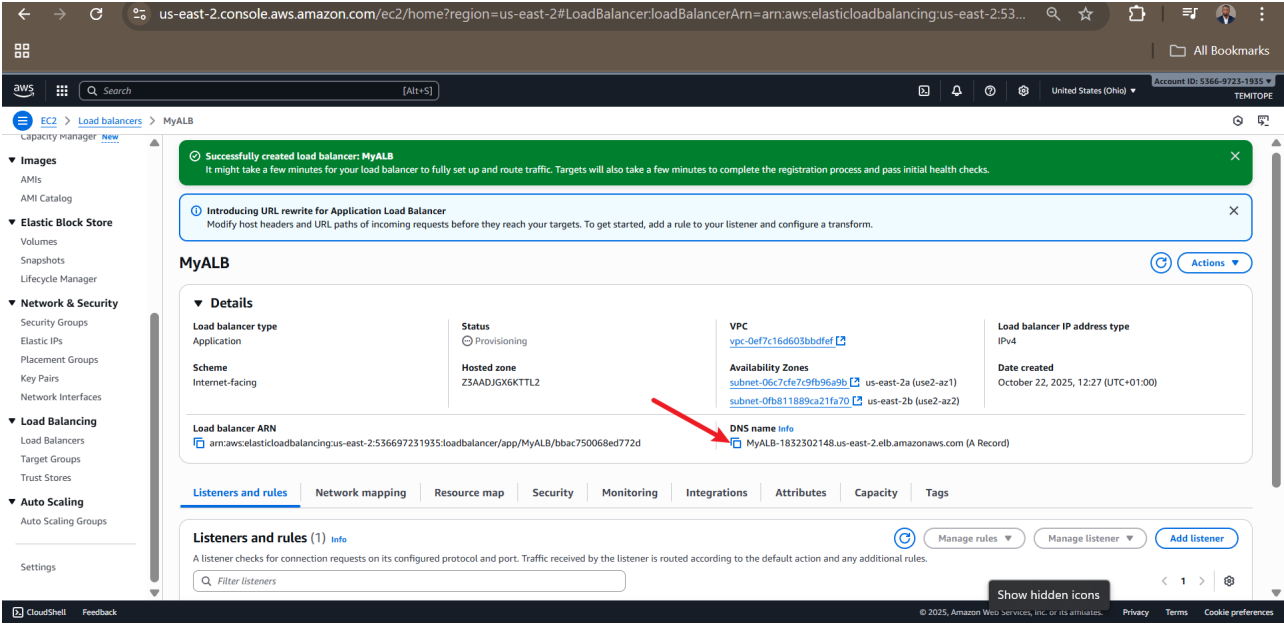
63. Leave every other thing at default and click create load balancer.



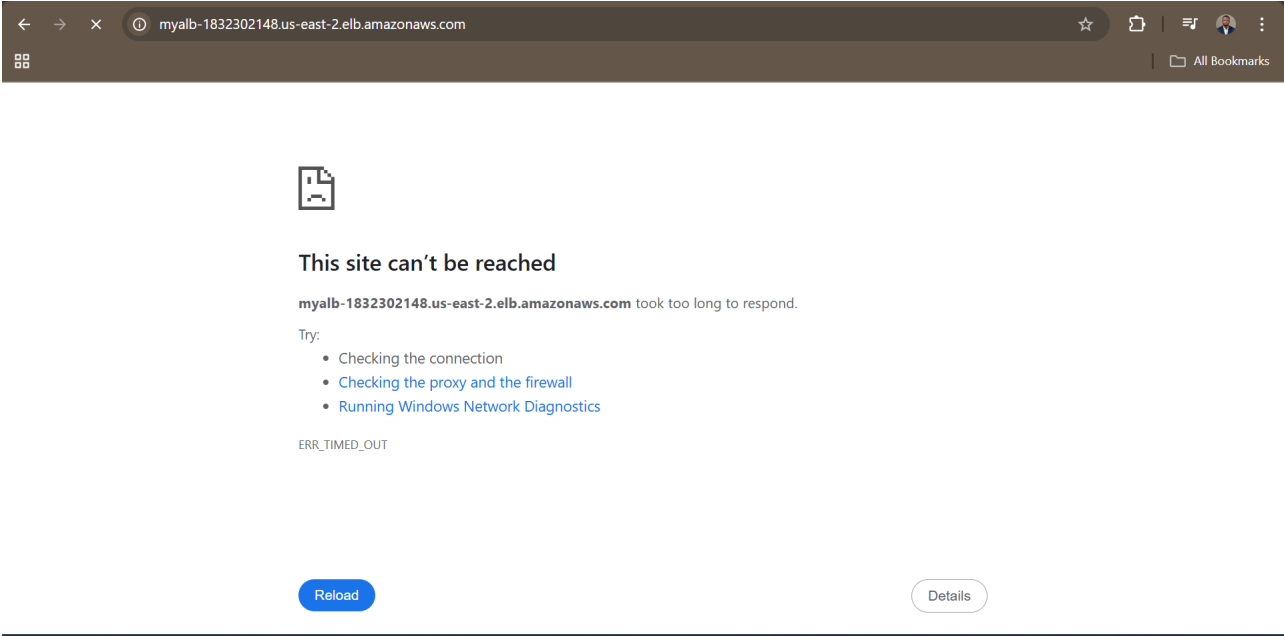
64. ALB successfully created and in provisioning state.



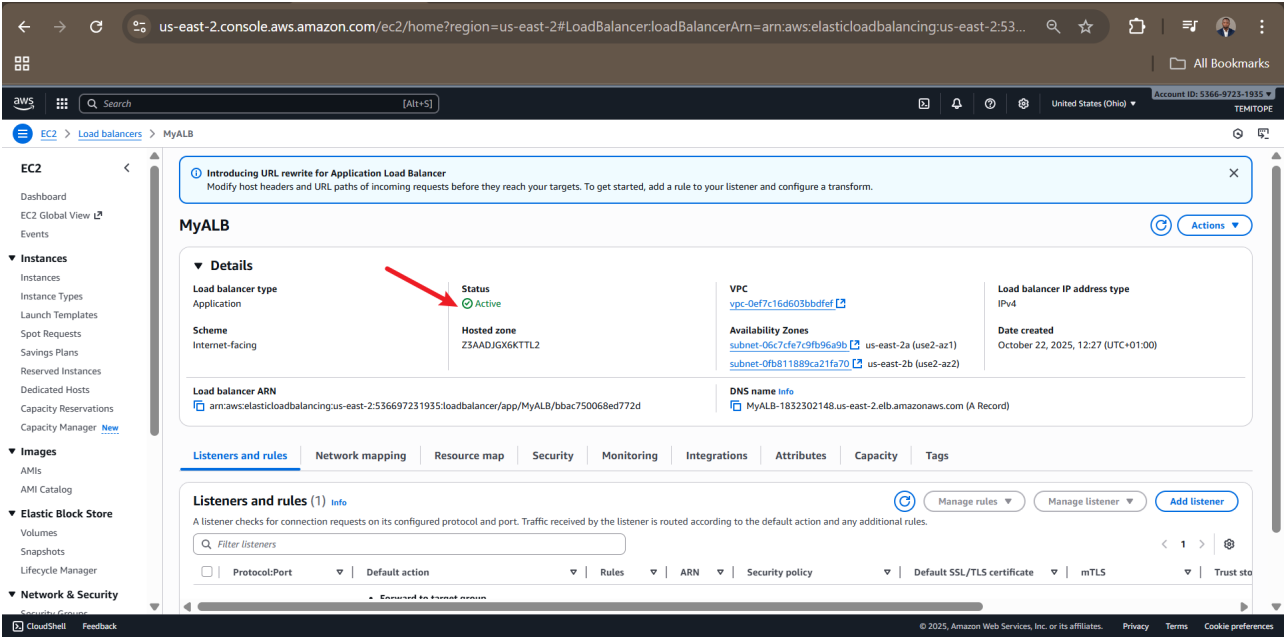
65. Copy the DNS name to paste it on a browser.



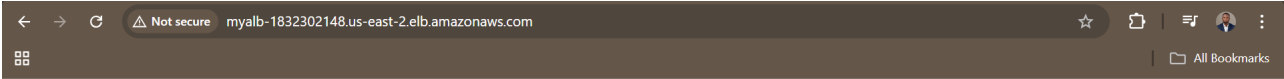
66. Site can not be reached because ALB is still in provisioning state.



67. Refresh after a while and ALB now active.



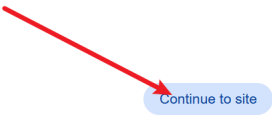
68. DNS now work click continue to site.



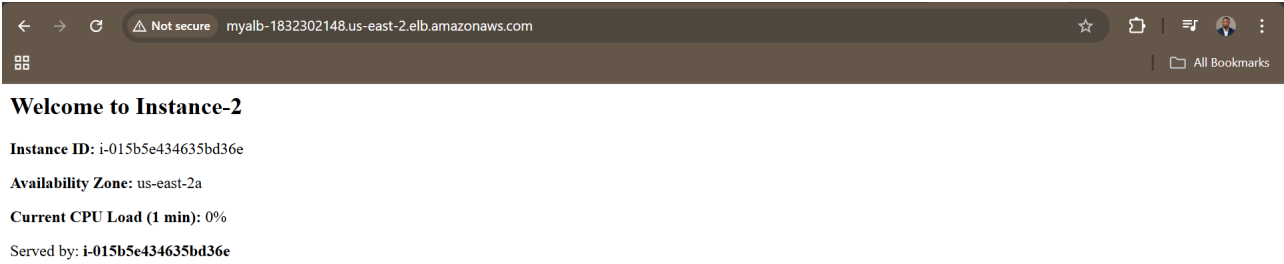
myalb-1832302148.us-east-2.elb.amazonaws.com doesn't support a secure connection with HTTPS

- **Attackers can see and change** information you send or receive from the site.
- **It's safest to visit this site later** if you're using a public network. There is less risk from a trusted network, like your home or work Wi-Fi.

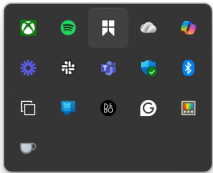
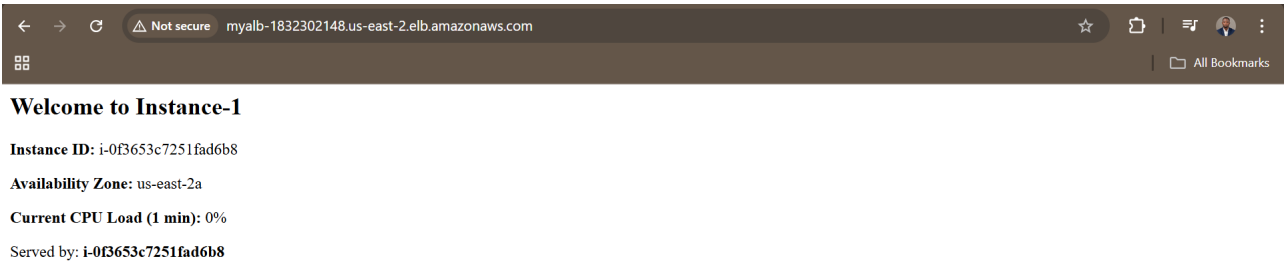
You might also contact the site owner and suggest they upgrade to HTTPS. [Learn more about this warning](#)



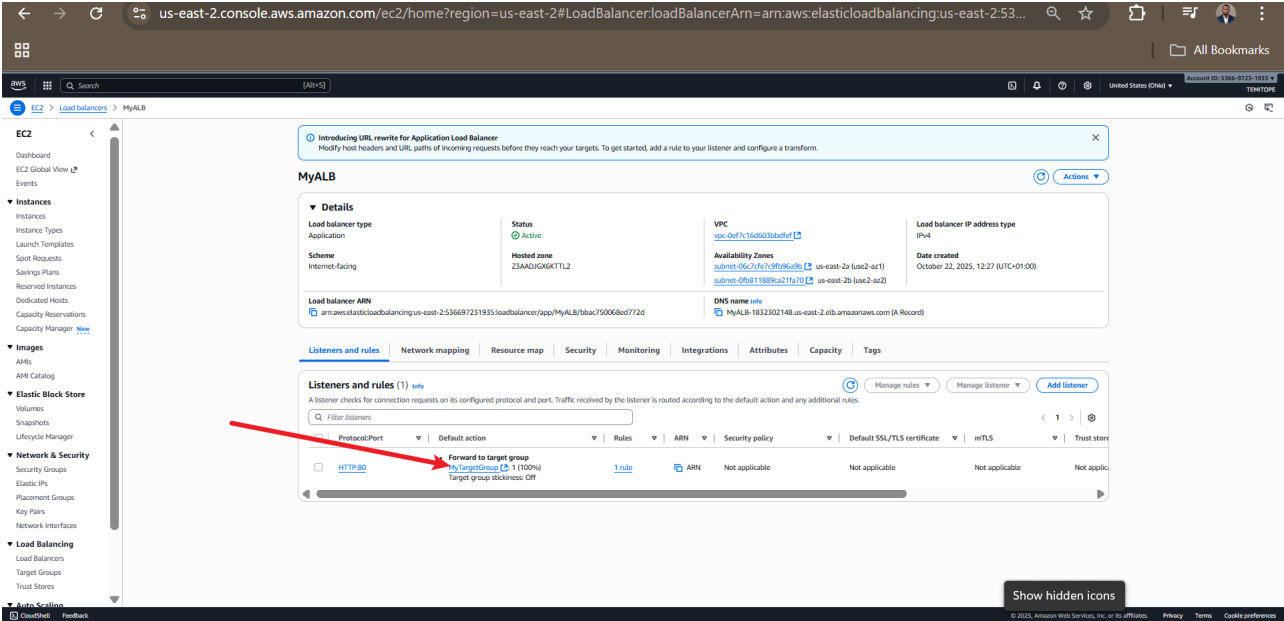
69. Instance-2 webpage loaded using the ALB DNS link.



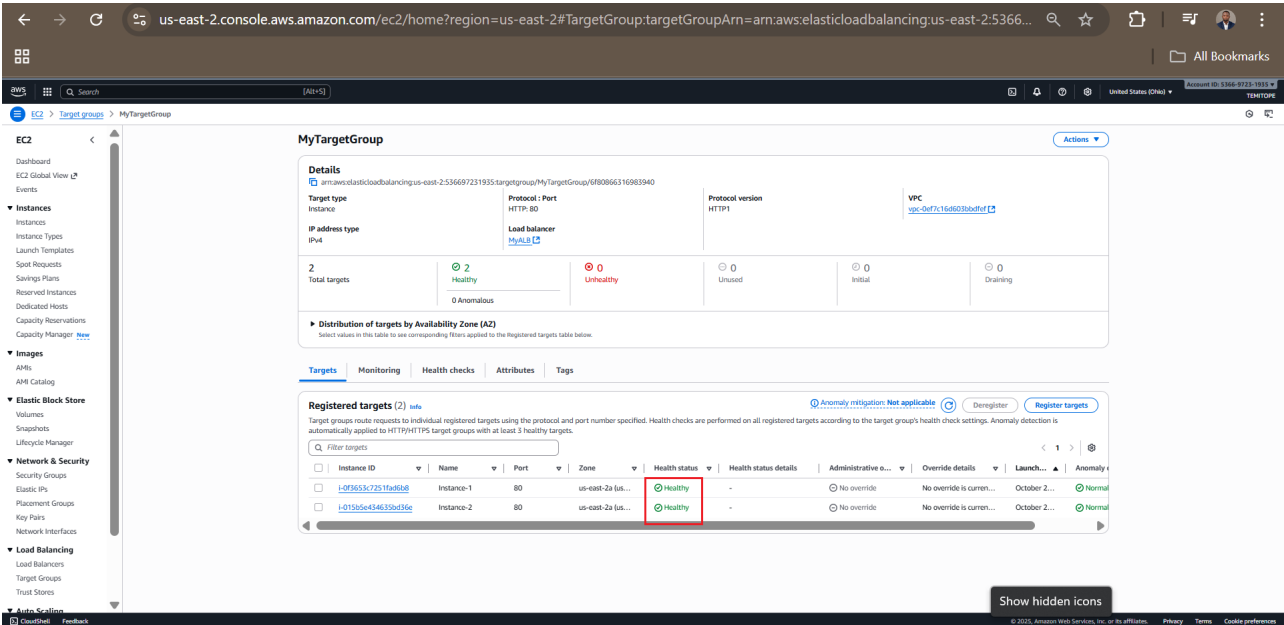
70. After reload instance-1 webpage loaded using the ALB DNS link (round robin).



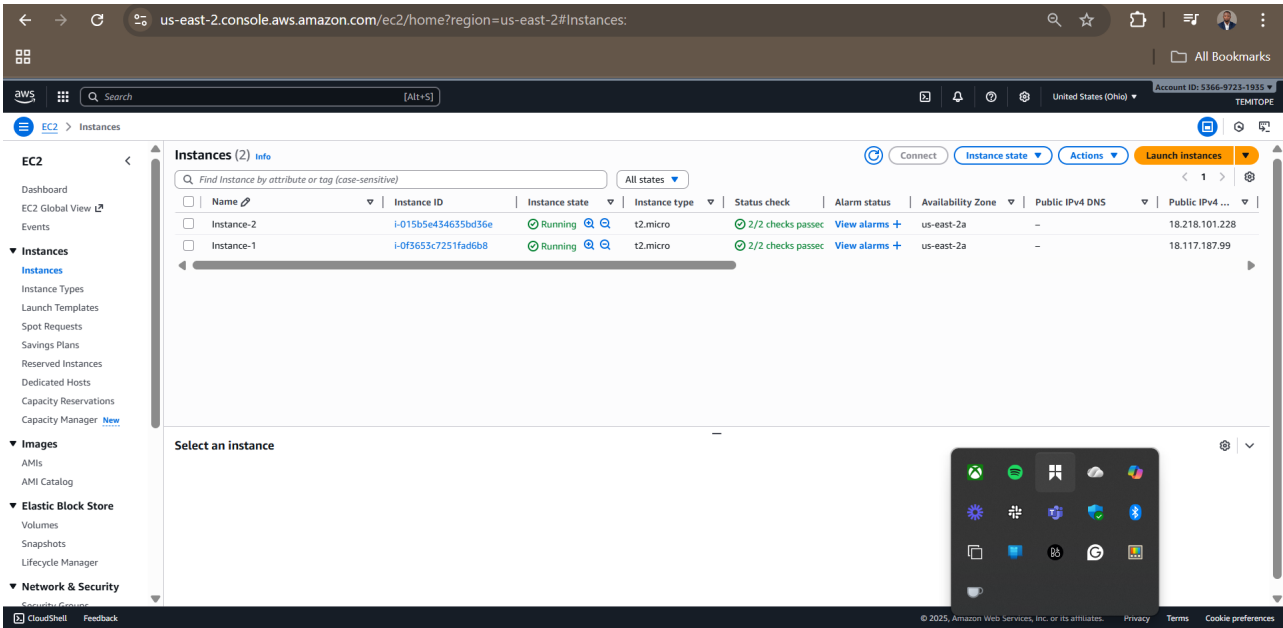
71. In case of problem click target group to inspect the target health.



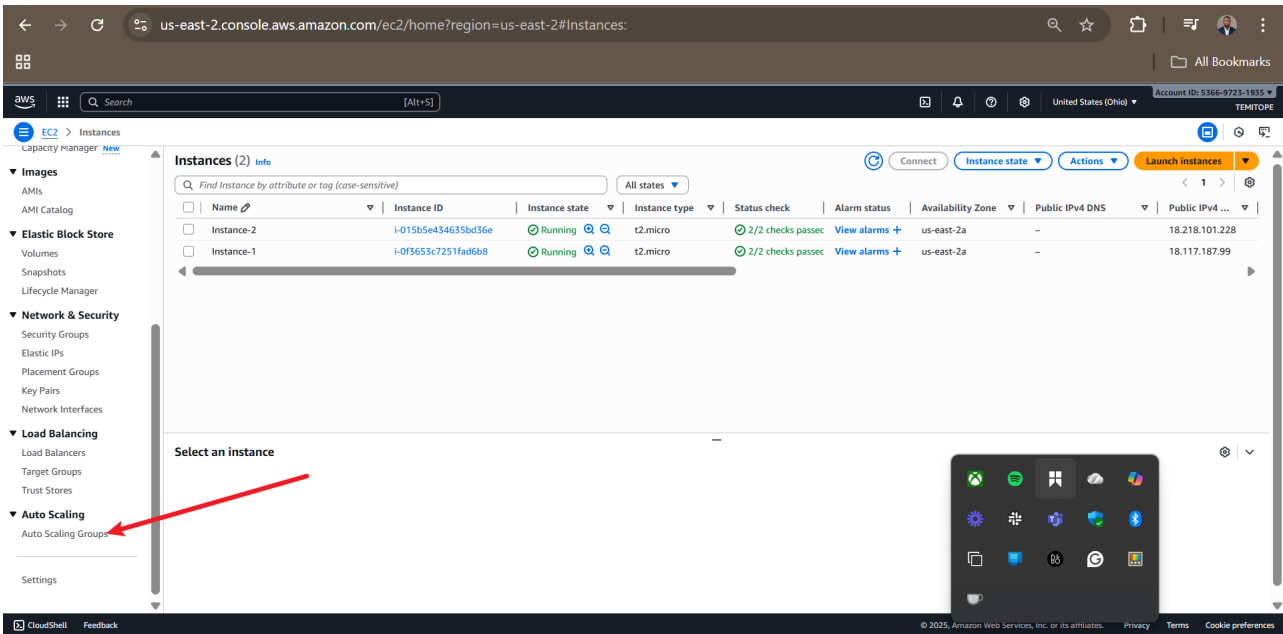
72. Both registered target are healthy for now.



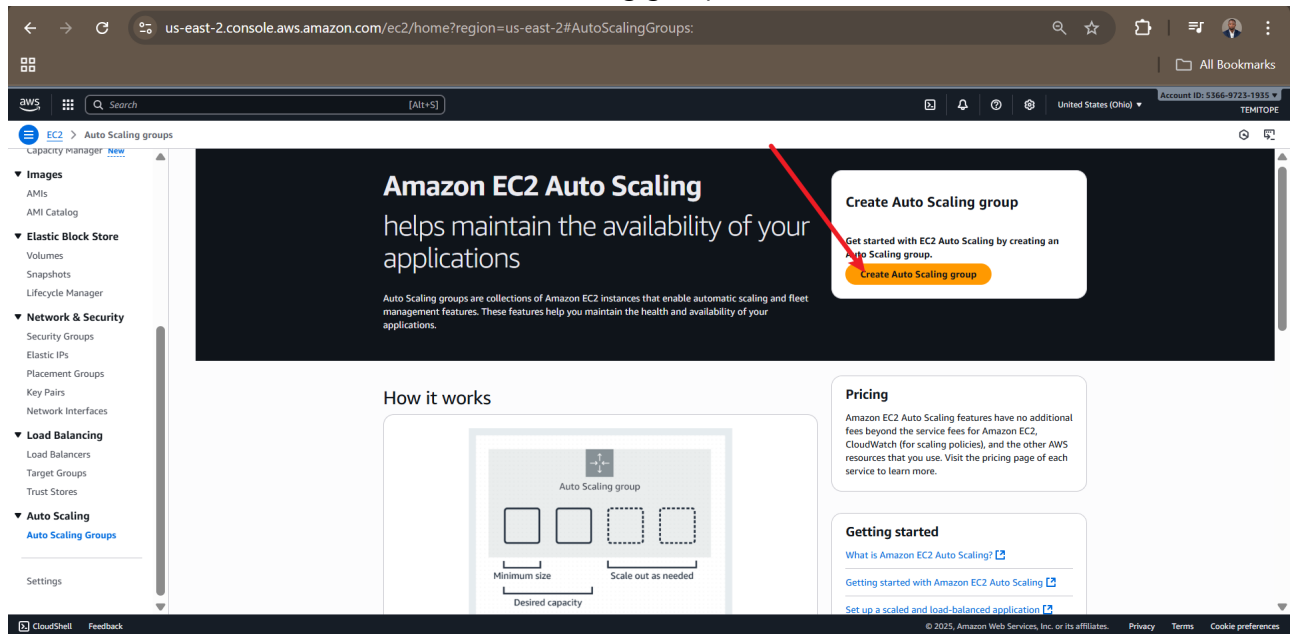
73. To auto scale the servers navigate to instances dashboard.



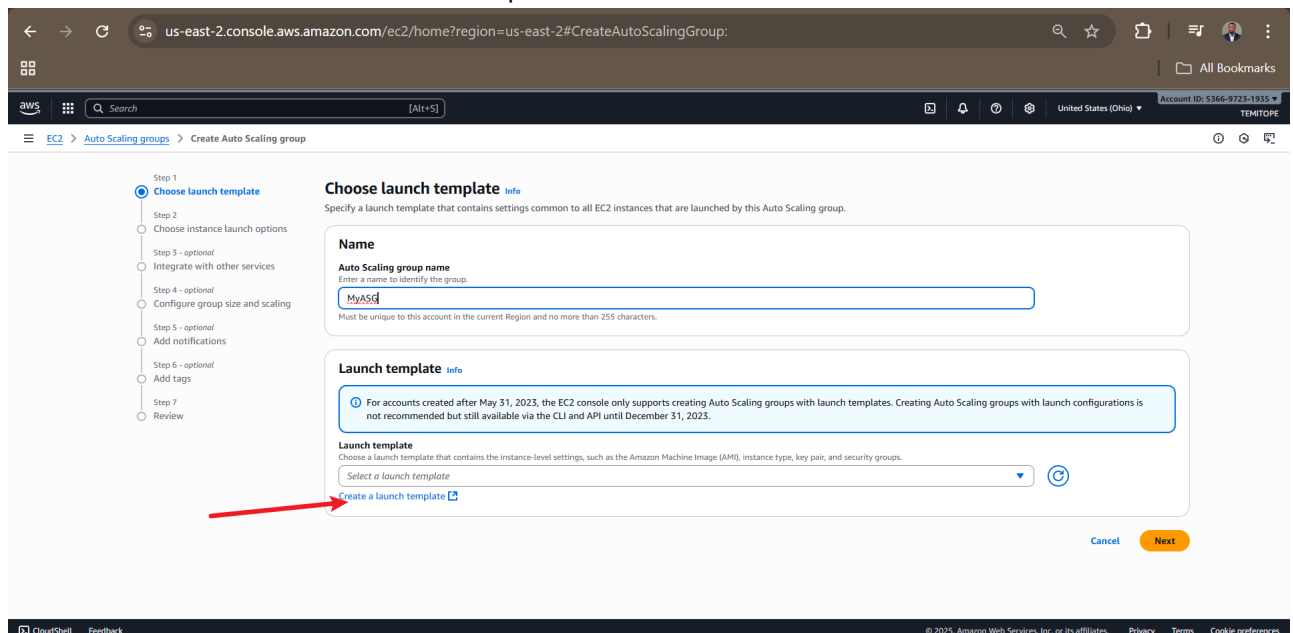
74. On the side menu scroll down to auto scaling menu and click on auto scaling groups.



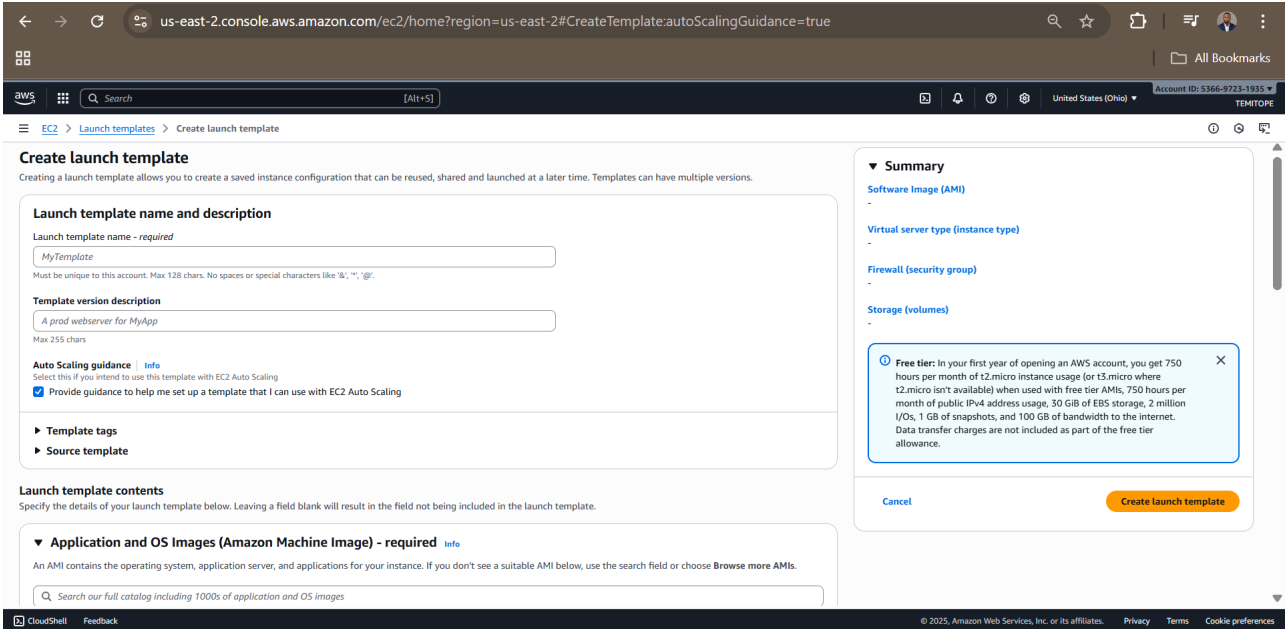
75. On EC2 ASG dashboard click on create auto scaling group.



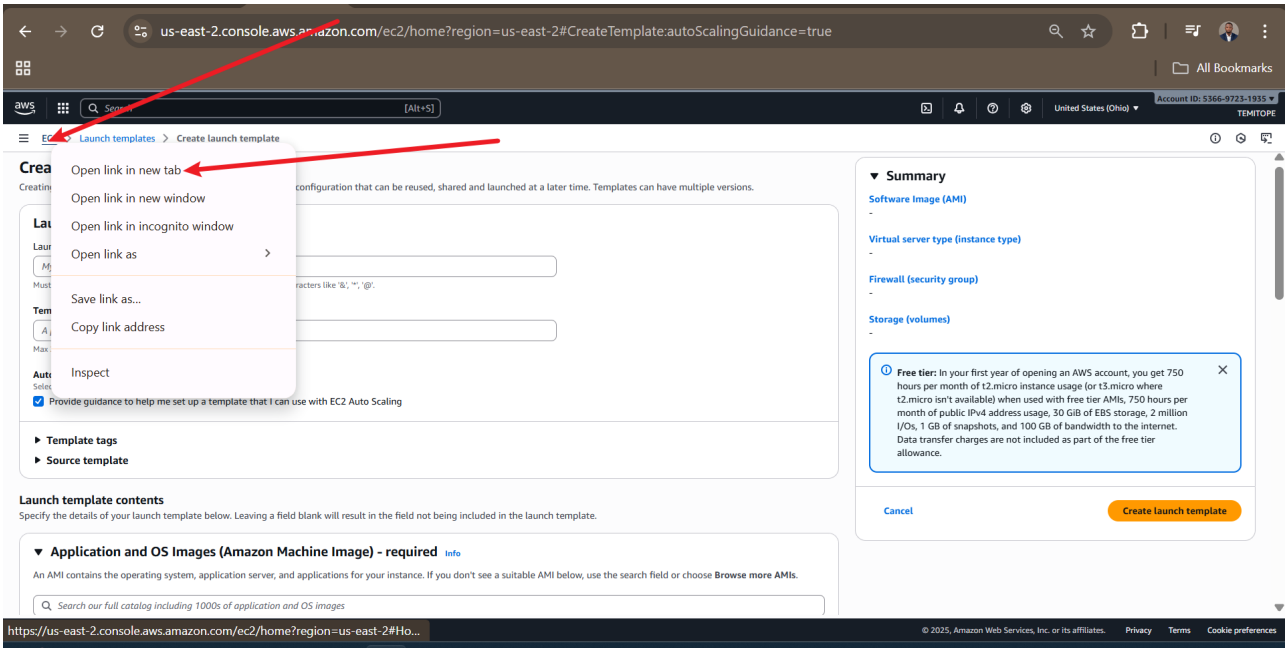
76. Name ASG and click create a launch template.



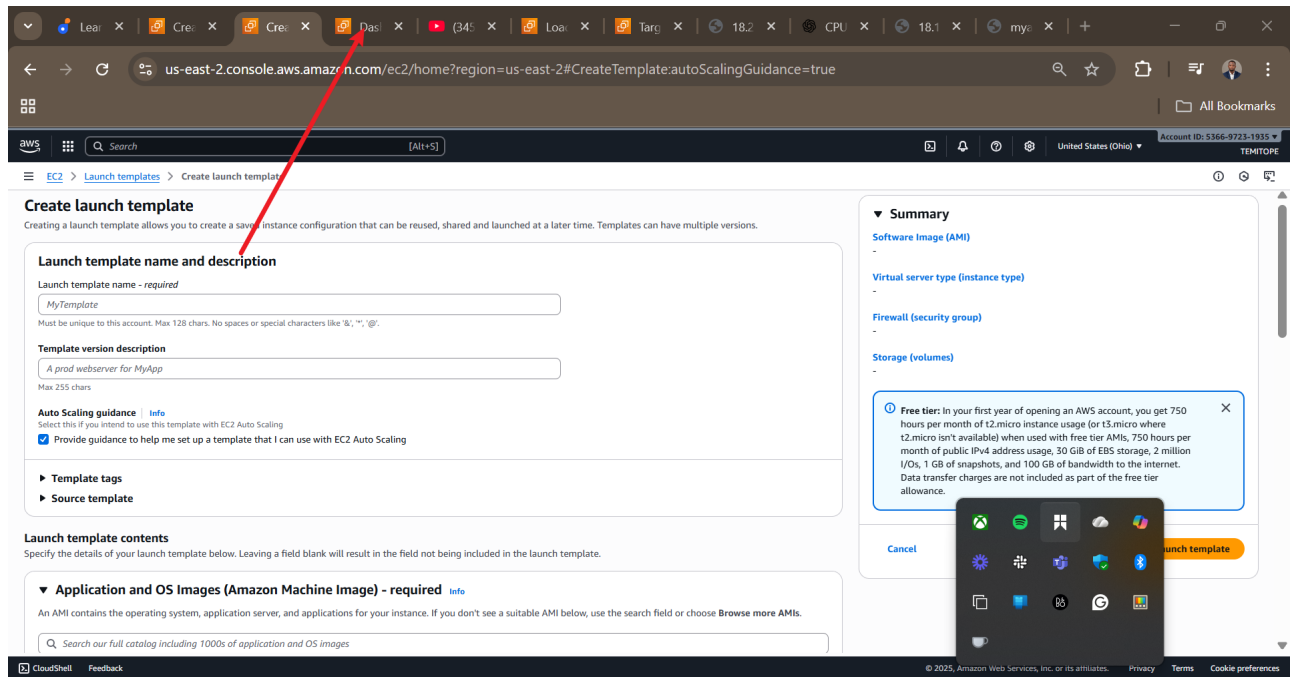
77. Create launch template page open let us create AMI image of the server.



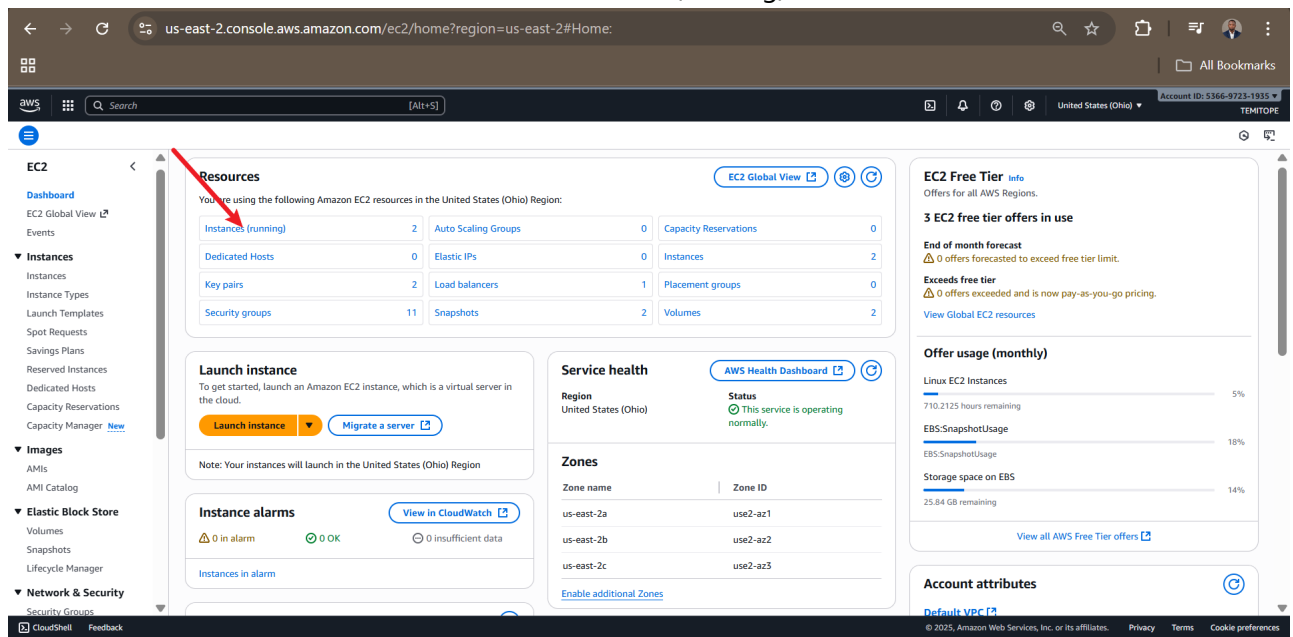
78. Right-click on EC2 from navigation menu and open link in a new tab.



79. Click on the new tab to switch to it.



80. On EC2 dashboard under resources click on instances (running).



81. Select instance-1 then click on actions then image and templates then create image.

The screenshot shows the AWS Management Console for the 'us-east-2' region. The 'Instances' page is active, displaying a list of EC2 instances. 'Instance-1' (ID: i-0f3653c7251fad6b8) is selected. The 'Actions' menu is open, and the 'Create image' option is highlighted. Red arrows indicate the sequence of actions: 1. Select 'Instance-1', 2. Click 'Actions', 3. Click 'Image and templates', and 4. Click 'Create image'.

Name	Instance ID	Instance state	Instance type	Status check	Alarm status	Availability Zone	Public IP address
Instance-2	i-015b5e434635bd36e	Running	t2.micro	2/2 checks passed	View alarm	us-east-2a	18.117.187.99
Instance-1	i-0f3653c7251fad6b8	Running	t2.micro	2/2 checks passed	View alarm	us-east-2a	18.117.187.99

i-0f3653c7251fad6b8 (Instance-1)

Instance summary

Instance ID: i-0f3653c7251fad6b8
IPV6 address: -
Public IPv4 address: 18.117.187.99 | open address
Instance state: Running
Private IP DNS name (IPv4 only): -

82. Name image add description and scroll down.

The screenshot shows the 'Create image' page in the AWS Management Console. The 'Image name' field is filled with 'LaunchImage' and the 'Image description' field is filled with 'Image for AutoScaling Group'. The 'Reboot instance' checkbox is checked. The 'Instance volumes' section shows a table with columns for Storage type, Device, Snapshot, Size, Volume type, IOPS, Throughput, Delete on termination, and Encrypted.

Create image

An image (also referred to as an AMI) defines the programs and settings that are applied when you launch an EC2 instance. You can create an image from the configuration of an existing instance.

Image details

Instance ID: i-0f3653c7251fad6b8 (Instance-1)

Image name: LaunchImage
Maximum 127 characters. Can't be modified after creation.

Image description - optional: Image for AutoScaling Group
Maximum 255 characters

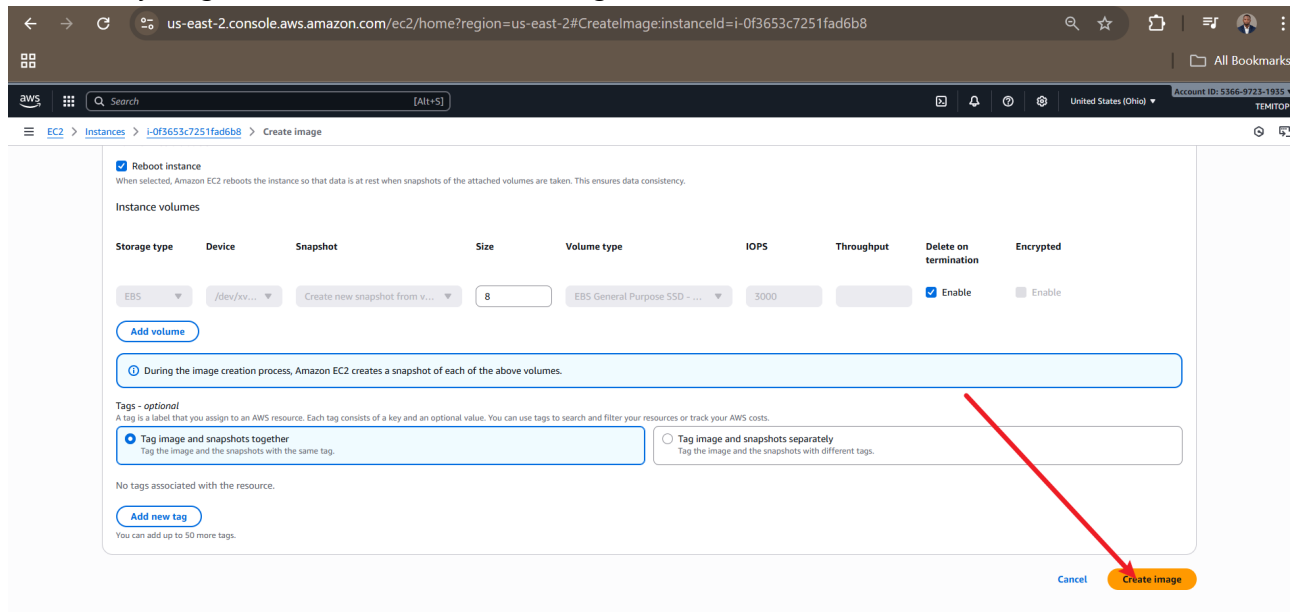
☒ Reboot instance
When selected, Amazon EC2 reboots the instance so that data is at rest when snapshots of the attached volumes are taken. This ensures data consistency.

Instance volumes

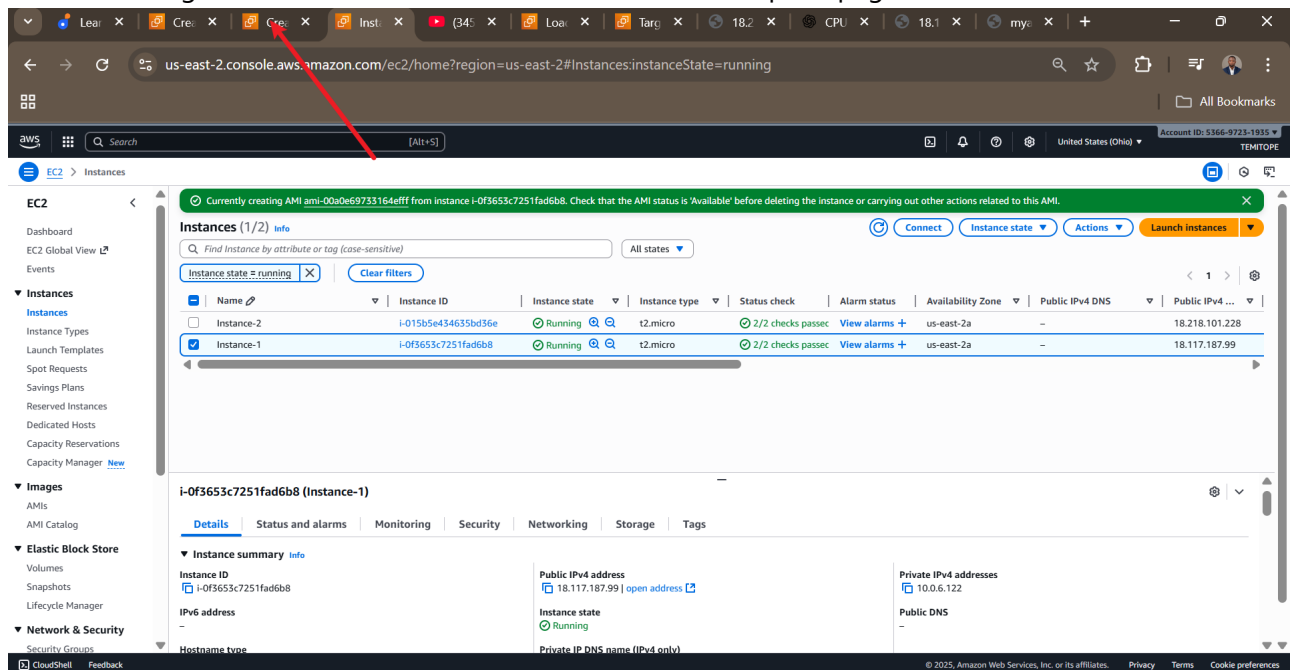
Storage type	Device	Snapshot	Size	Volume type	IOPS	Throughput	Delete on termination	Encrypted
EBS	/dev/xv...	Create new snapshot from v...	8	EBS General Purpose SSD - ...	3000		☒ Enable	☐ Enable

[Add volume](#)

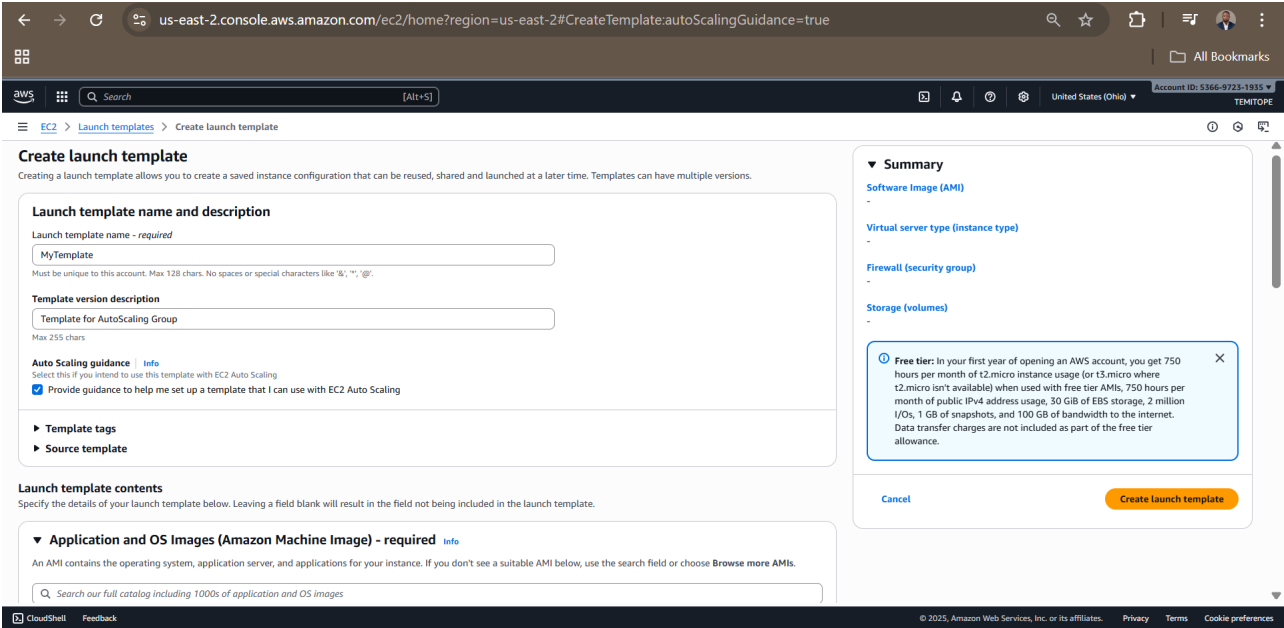
83. Leave everything at default and click create image.



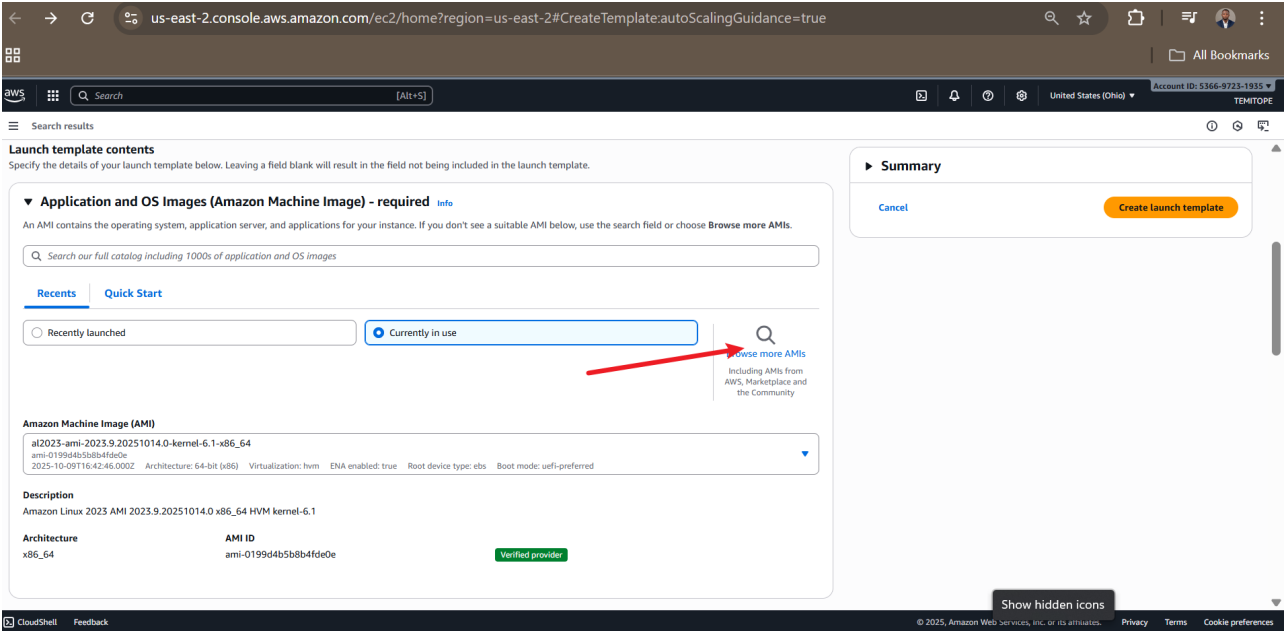
84. AMI now being created now switch back to create launch template page to continue.



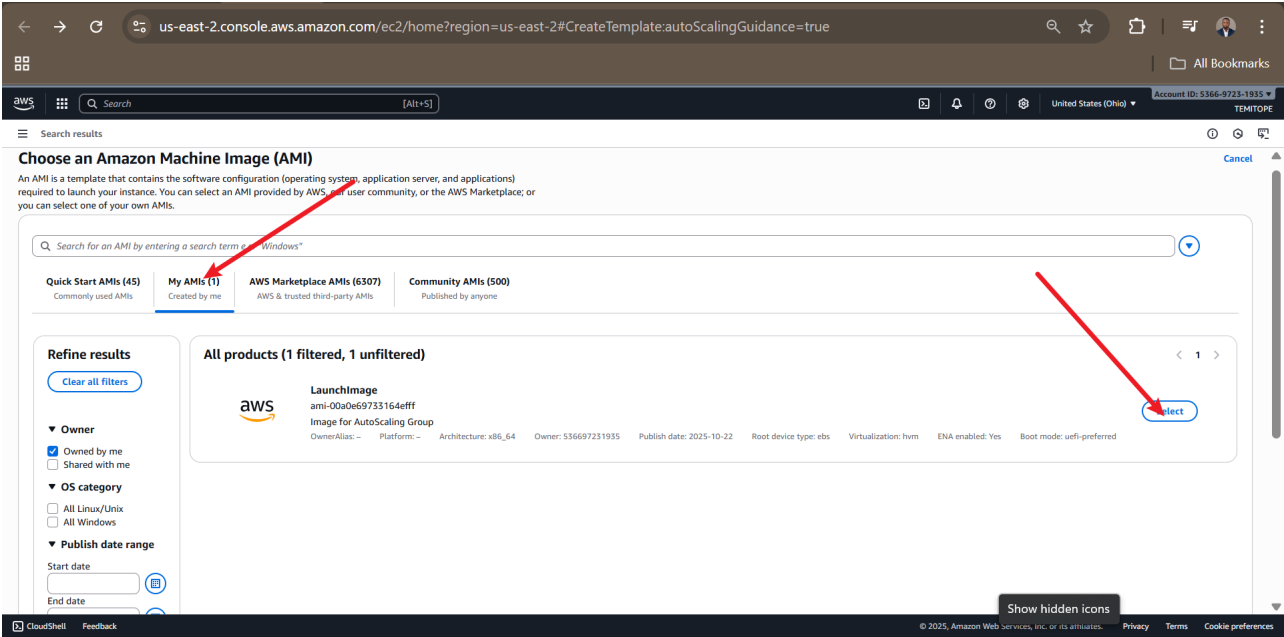
85. Name template describe template and scroll down.



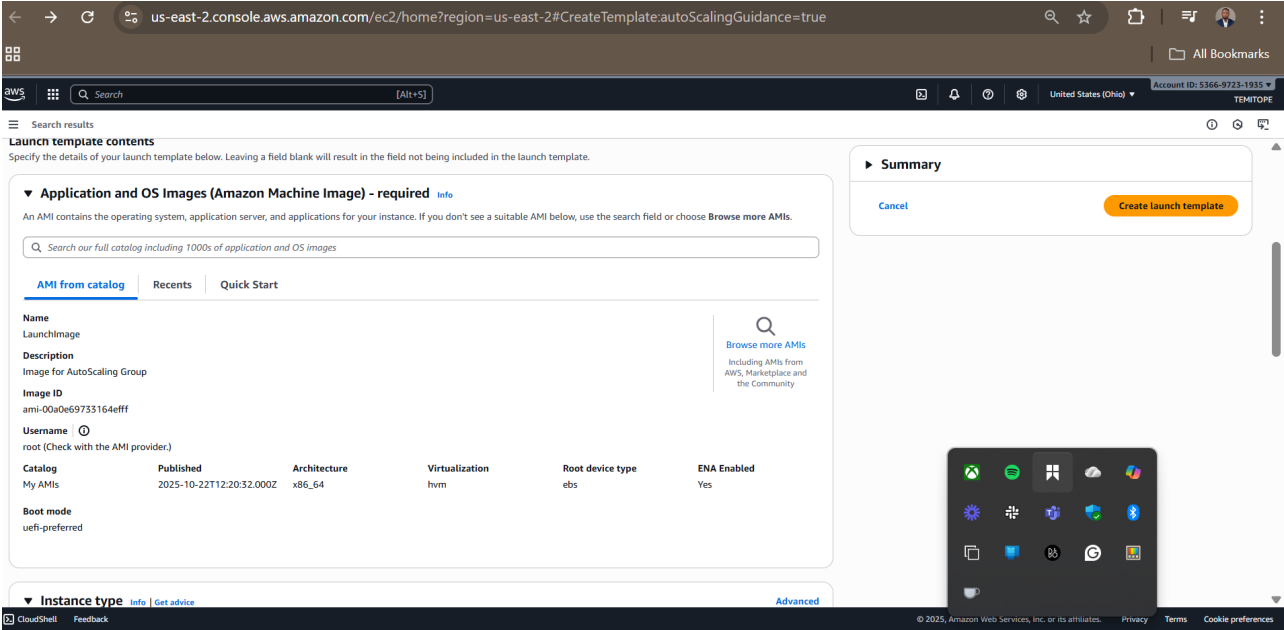
86. Under Application and OS click browse more AMIs.



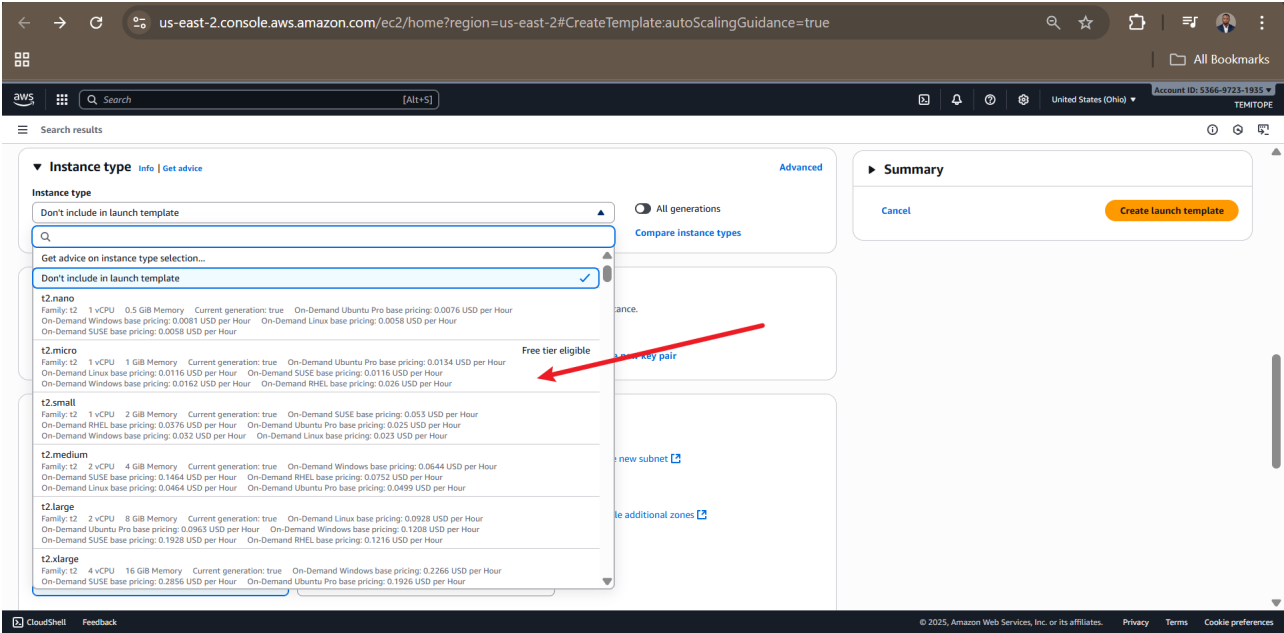
87. Click My AMI and select the just created AMI.



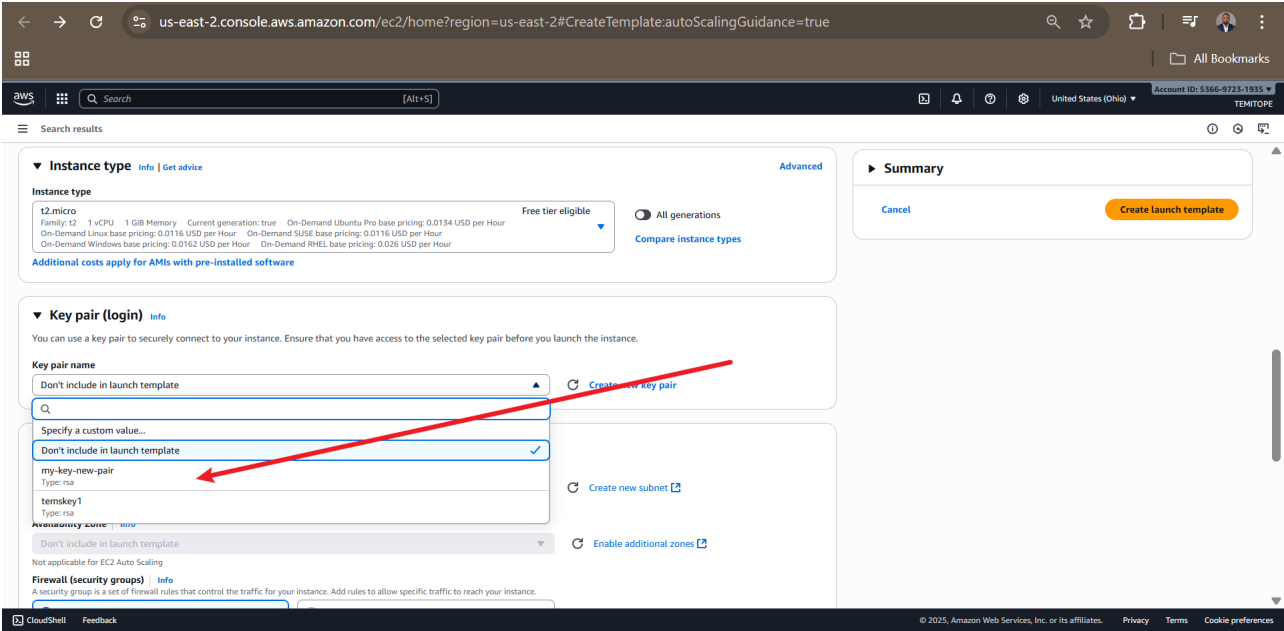
88. Image now selected scroll down.



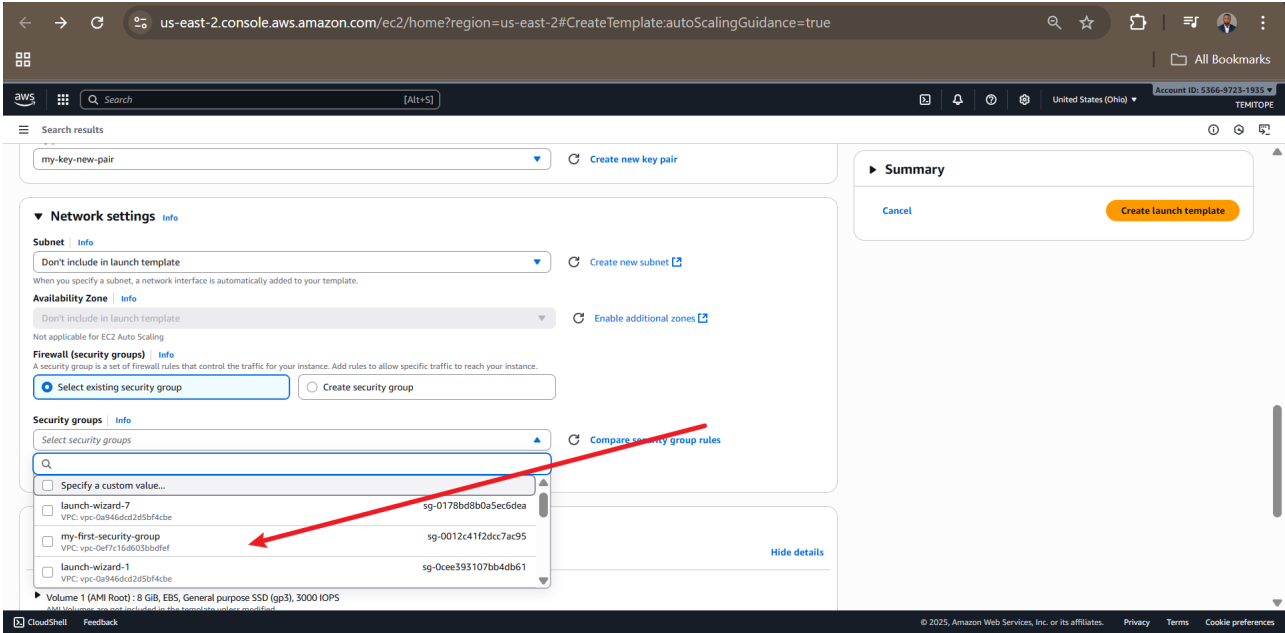
89. For instance type select t2 micro and scroll down.



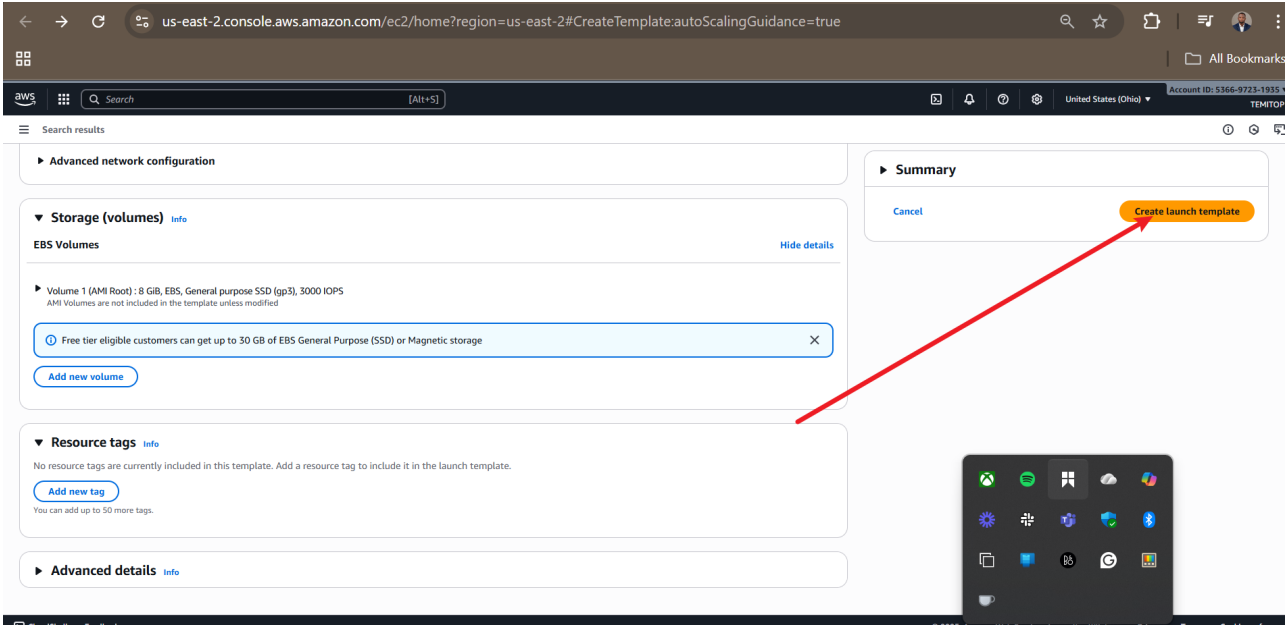
90. Select the key pair for all the instances.



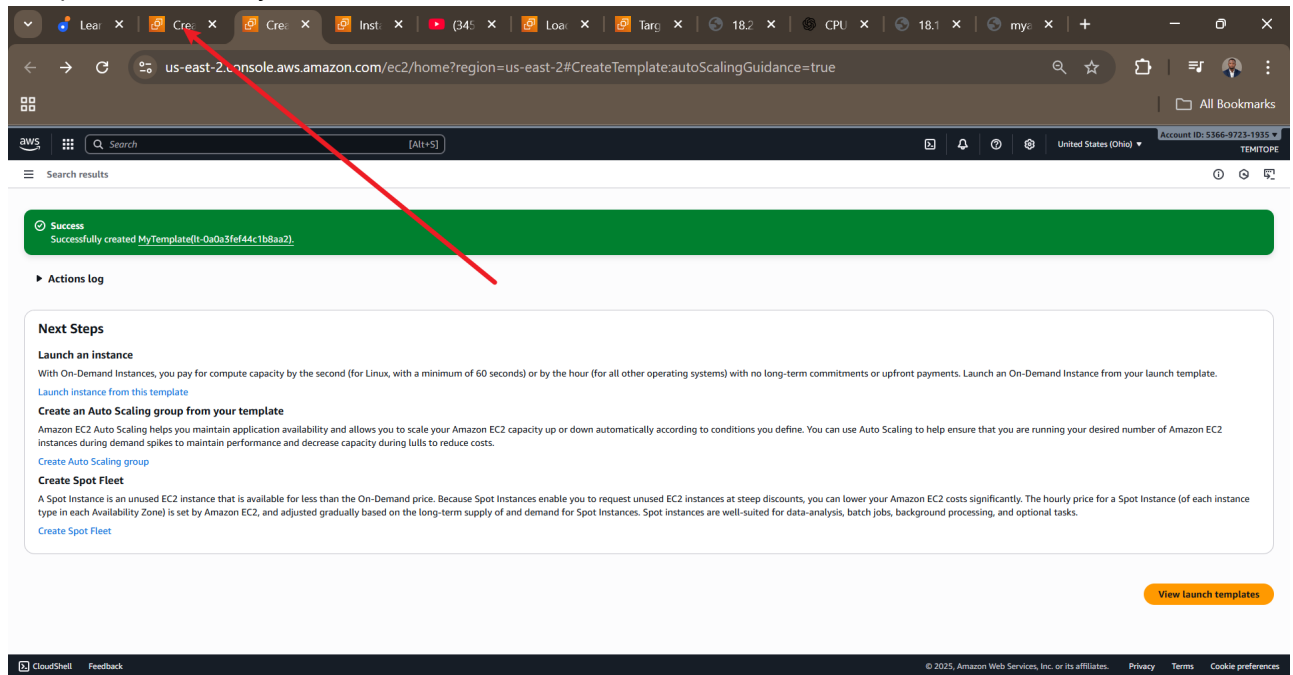
91. Select security group for all instances or create new.



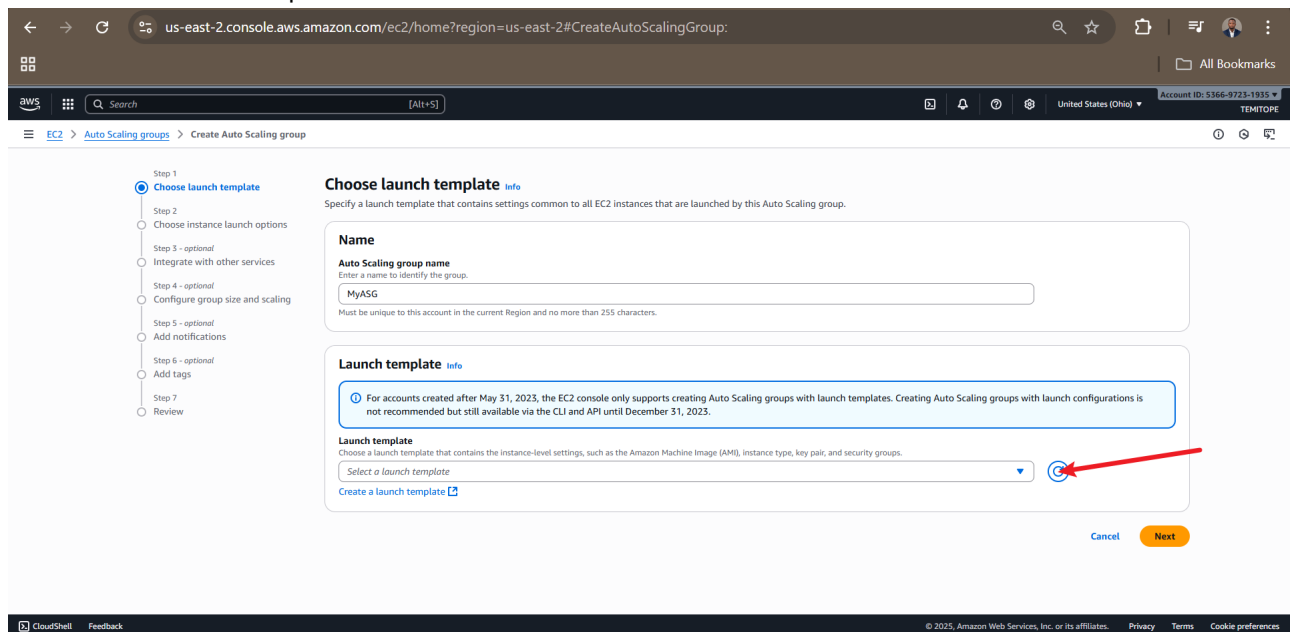
92. Leave everything and create launch template.



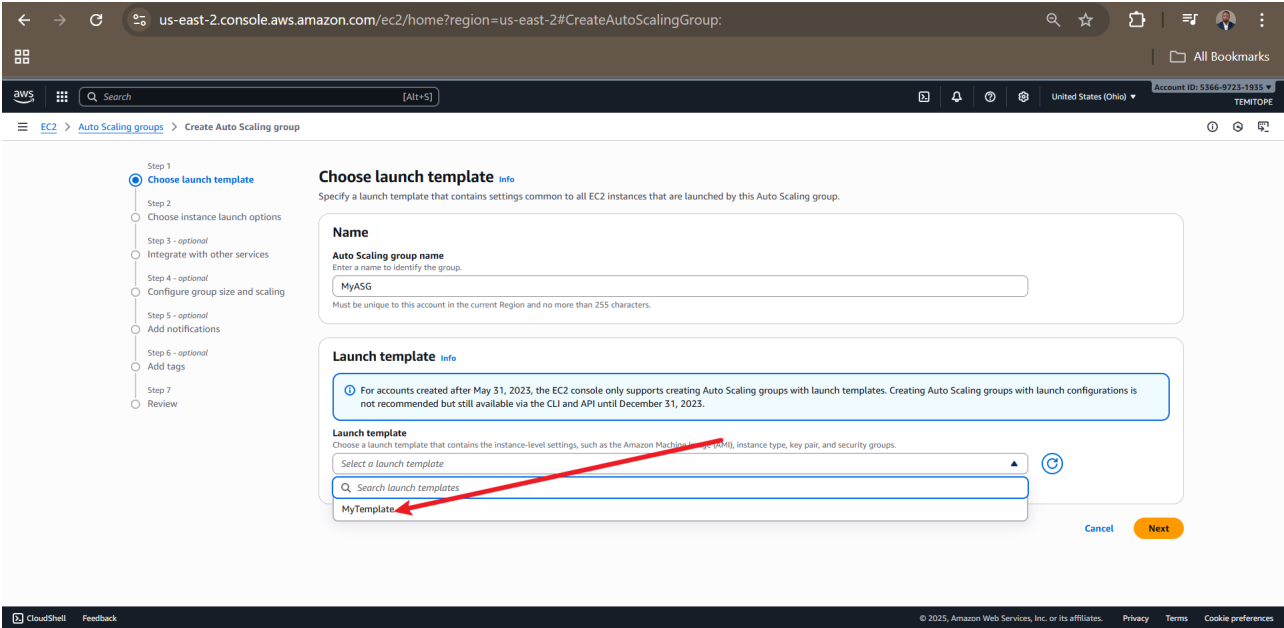
93. Template successfully created now continue to ASG creation tab.



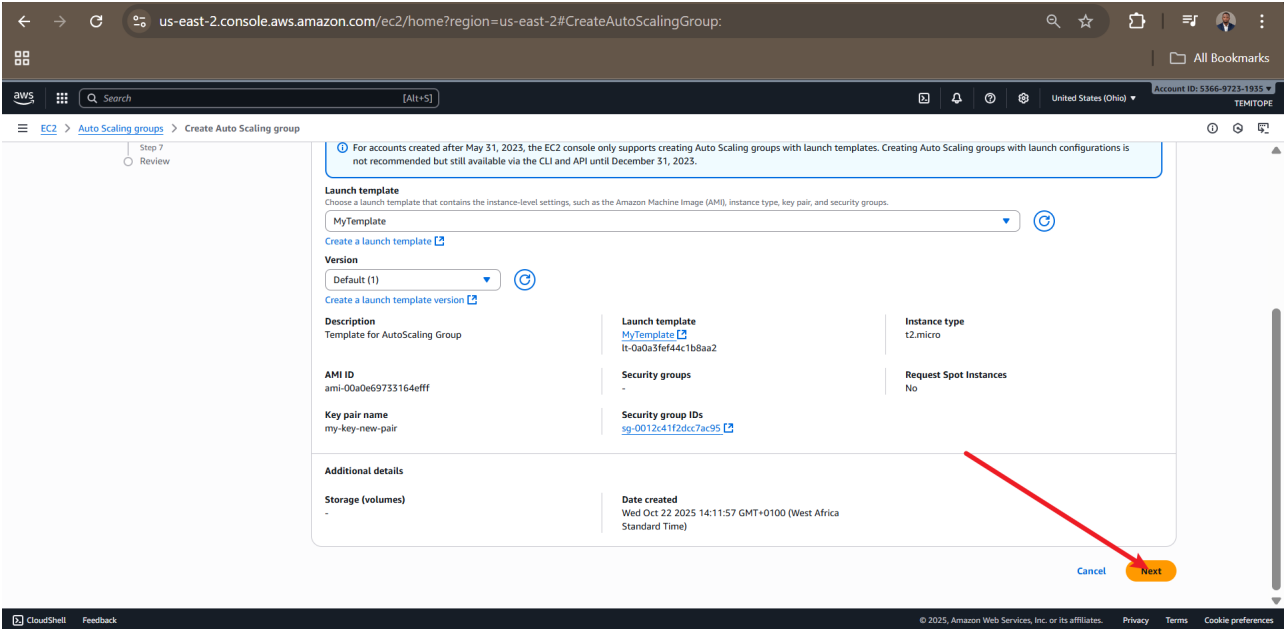
94. To choose created template click the refresh button.



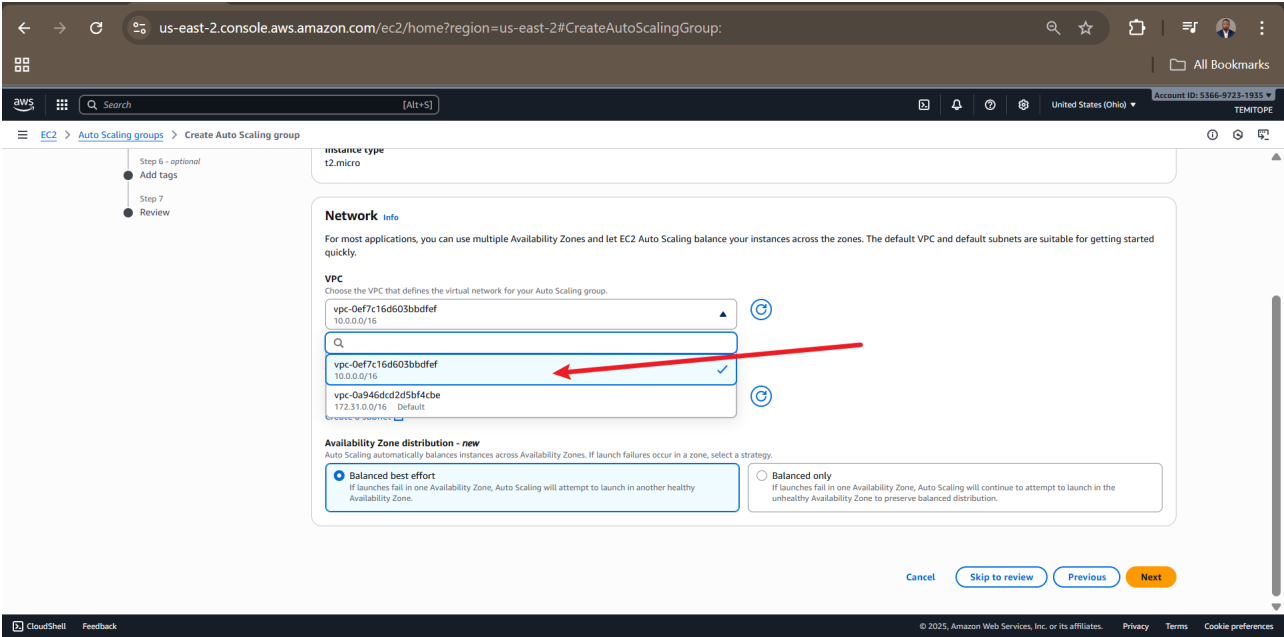
95. Search refreshed now select created template.



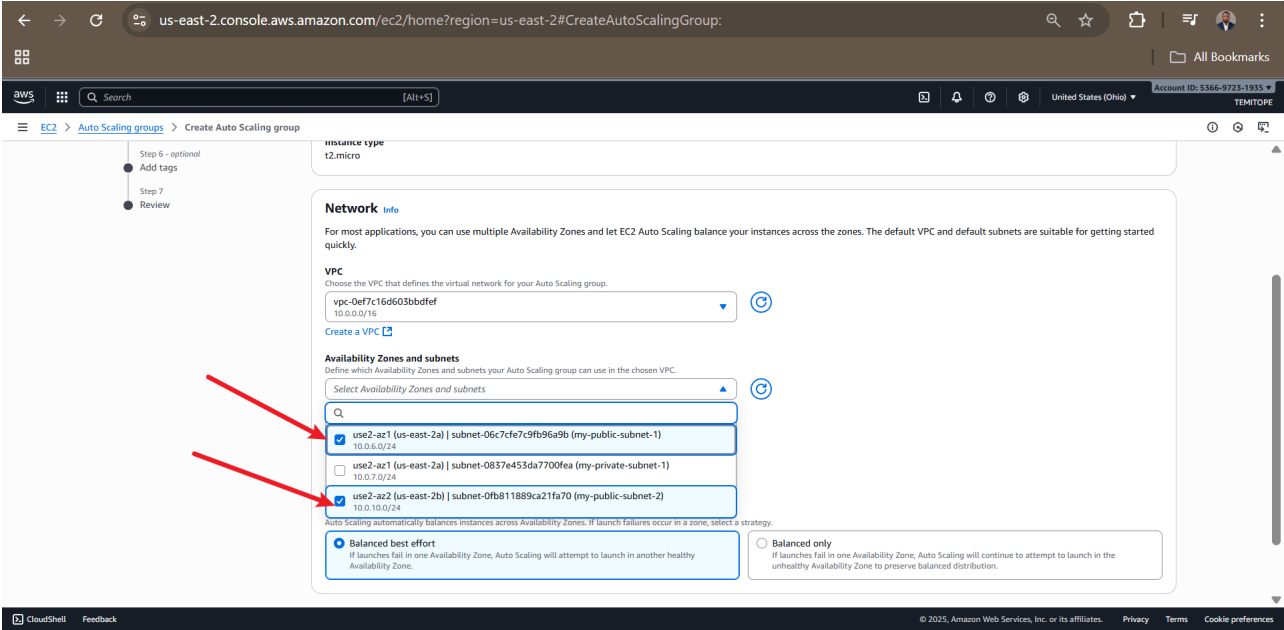
96. Template selected click next.



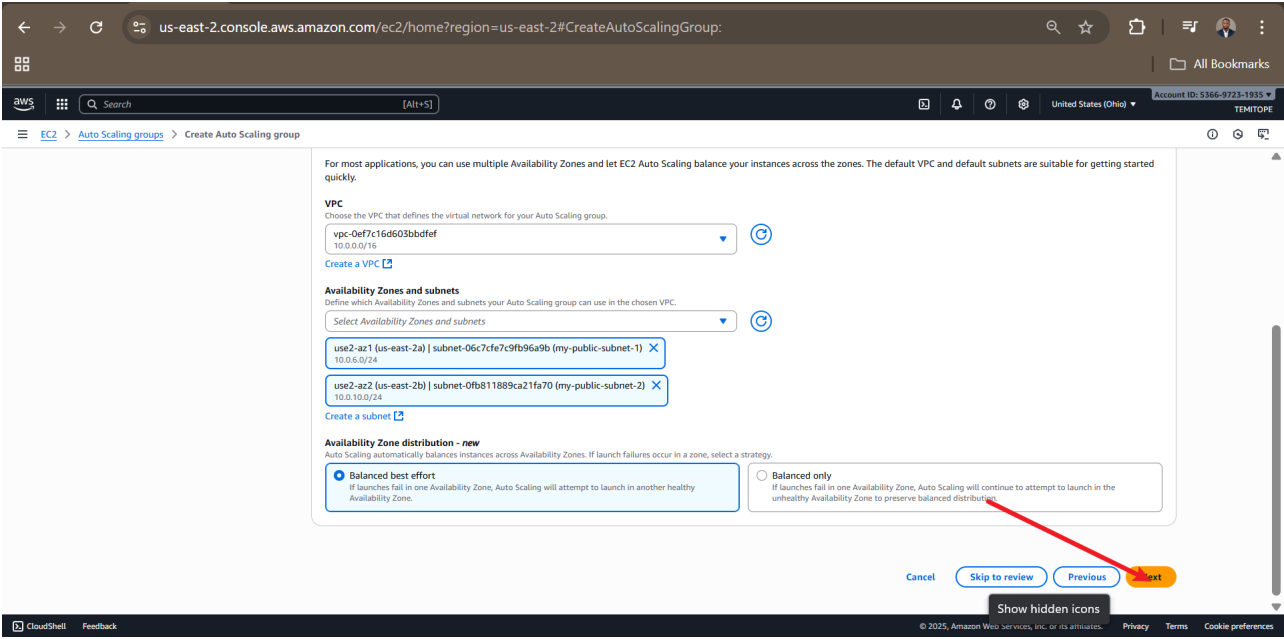
97. Choose VPC where servers are located.



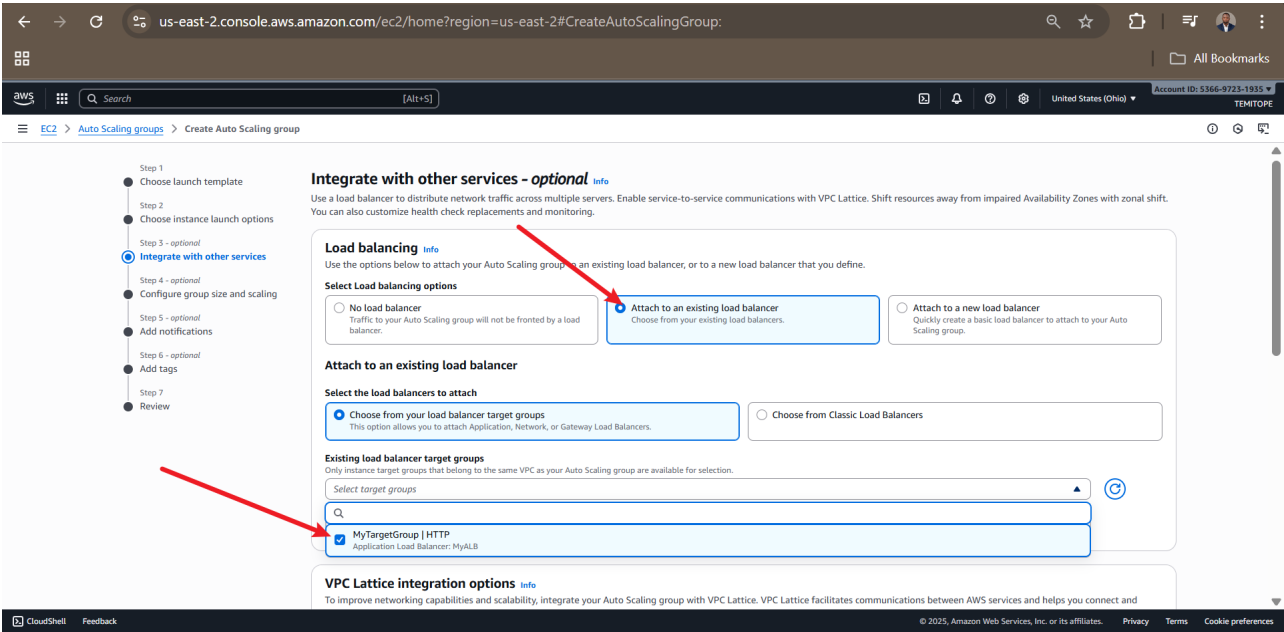
98. Select all the public availability zones if you have only one public create another one.



99. At least two public AZs selected click next.



100. Click attach to an existing load balancer and select created target group.



101. Turn on elastic load balancer healthcheck and click next.

us-east-2.console.aws.amazon.com/ec2/home?region=us-east-2#CreateAutoScalingGroup:

Application Recovery Controller (ARC) console - new info

During an Availability Zone impairment, target instance launches towards other healthy Availability Zones.

☐ Enable zonal shift

New instance launches will be retargeted towards healthy Availability Zones until the zonal shift is canceled.

Health checks

Health checks increase availability by replacing unhealthy instances. When you use multiple health checks, all are evaluated, and if at least one fails, instance replacement occurs.

EC2 health checks

[Always enabled](#)

Additional health check types - optional | info

☒ Turn on Elastic Load Balancing health checks **Recommended**

Elastic Load Balancing monitors whether instances are available to handle requests. When it reports an unhealthy instance, EC2 Auto Scaling can replace it on its next periodic check.

EC2 Auto Scaling will start to detect and act on health checks performed by Elastic Load Balancing. To avoid unexpected terminations, first verify the settings of these health checks in the Load Balancer console.

☐ Turn on VPC Lattice health checks

VPC Lattice can monitor whether instances are available to handle requests. If it considers a target as failed a health check, EC2 Auto Scaling replaces it after its next periodic check.

☐ Turn on Amazon EBS health checks

EBS monitors whether an instance's root volume or attached volume stalls. When it reports an unhealthy volume, EC2 Auto Scaling can replace the instance on its next periodic health check.

Health check grace period | info

This time period delays the first health check until your instances finish initializing. It doesn't prevent an instance from terminating when placed into a non-running state.

300 seconds

Cancel Skip to review Previous **Next**

102. Set desired capacity scaling limit (min and max) to manage cost.

us-east-2.console.aws.amazon.com/ec2/home?region=us-east-2#CreateAutoScalingGroup:

Step 1: Choose launch template

Step 2: Choose instance launch options

Step 3 - optional: Integrate with other services

Step 4 - optional: **Configure group size and scaling**

Step 5 - optional: Add notifications

Step 6 - optional: Add tags

Step 7: Review

Configure group size and scaling - optional | info

Define your group's desired capacity and scaling limits. You can optionally add automatic scaling to adjust the size of your group.

Group size | info

Set the initial size of the Auto Scaling group. After creating the group, you can change its size to meet demand, either manually or by using automatic scaling.

Desired capacity type

Choose the unit of measurement for the desired capacity value. vCPUs and Memory (GiB) are only supported for mixed instances groups configured with a set of instance attributes.

Units (number of instances)

Desired capacity

Specify your group size.

1

Scaling | info

You can resize your Auto Scaling group manually or automatically to meet changes in demand.

Scaling limits

Set limits on how much your desired capacity can be increased or decreased.

Min desired capacity

1

Equal or less than desired capacity

Max desired capacity

5

Equal or greater than desired capacity

103. Choose target tracking scaling policy and set name metric target value.

us-east-2.console.aws.amazon.com/ec2/home?region=us-east-2#CreateAutoScalingGroup:

Automatic scaling - optional | info

Choose whether to use a target tracking policy | info

You can set up other metric-based scaling policies and scheduled scaling after creating your Auto Scaling group.

☐ No scaling policies

Your Auto Scaling group will remain at its initial size and will not dynamically resize to meet demand.

☒ Target tracking scaling policy

Choose a CloudWatch metric and target value and let the scaling policy adjust the desired capacity in proportion to the metric's value.

Scaling policy name

Target Tracking Policy

Metric type | info

Monitored metric that determines if resource utilization is too low or high. If using EC2 metrics, consider enabling detailed monitoring for better scaling performance.

Average CPU utilization

Target value

75%

Instance warmup | info

300 seconds

☐ Disable scale in to create only a scale-out policy

Instance maintenance policy | info

Control your Auto Scaling group's availability during instance replacement events. This includes health checks, instance refreshes, maximum instance lifetime features and events that happen automatically to keep your group balanced, called rebalancing events.

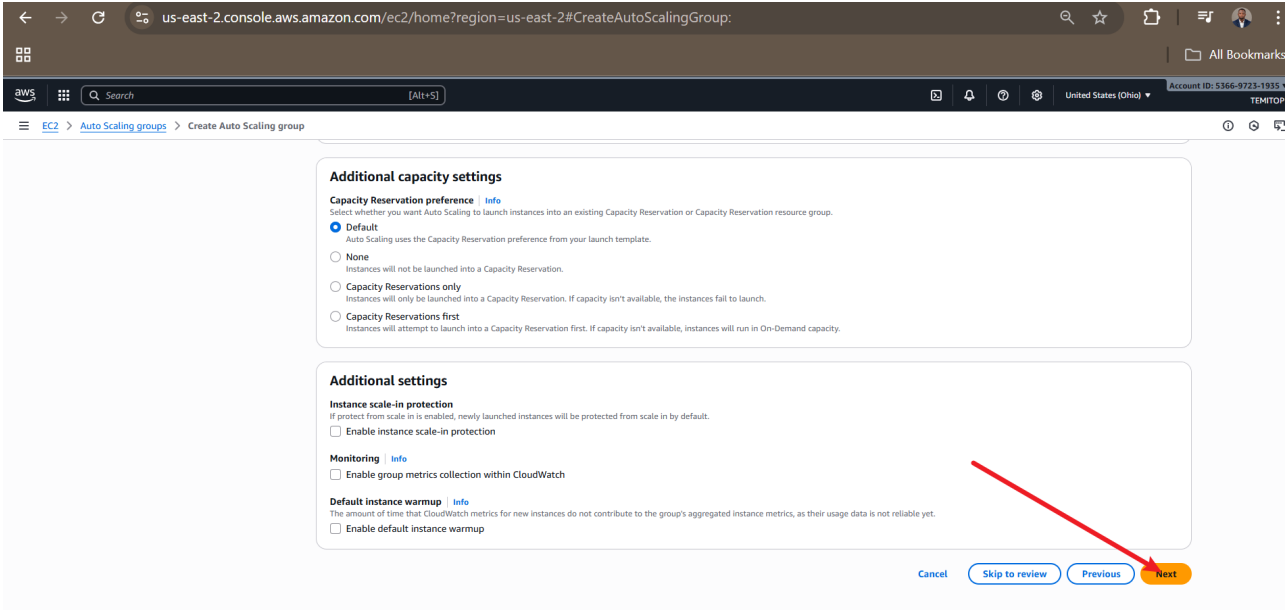
Choose a replacement behavior depending on your availability requirements

Mixed behavior | Prioritize availability | Control costs | Flexible

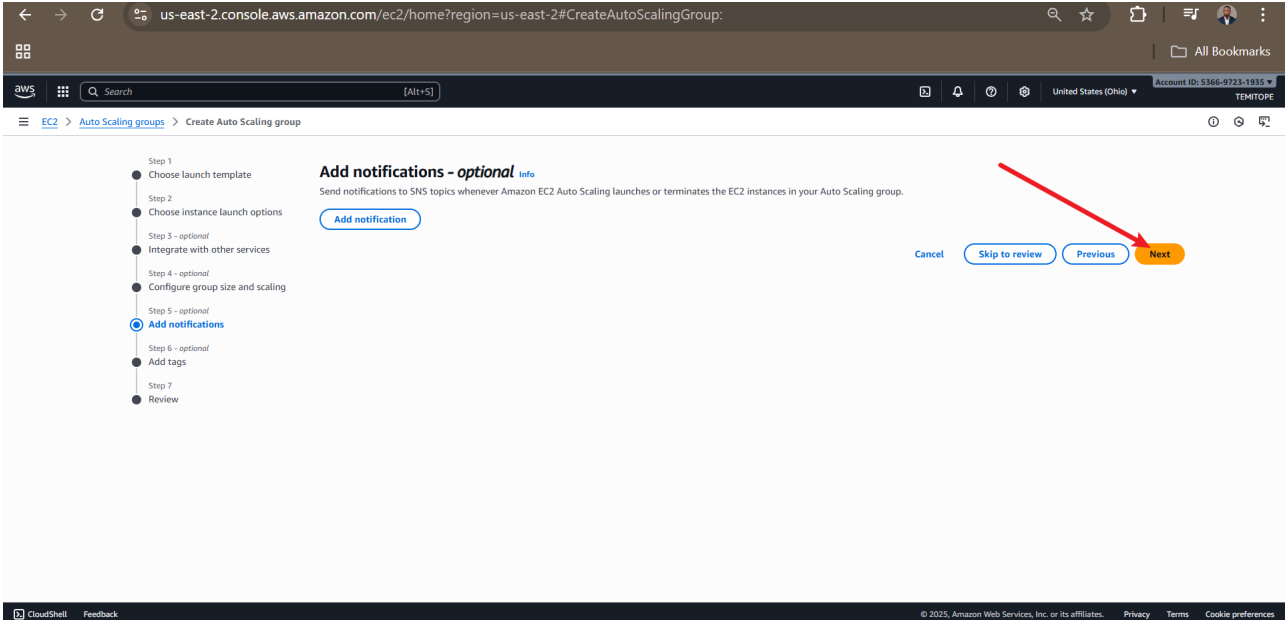
Show hidden icons

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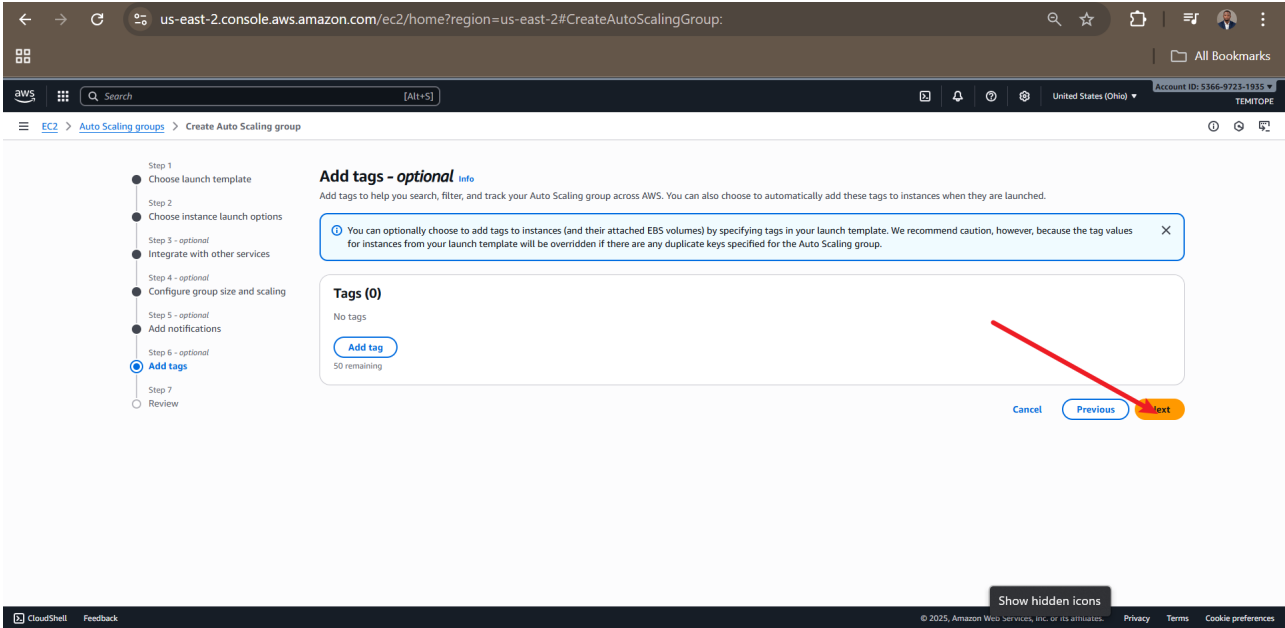
104. Leave everything at default and click next.



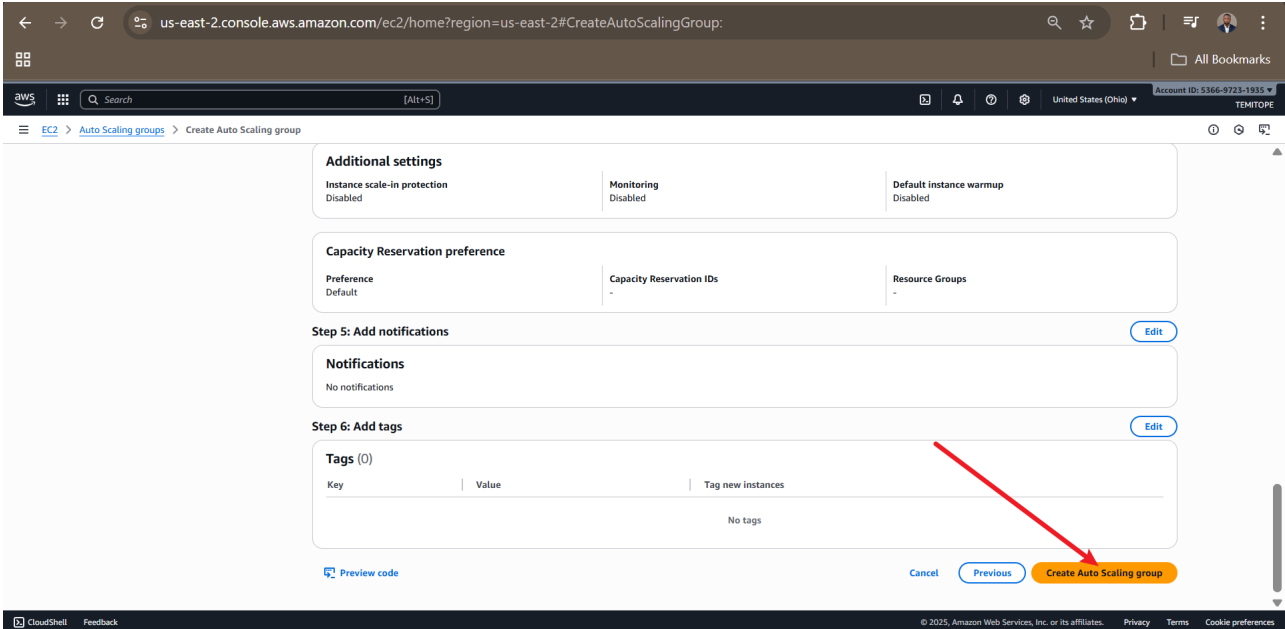
105. Leave add notification at default and click next.



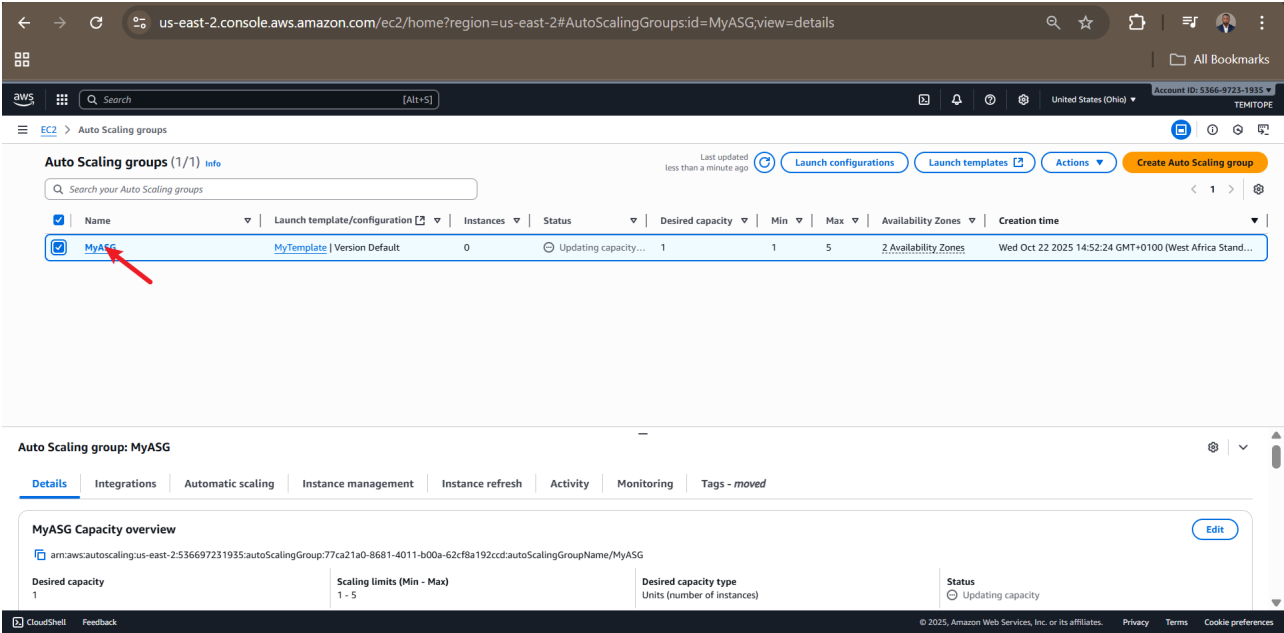
106. Leave add tags at default and click next.



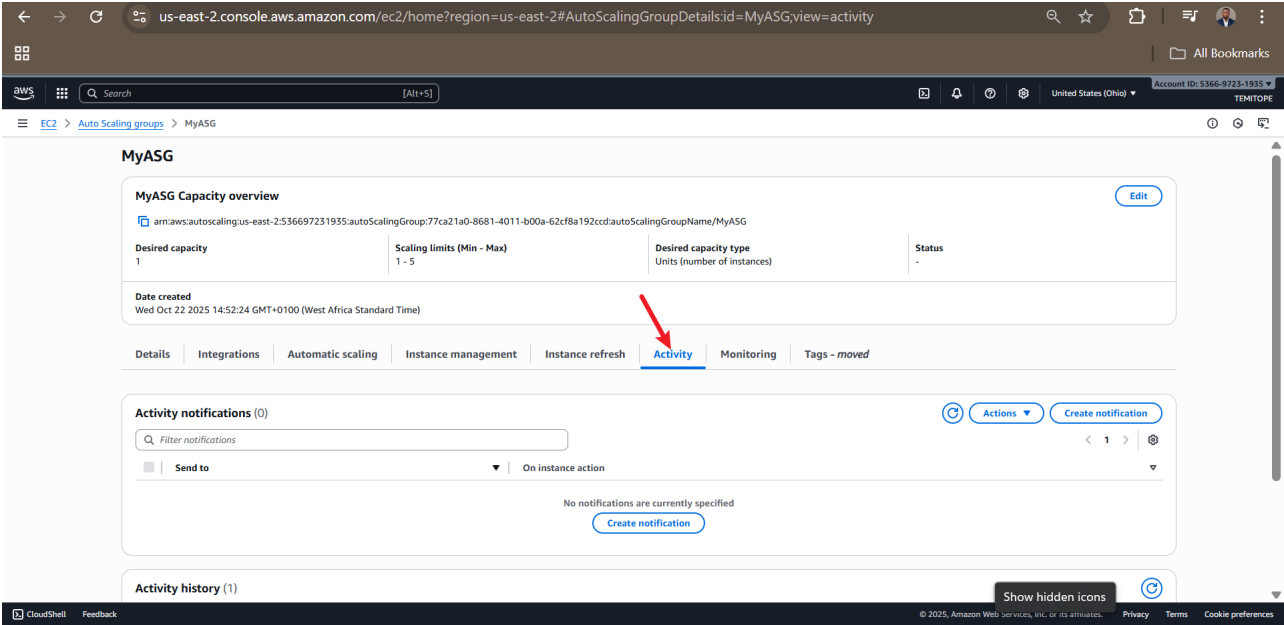
107. On the review page click Create Auto Scaling Group.



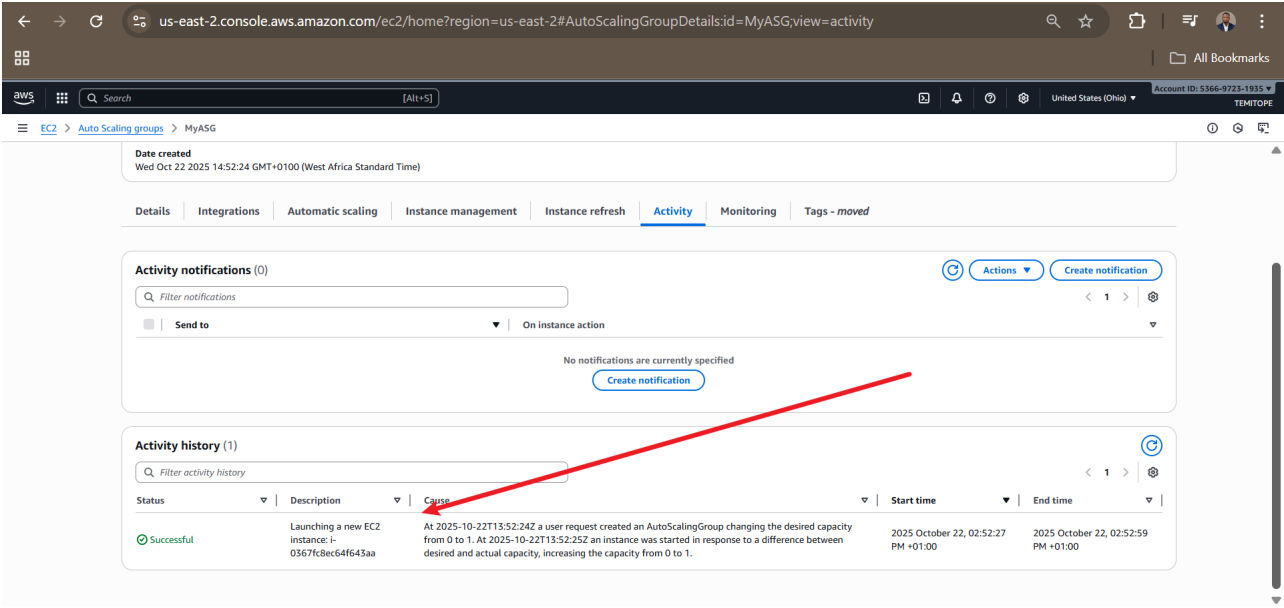
108. Auto scaling group successfully created click on it.



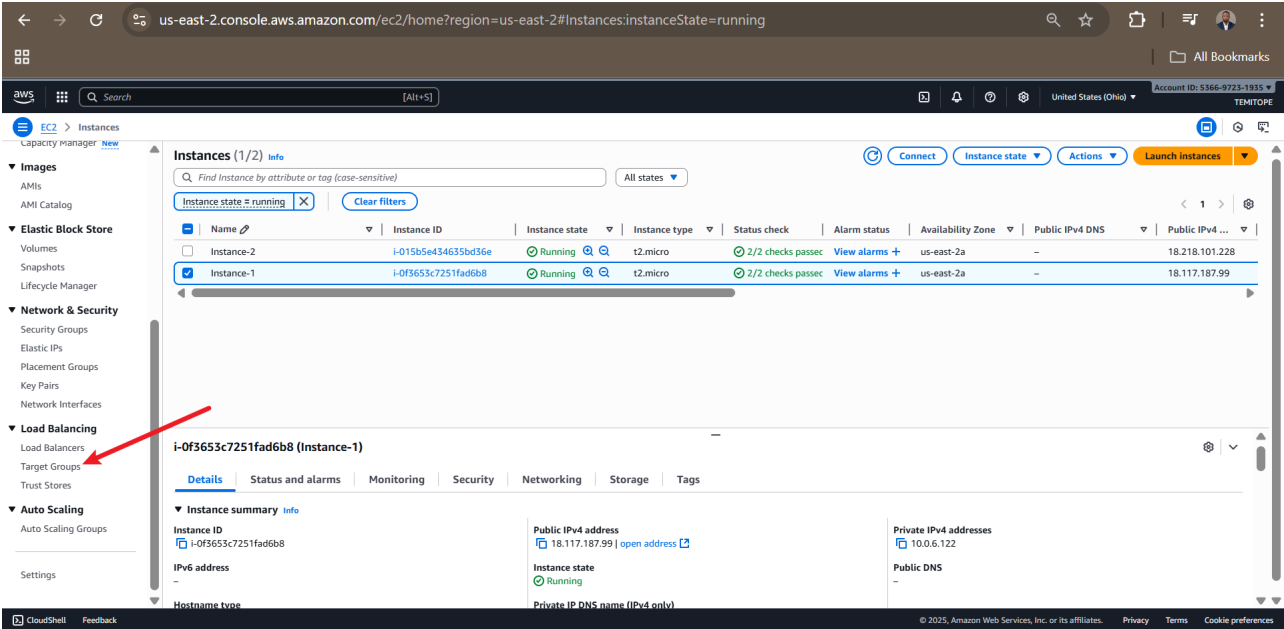
109. On created ASG dashboard click on activity tab and scroll down.



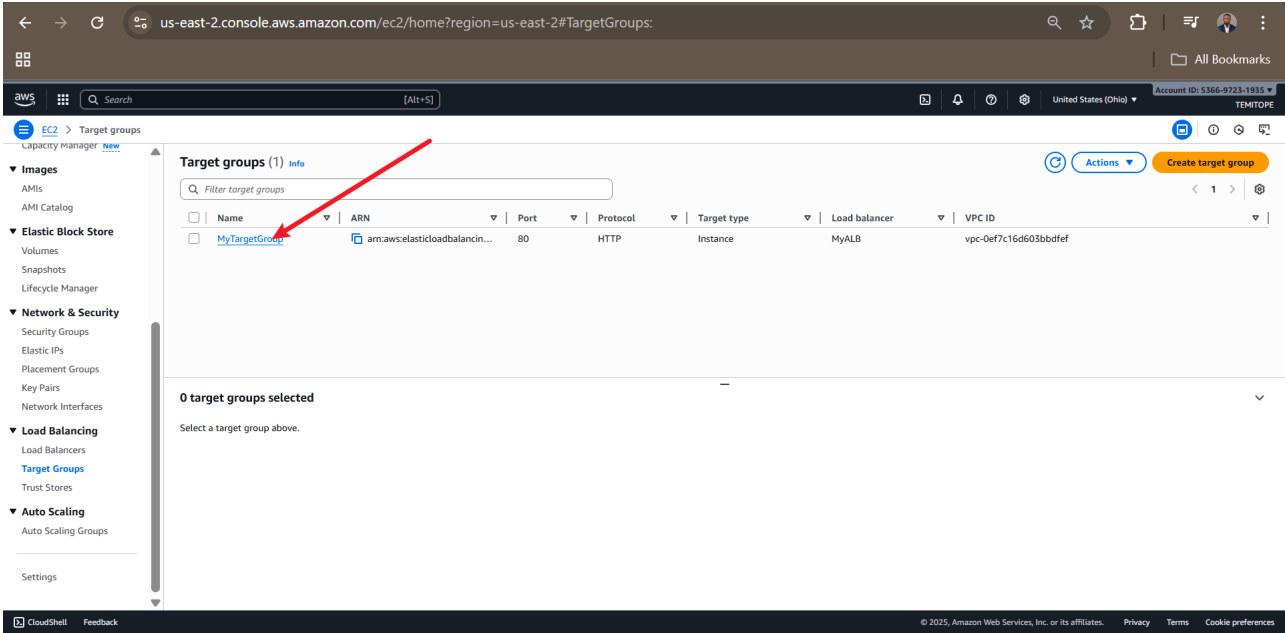
110. Activity history shows launching a new EC2 instance increasing capacity from 0-1.



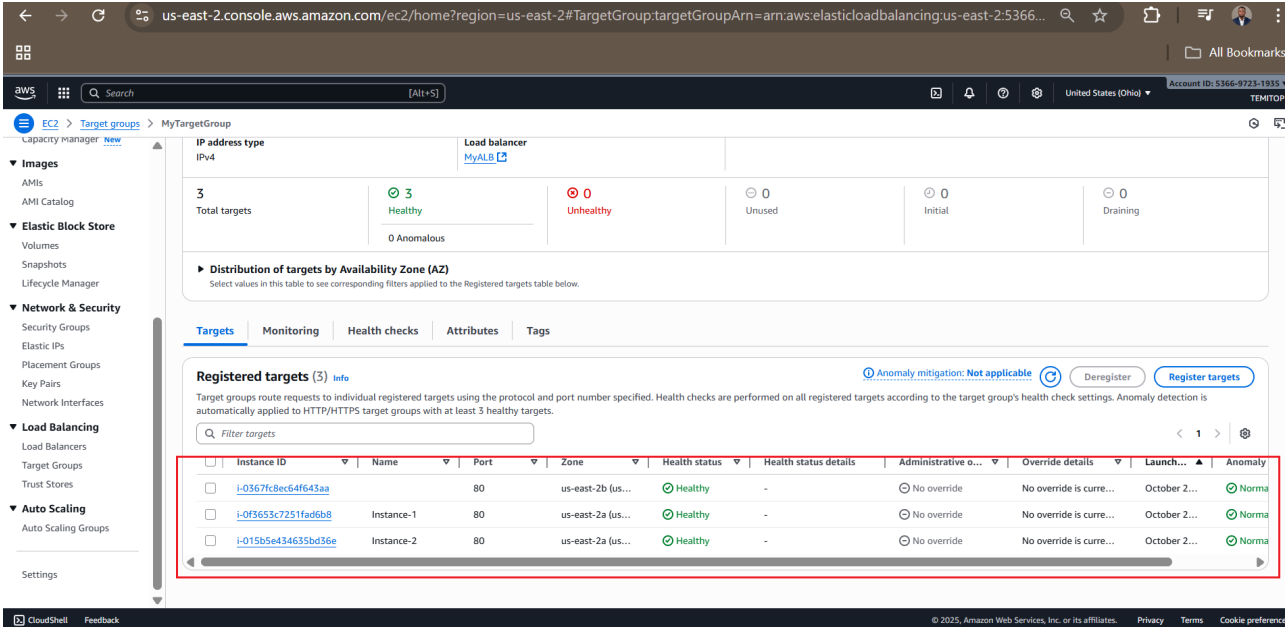
111. Navigate to instances page and under load balancing click on target group.



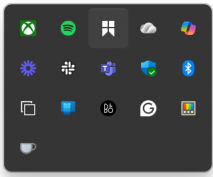
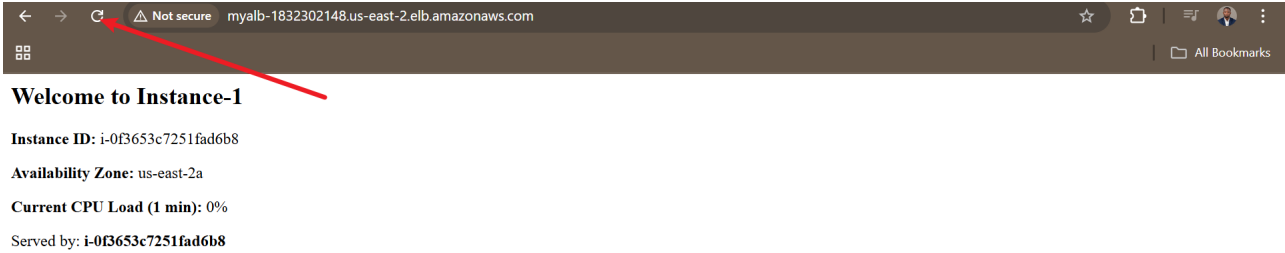
112. Click on the target group for the ALB and ASG.



113. Observe that there are now three instances under targets.



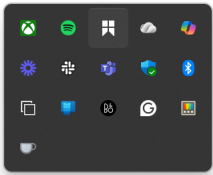
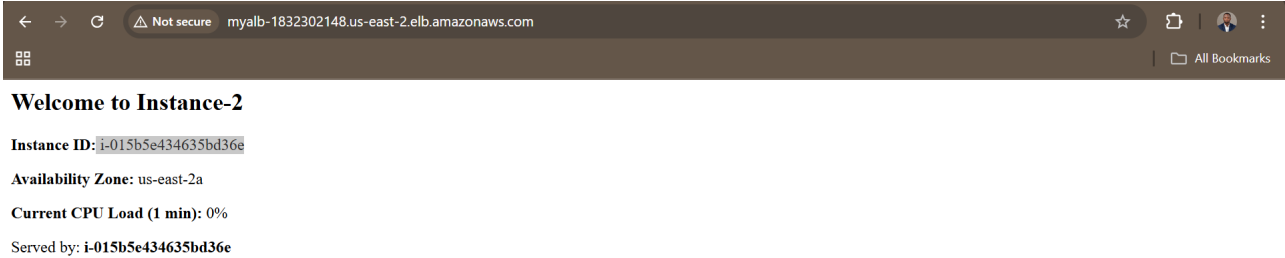
114. Proceed to the webpage and reload page.



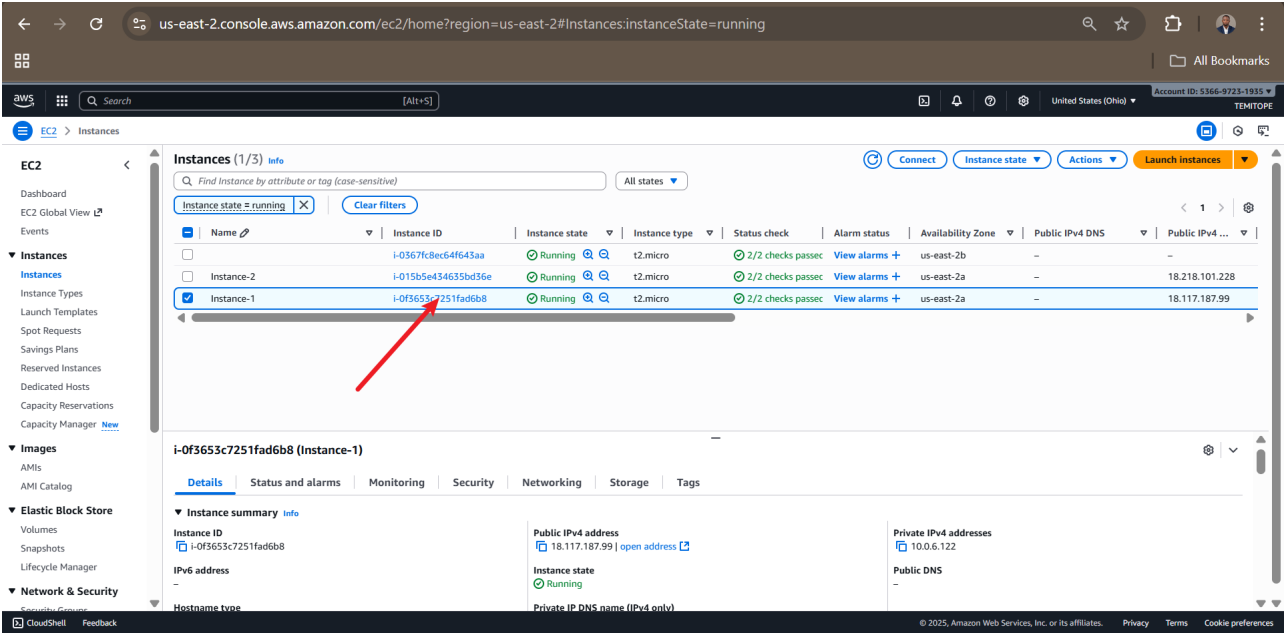
115. Instance with ID i-015b5e434635bd36e now displays reload again.
![[Instance with ID i-015b5e434635bd36e now displays reload again](./img/114.instance_with_id_i-015b5e434635bd36e_now_displays_reload_again.png)]

116. Second instance with ID i-0367fc8ec64f643aa now displays reload again.
![[Second instance with ID i-0367fc8ec64f643aa now displays reload again](./img/115.second_instance_with_id_i-0367fc8ec64f643aa_now_displays_reload_again.png)]

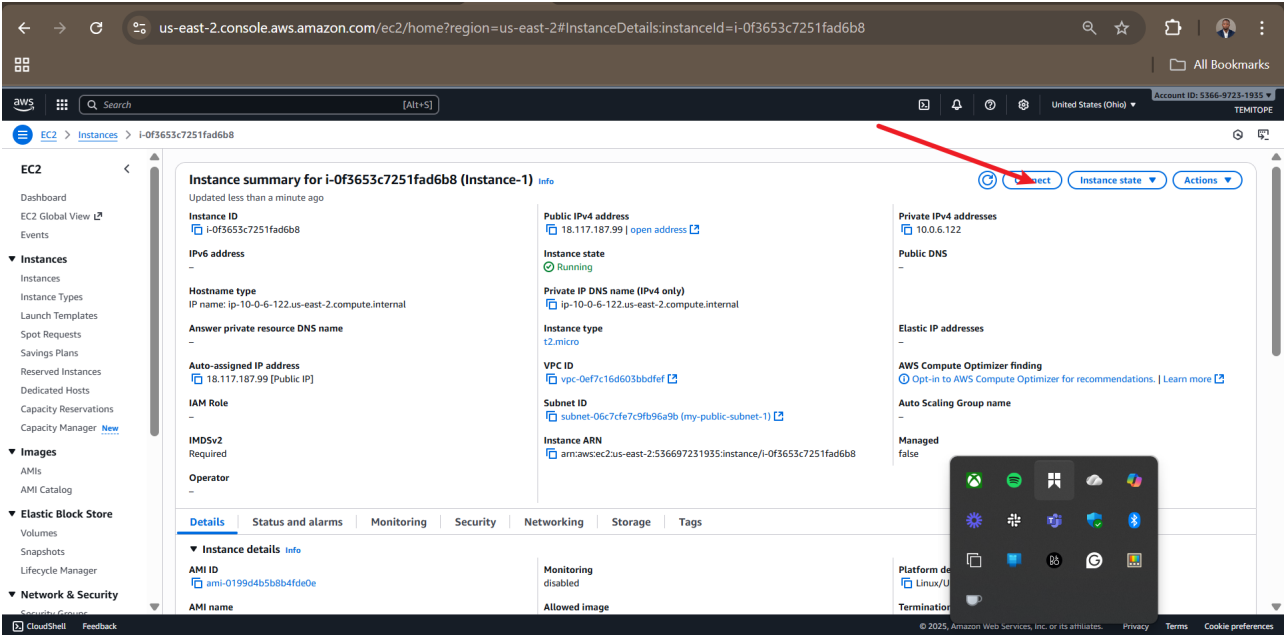
117. Third instance with ID i-015b5e434635bd36e now displays reload again.



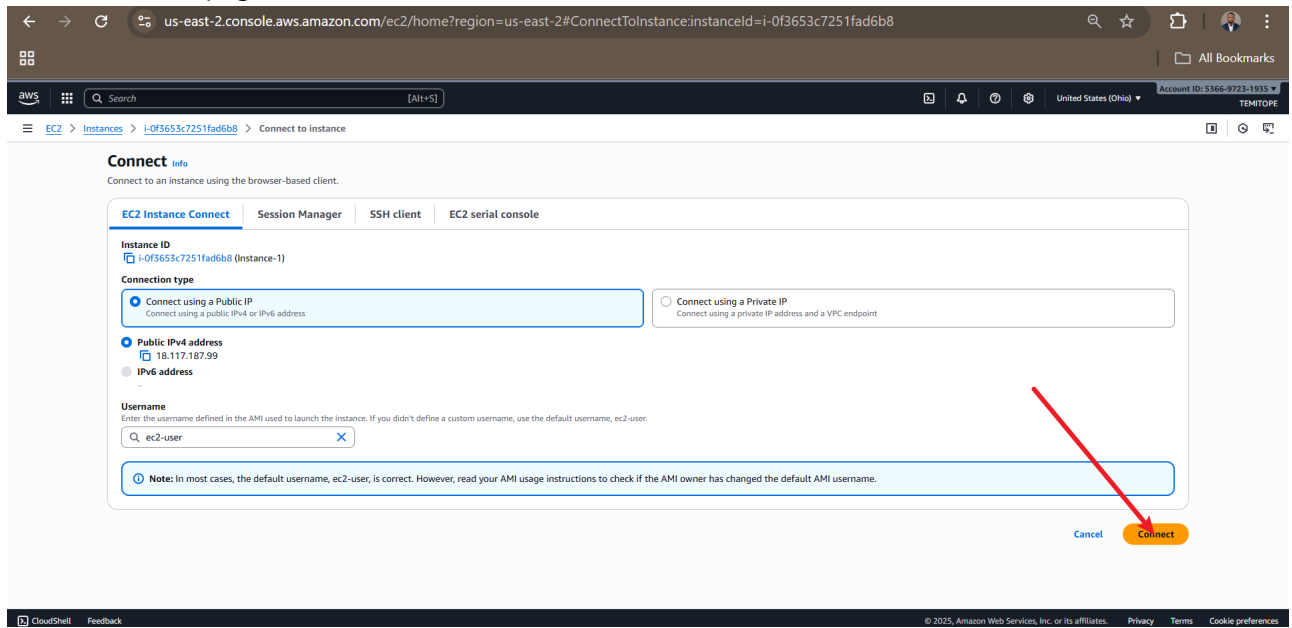
118. Navigate to the instances page select one of the original instances and click on it.



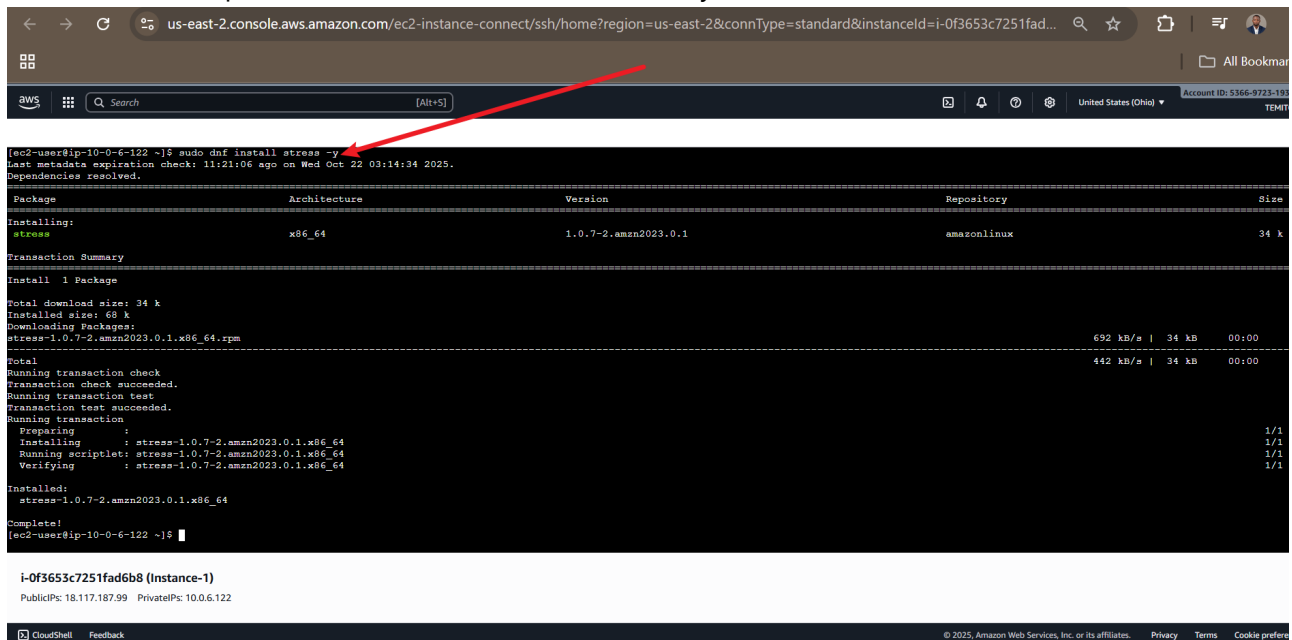
119. On the chosen original instance dashboard click connect.



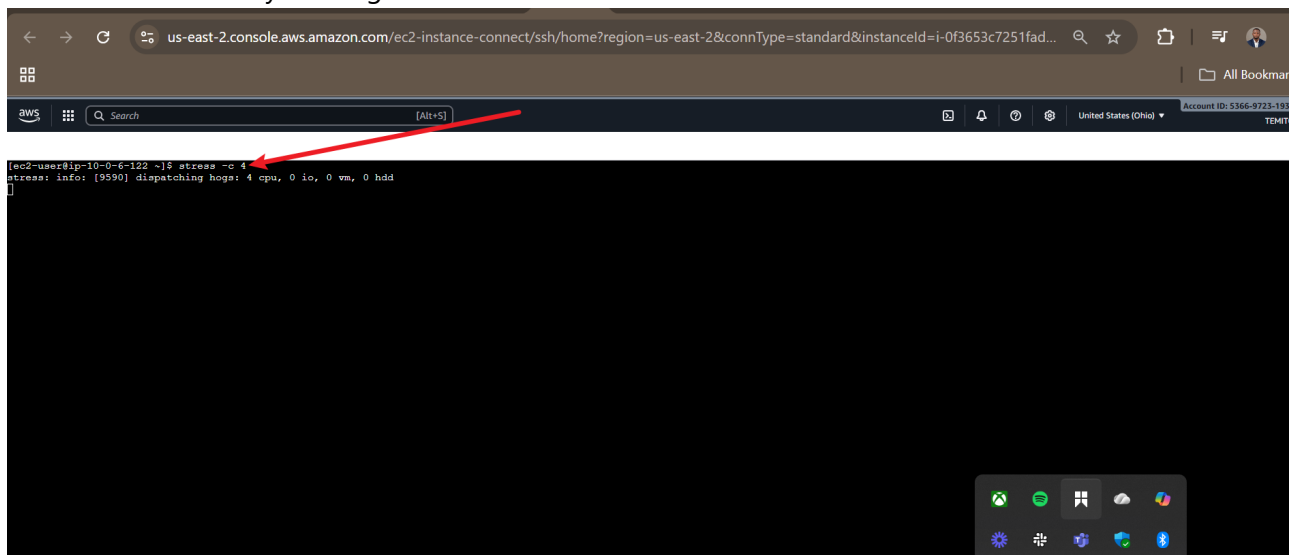
120. On the connect page under the EC2 instance connect tab click connect.



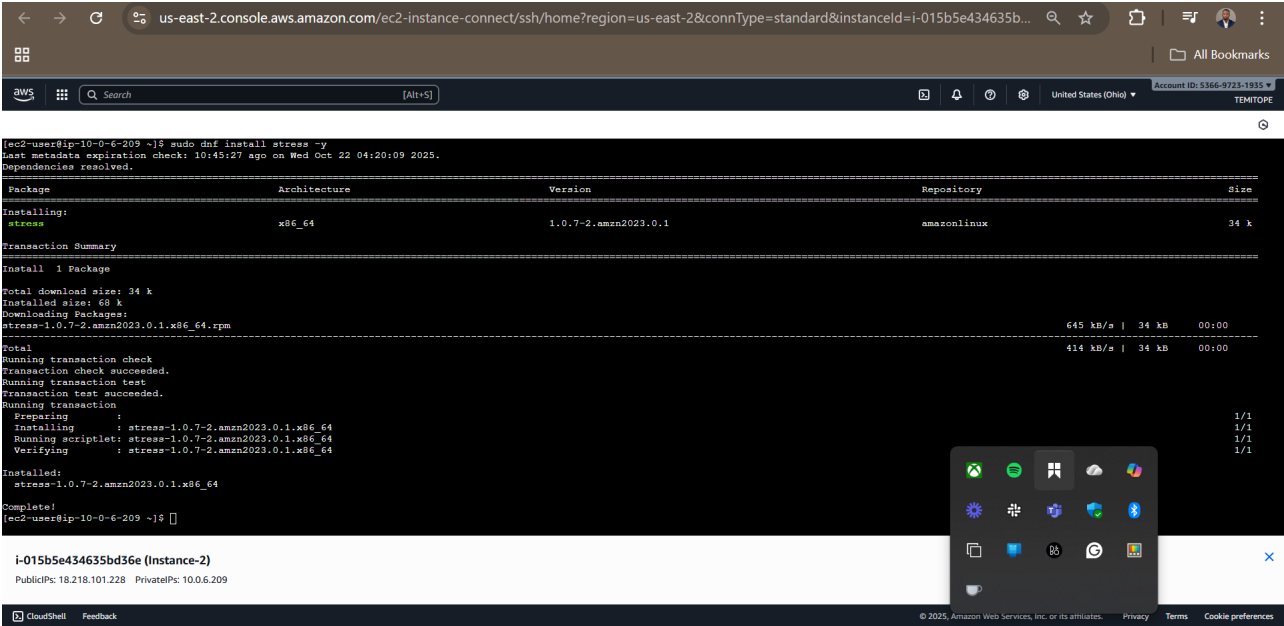
121. On the terminal update then run `sudo dnf install stress -y`.



122. Generate CPU load by running `stress -c 4`.



123. Note that after reloading CPU utilization of i-0f3653c7251fad6b8 has now increased to 400 percent.
![Note that after reloading CPU utilization of i-0f3653c7251fad6b8 has now increased to 400 percent]
(./img/122.note_that_after_reloading_cpu_utilization_of_i-0f3653c7251fad6b8_has_now_increased_to_400_percent.png)
124. Install stress on the second original instance as well.



125. Stress the second original instance.
- A screenshot of the AWS CloudShell interface. The terminal window shows the command `stress -c 4` being executed. The output displays the stress test configuration: `stress: info: [177112] dispatching hogs: 4 cpu, 0 io, 0 vm, 0 hdd`. The stress test is running, and the terminal output shows the progress of the test. A system tray with various application icons is overlaid on the bottom right of the terminal window.

124b. Note that the ASG created instance can not connect like the originals.

Connect Info

Connect to an instance using the browser-based client.

EC2 Instance Connect

Session Manager

SSH client

EC2 serial console

No public IPv4 or IPv6 address assigned

With no public IPv4 or IPv6 address, you can't use EC2 Instance Connect. Alternatively, you can try connecting using [EC2 Instance Connect Endpoint](#).

Instance ID

i-0367fc8ec64f643aa

Connection type

Connect using a Public IP

Connect using a public IPv4 or IPv6 address

Connect using a Private IP

Connect using a private IP address and a VPC endpoint

Public IPv4 address

IPv6 address

Username

Enter the username defined in the AMI used to launch the instance. If you didn't define a custom username, use the default username, root.

root

Note: In most cases, the default username, root, is correct. However, read your AMI usage instructions to check if the AMI owner has changed the default AMI username.

125. Navigate to instances page and click on launch template.

us-east-2.console.aws.amazon.com/ec2/home?region=us-east-2#Instances:instanceState=running

EC2 > Instances

Instances (1/3) Info

Instance state = running

Clear filters

Name	Instance ID	Instance state	Instance type	Status check	Alarm status	Availability Zone	Public IPv4 DNS	Public IPv4 ...
Instance-2	i-015b5e434635bd36e	Running	t2.micro	2/2 checks passed	View alarms +	us-east-2b	-	18.218.101.228
Instance-1	i-0f3653c7251fad6b8	Running	t2.micro	2/2 checks passed	View alarms +	us-east-2a	-	18.117.187.99

i-0367fc8ec64f643aa

Details

Status and alarms

Monitoring

Security

Networking

Storage

Tags

Instance summary

Instance ID

i-0367fc8ec64f643aa

IPv6 address

-

Public IPv4 address

-

Instance state

Running

Private IPv4

10.0.10.0

Public DNS

-

Private DNS name (IPv4 only)

-

126. Click on the created launch template.

us-east-2.console.aws.amazon.com/ec2/home?region=us-east-2#LaunchTemplates:

EC2 > Launch Templates

Launch Templates (1) Info

Launch Template ID

Launch Template Name

Default Version

Latest Version

Create Time

Created By

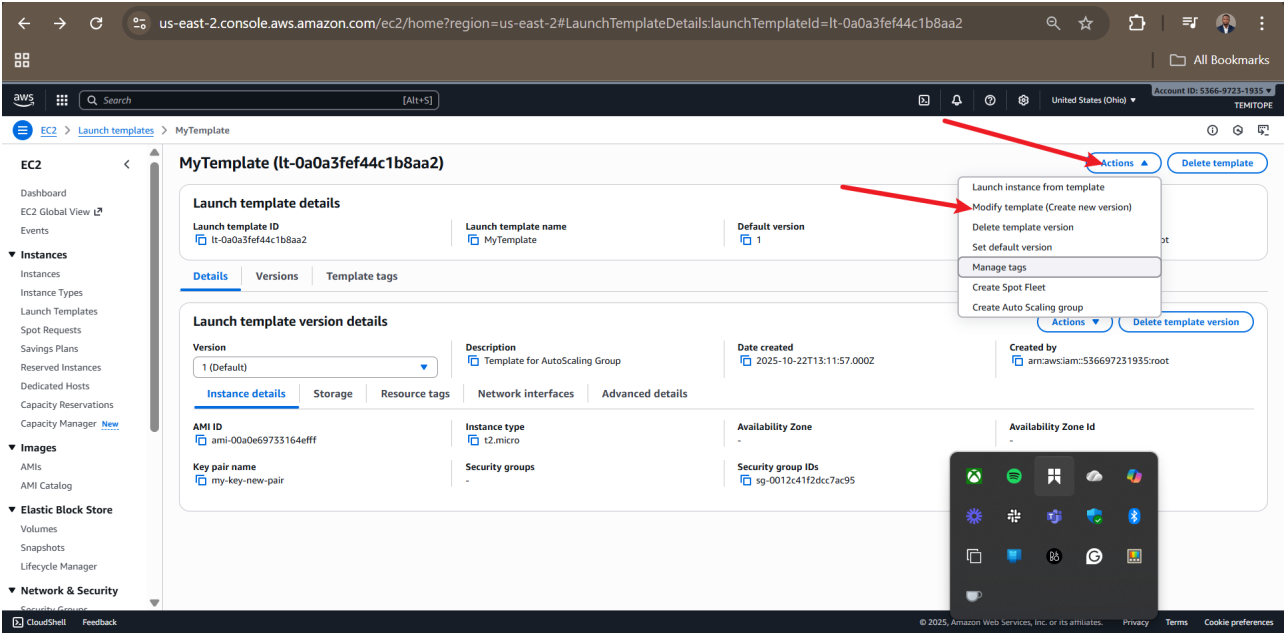
Managed

Operator

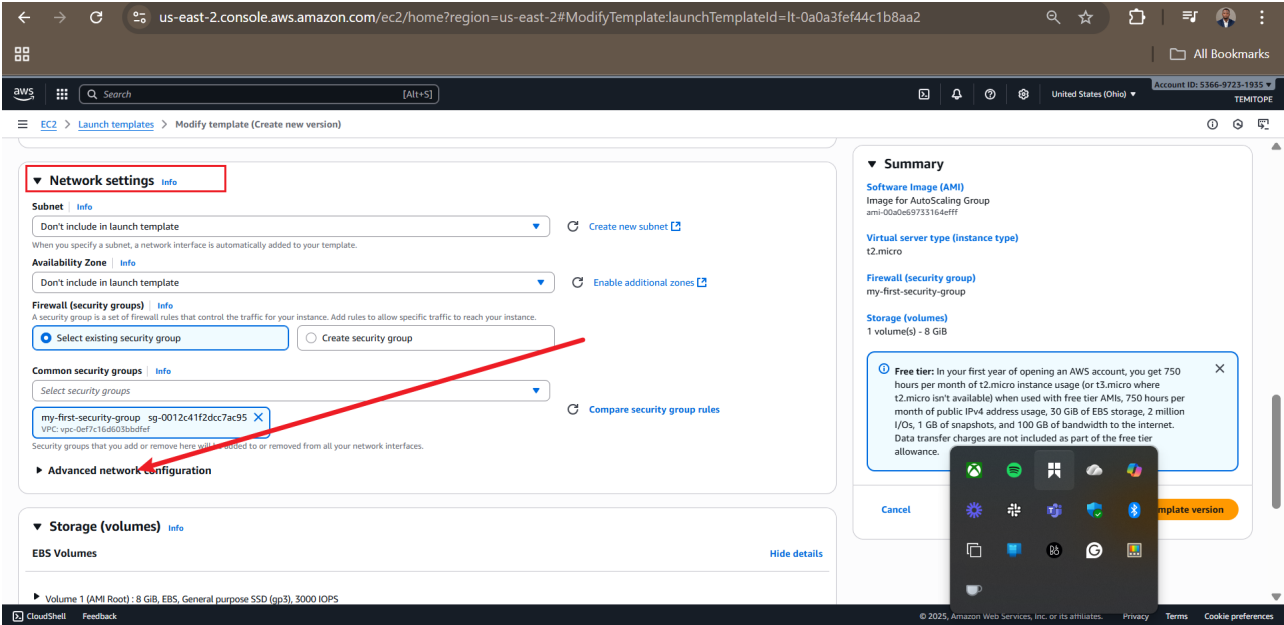
lt-0a0a3fef44c1b8592	MyTemplate	1	1	2025-10-22T13:11:57.000Z	arn:aws:iam::536697231935:root	false	-
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Select a launch template

127. Click on actions then modify template (create new version).



128. Now scroll down to network setting click on advanced network configuration.



129. Under Advanced network configuration enable Auto Assign Public IP create new version.

The screenshot shows the AWS Management Console interface for modifying a launch template. The browser address bar shows the URL: `us-east-2.console.aws.amazon.com/ec2/home?region=us-east-2#ModifyTemplate:launchTemplateId=lt-0a0a3fef44c1b8aa2`. The page title is "Modify template (Create new version)".

In the "Advanced network configuration" section, the "Auto-assign public IP" dropdown is set to "Enable". A red box highlights this dropdown, and a red arrow points to the "Create template version" button in the Summary panel on the right.

The Summary panel on the right shows the following details:

- Software Image (AMI):** Image for AutoScaling Group ami-0b0e469735164aff7
- Virtual server type (instance type):** t2.micro
- Firewall (security group):** my-first-security-group
- Storage (volumes):** 1 volume(s) - 8 GB

A free tier notification is also visible: "Free tier: In your first year of opening an AWS account, you get 750 hours per month of t2.micro instance usage (or t3.micro where t2.micro isn't available) when used with free tier AMIs, 750 hours per month of public IPv4 address usage, 30 GiB of EBS storage, 2 million I/Os, 1 GiB of snapshots, and 100 GiB of bandwidth to the Internet. Data transfer charges are not included as part of the free tier allowance."

130. Successfully modified template click view launch templates.

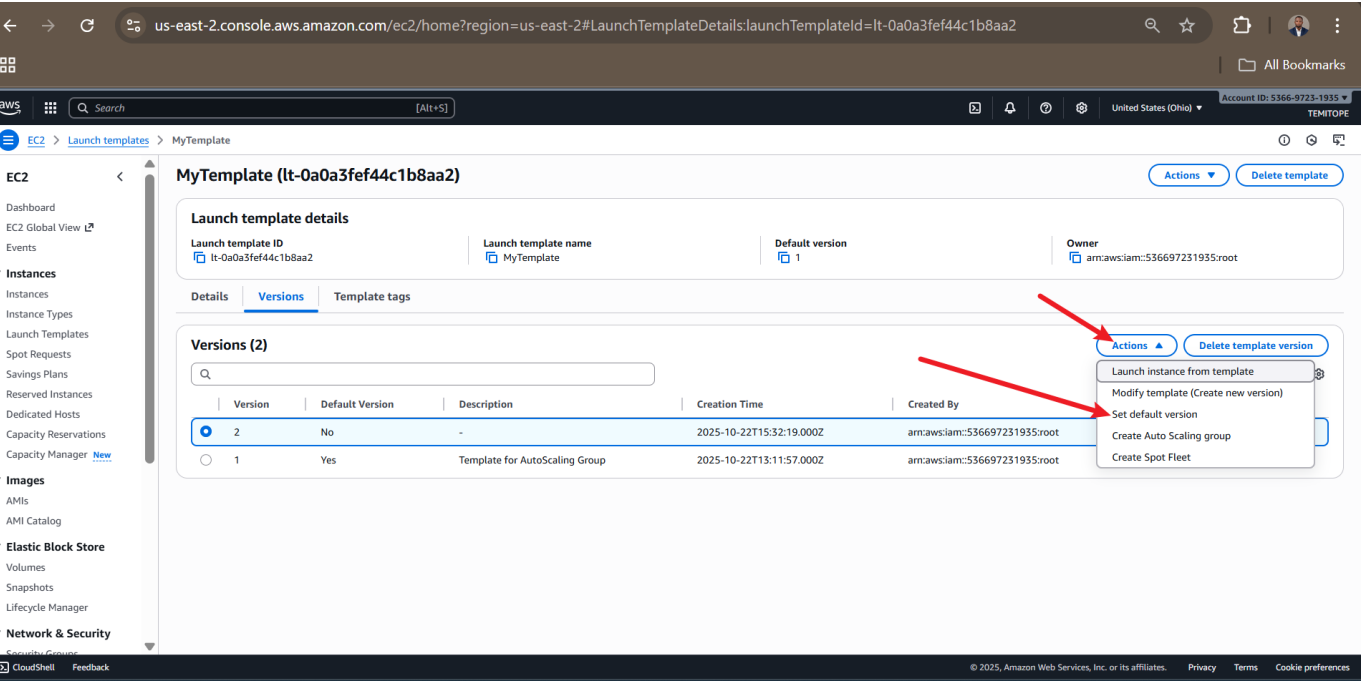
The screenshot shows the AWS Management Console interface for the "Next Steps" page after successfully modifying a launch template. The browser address bar shows the URL: `us-east-2.console.aws.amazon.com/ec2/home?region=us-east-2#ModifyTemplate:launchTemplateId=lt-0a0a3fef44c1b8aa2`. The page title is "Modify template (Create new version)".

The "Next Steps" section includes the following links and descriptions:

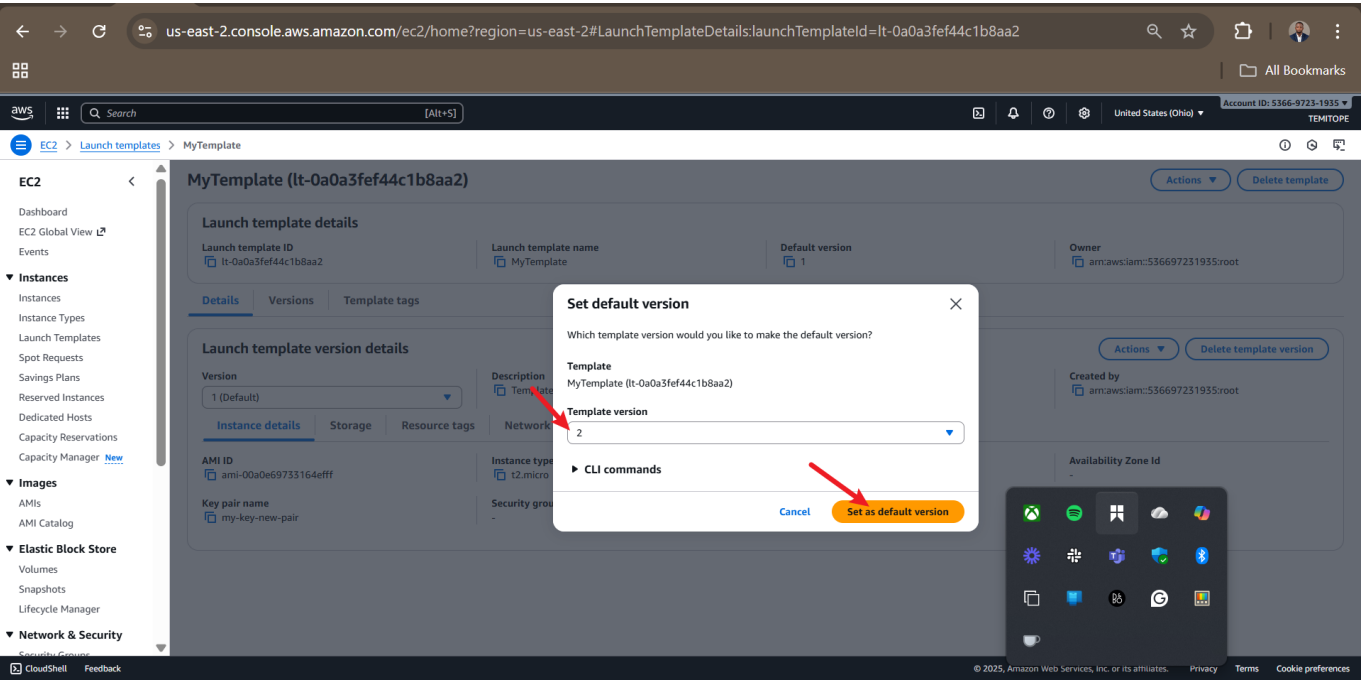
- Launch an instance:** With On-Demand Instances, you pay for compute capacity by the second (or by the hour for all other operating systems) with a minimum of 60 seconds or by the hour (for all other operating systems) with no long-term commitments or upfront payments. Launch an On-Demand Instance from your launch template.
[Launch instance from this template](#)
- Create an Auto Scaling group from your template:** Amazon EC2 Auto Scaling helps you maintain application availability and allows you to scale your Amazon EC2 capacity up or down automatically according to conditions you define. You can use Auto Scaling to help ensure that you are running your desired number of Amazon EC2 instances during demand spikes to maintain performance and decrease capacity during lulls to reduce costs.
[Create Auto Scaling group](#)
- Create Spot Fleet:** A Spot Instance is an unused EC2 instance that is available for less than the On-Demand price. Because Spot Instances enable you to request unused EC2 instances at steep discounts, you can lower your Amazon EC2 costs significantly. The hourly price for a Spot Instance in each Availability Zone is set by Amazon EC2, and adjusted gradually based on the long-term supply of and demand for Spot Instances. Spot instances are well-suited for data-analysis, batch jobs, background processing, and optional tasks.
[Create Spot Fleet](#)

A red arrow points to the "View launch templates" button at the bottom right of the page.

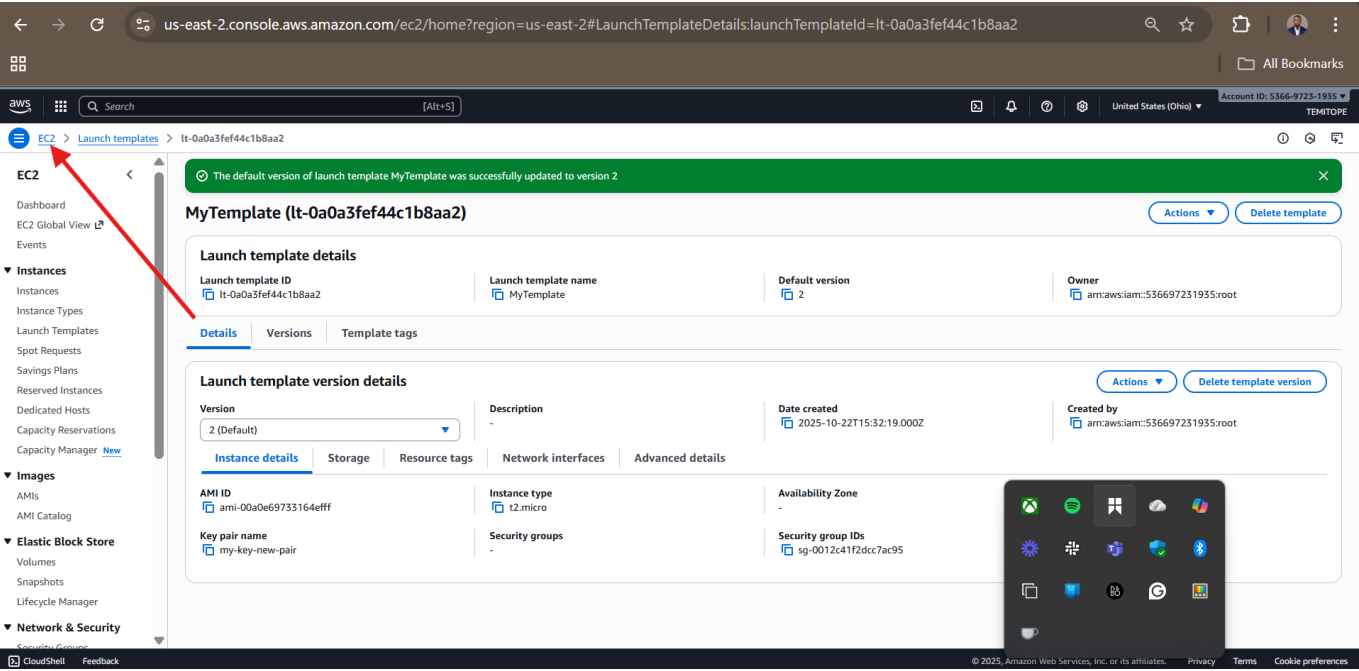
130b. Click versions tab then actions button then set default version.



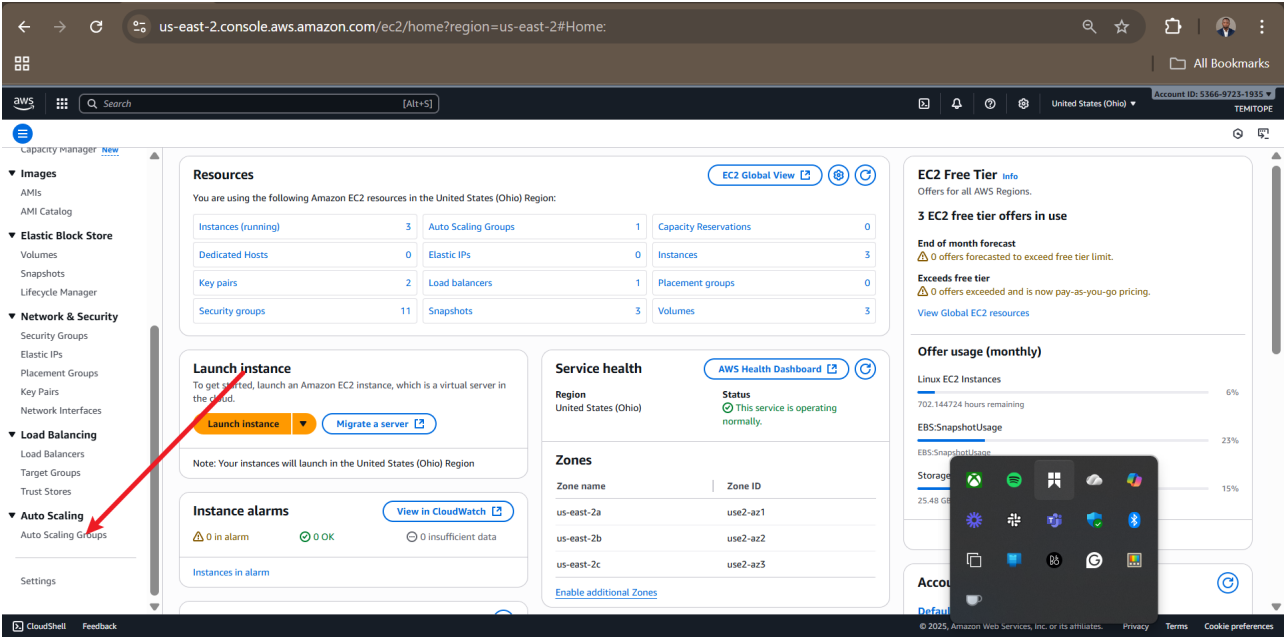
130c. Choose version 2 and click set as default version.



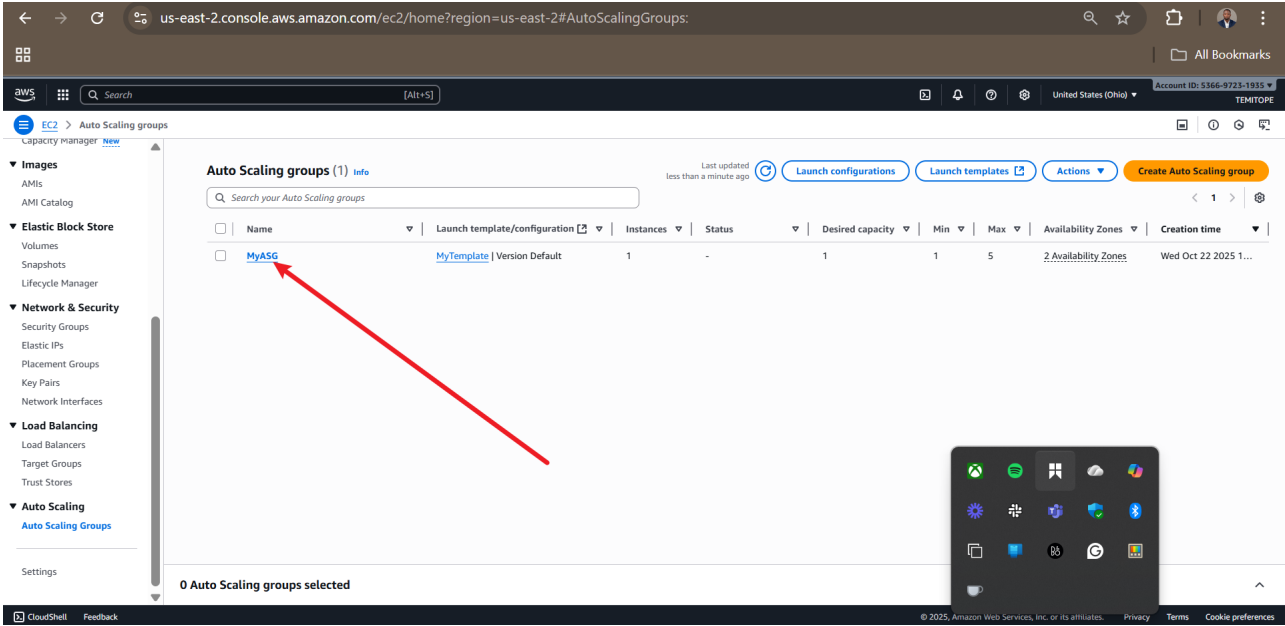
130d. Version 2 successfully set as default version click EC2 to navigate to EC2.



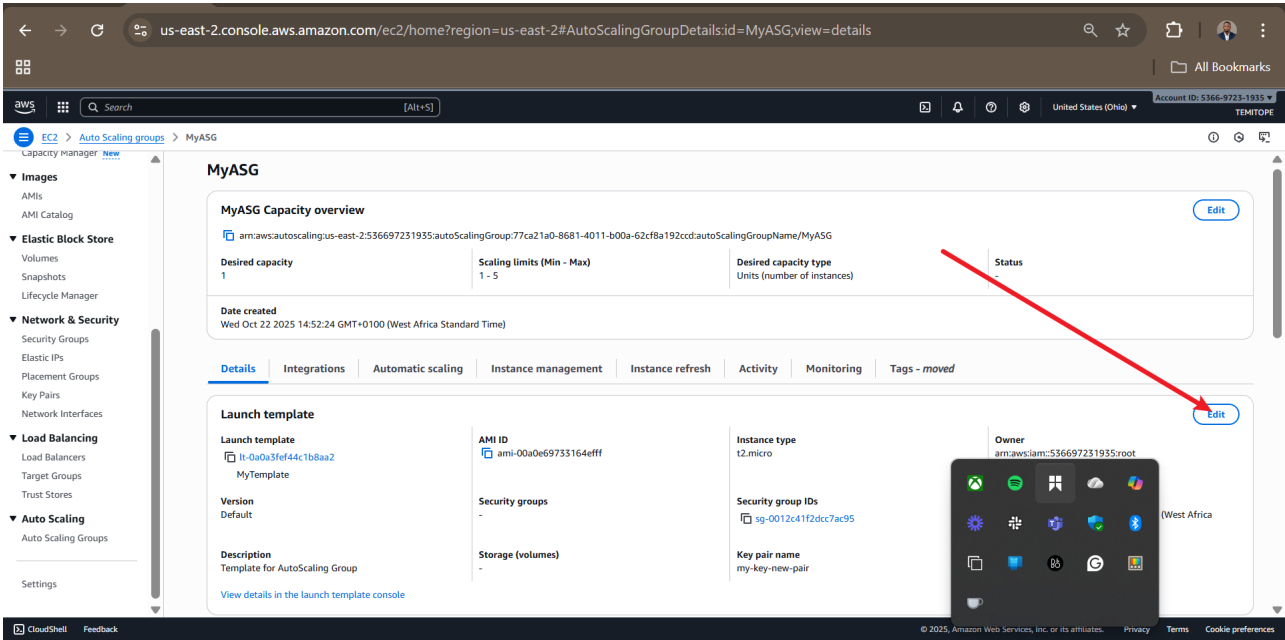
131. Navigate to EC2 page scroll to Auto scaling and click on auto scaling group.



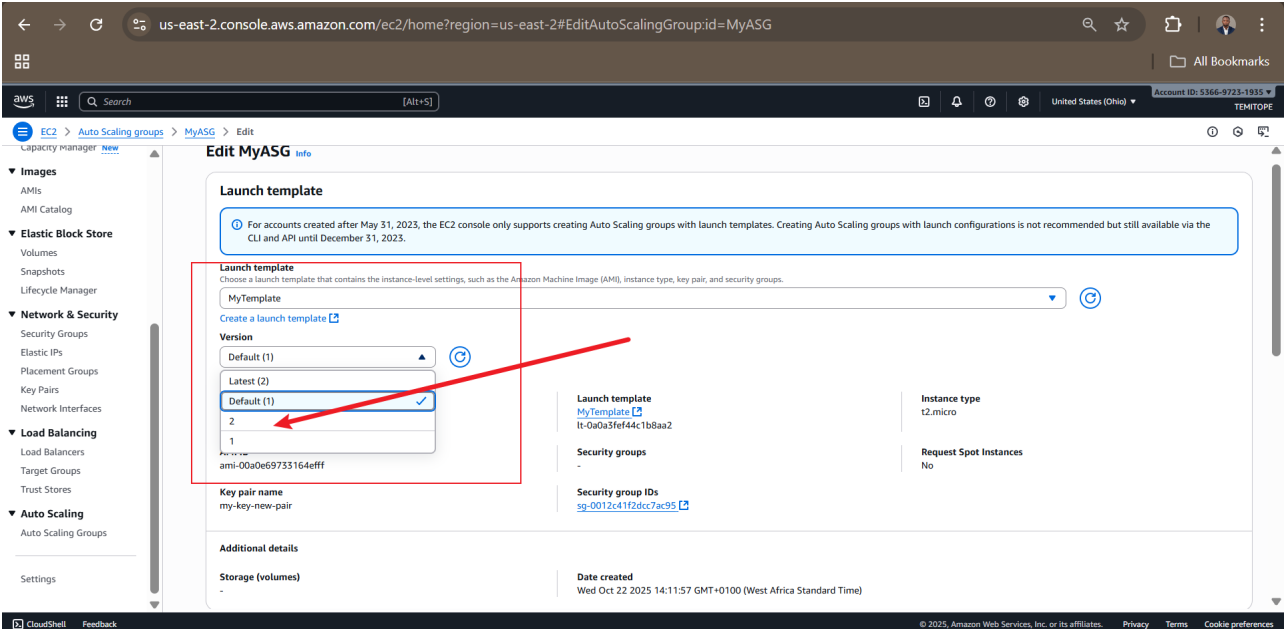
132. Click on the created AutoScaling Group.



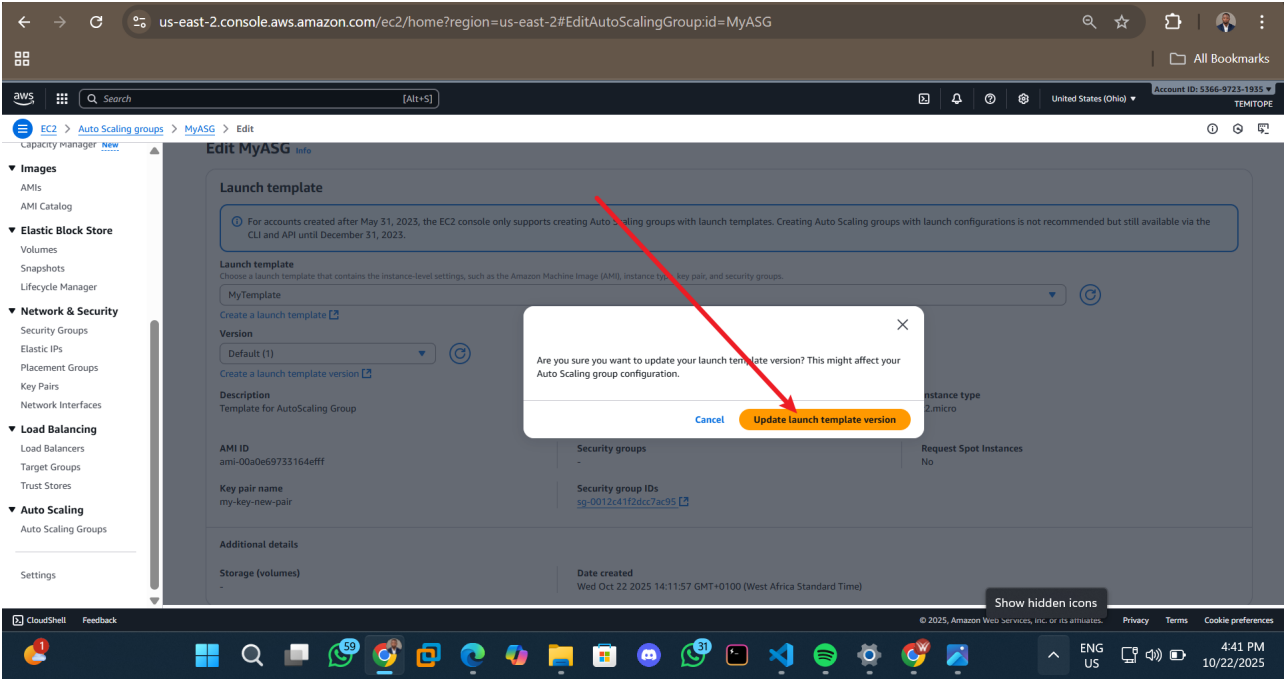
133. Click edit we are doing this to make scales public.



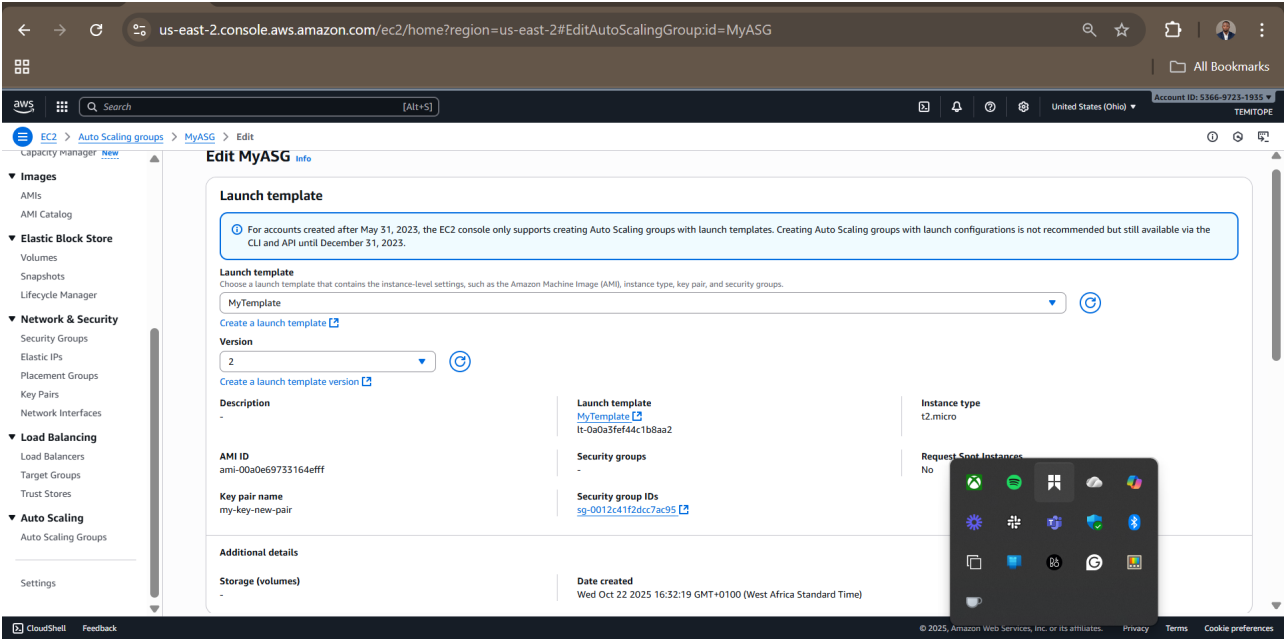
134. Under launch template version select 2 (new version).



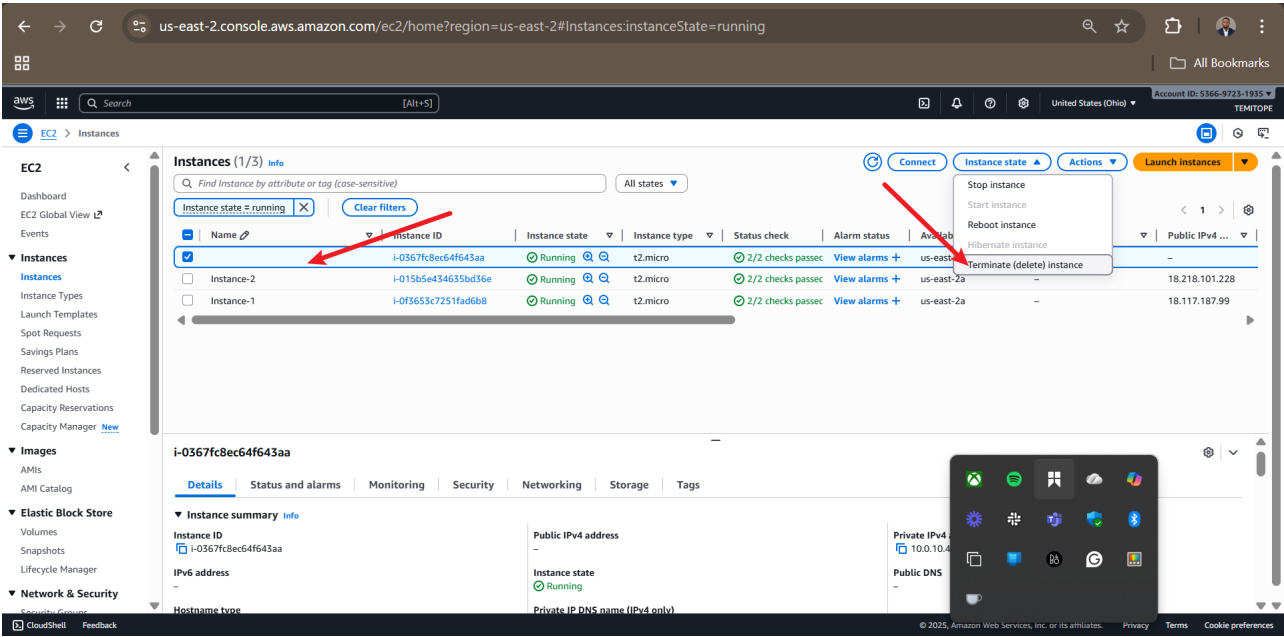
135. Click update launch template version on the pop-up.



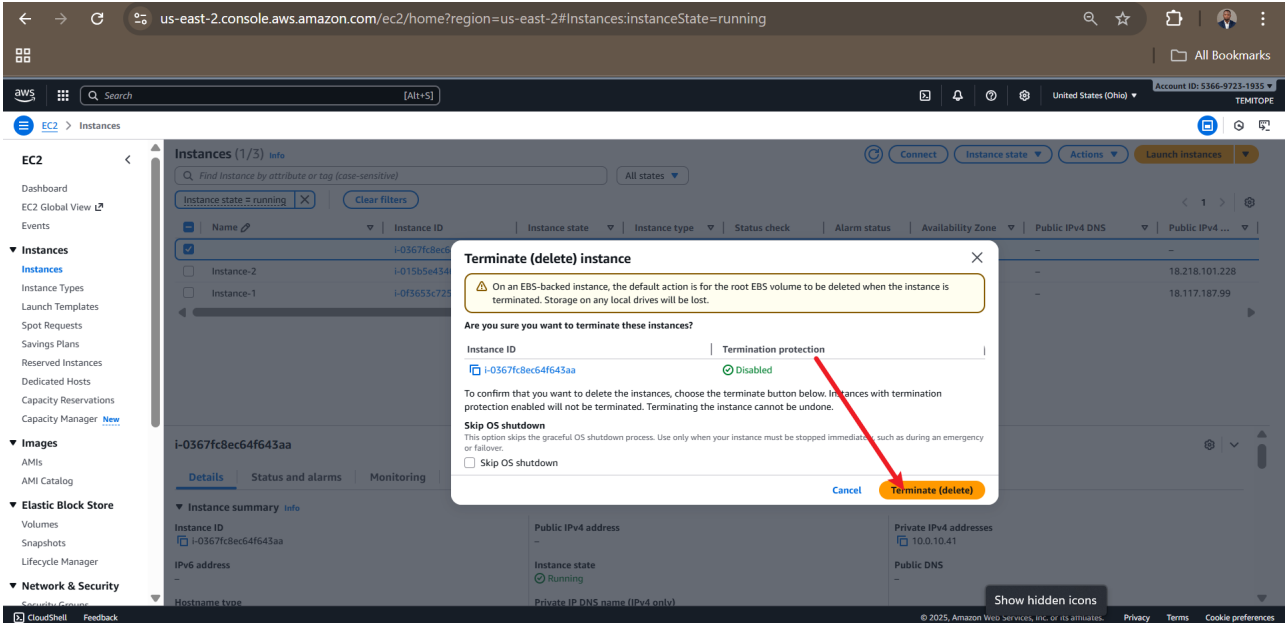
136. Version now updated so we can use EC2 instance connect to fuzz our scales.



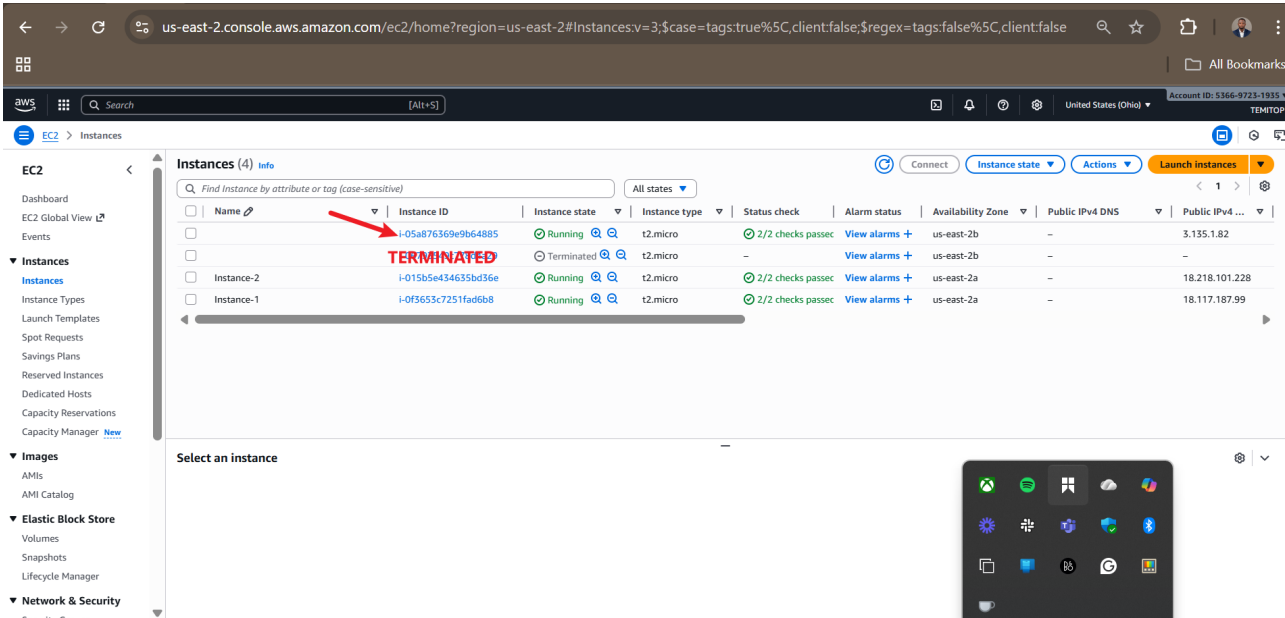
137. Select the EC2 created by ASG and terminate it.



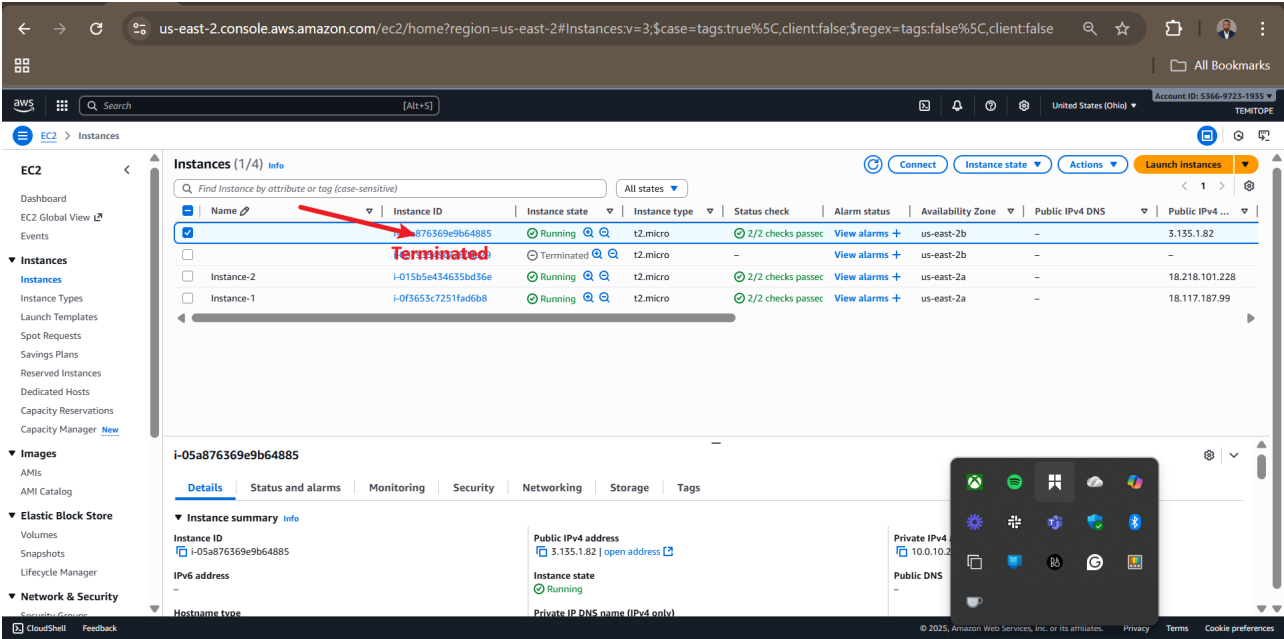
138. Click terminate ASG will create a new one that we can SSH into.



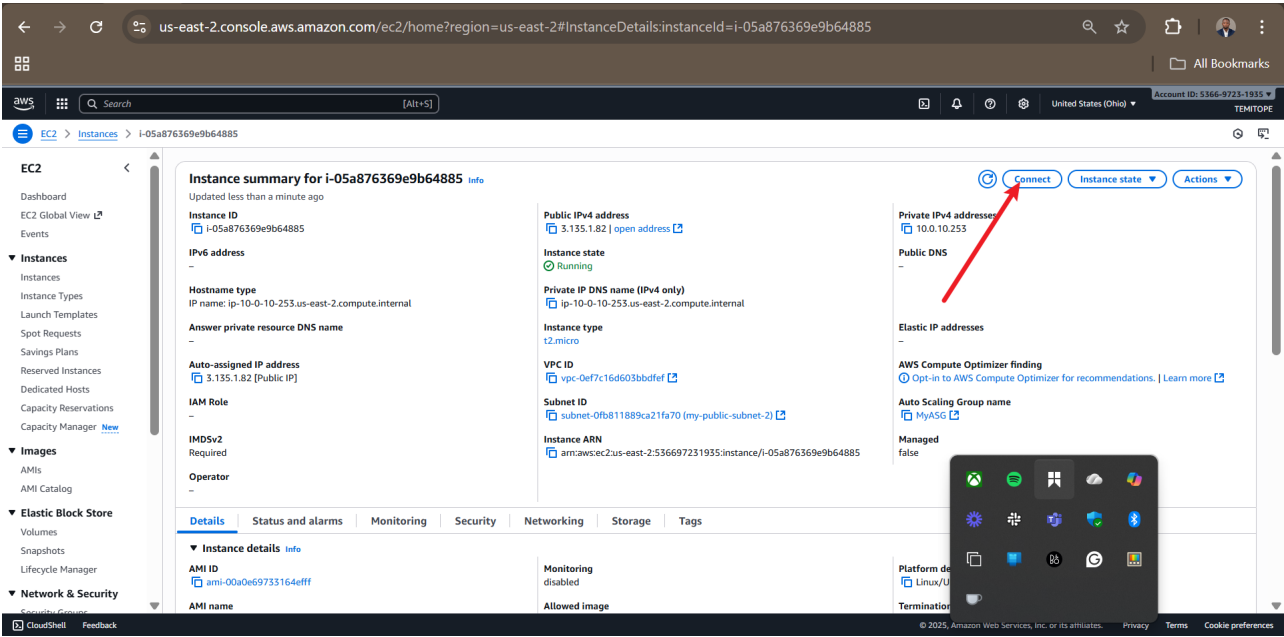
139. ASG has now spun up a new instance according to our policy (min of 1).



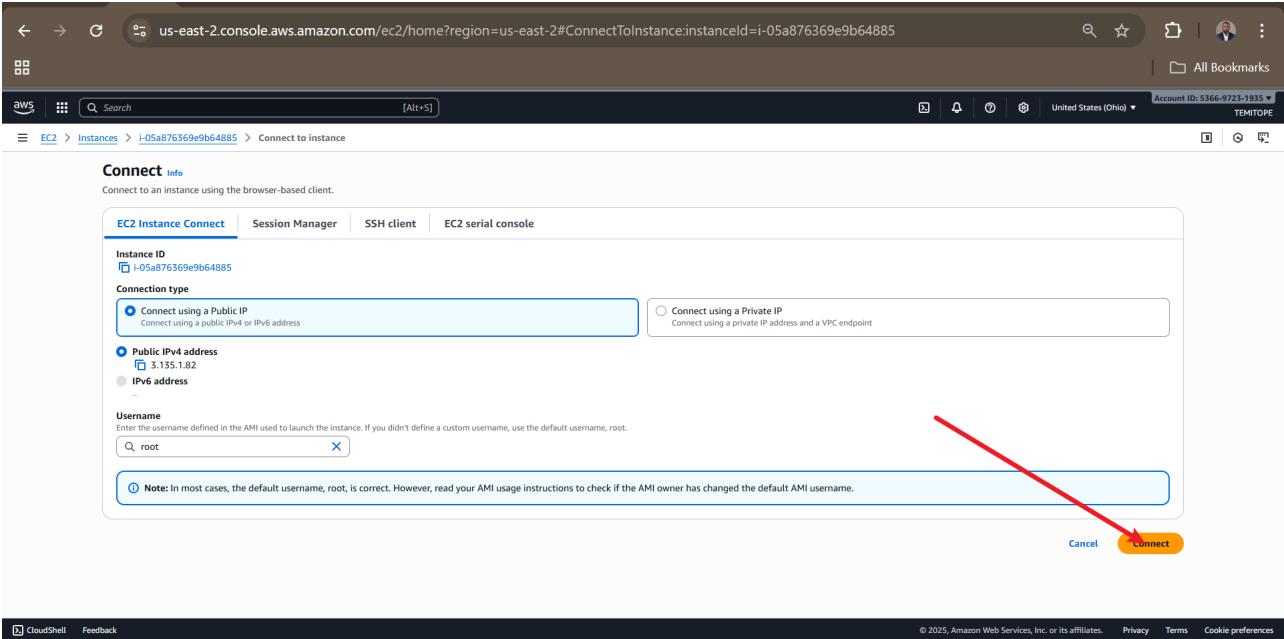
140. Click on the new instance to see if it can be connected to via SSH (internet).



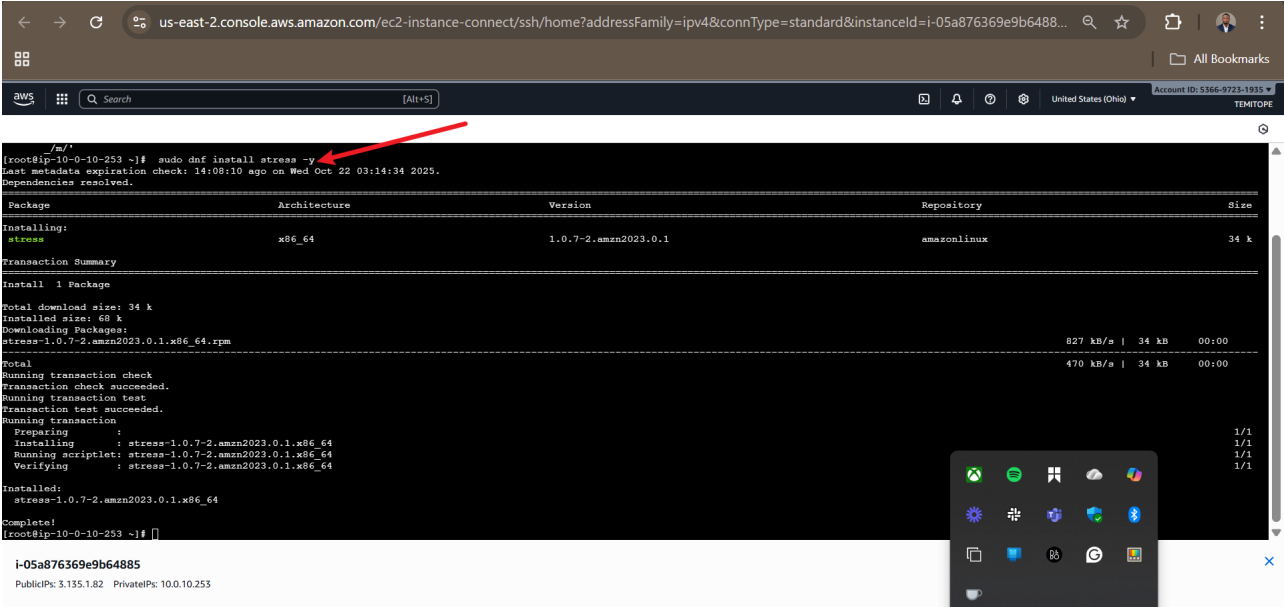
141. Now click connect to go to connect page.



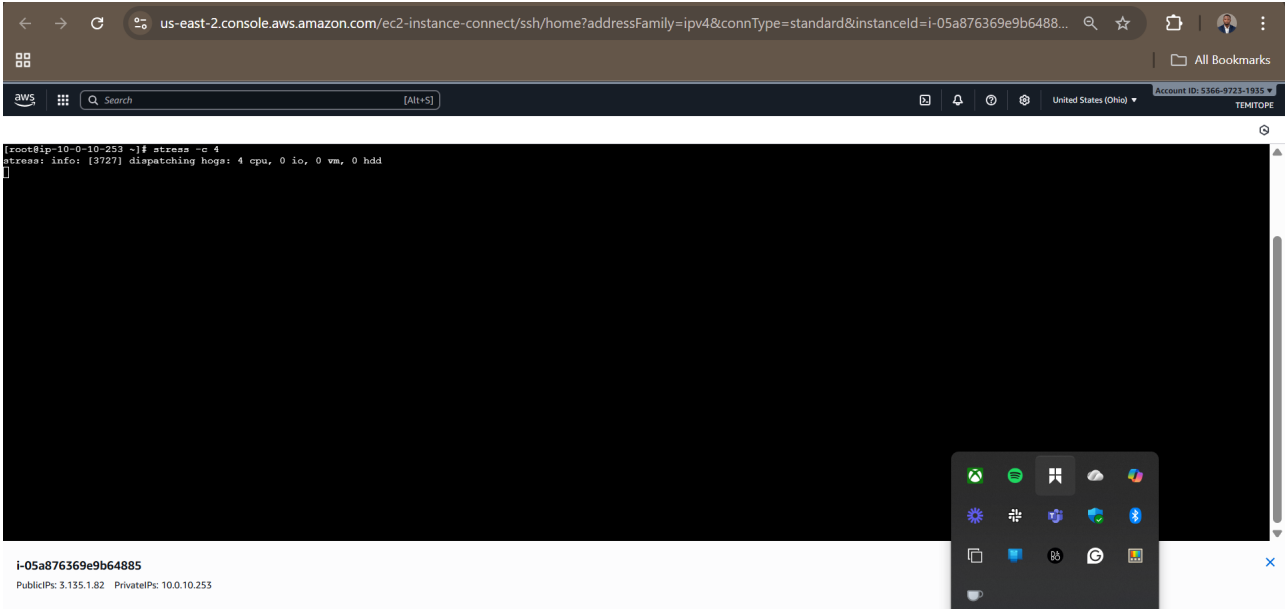
142. New auto scaling instance routable over the internet click connect.



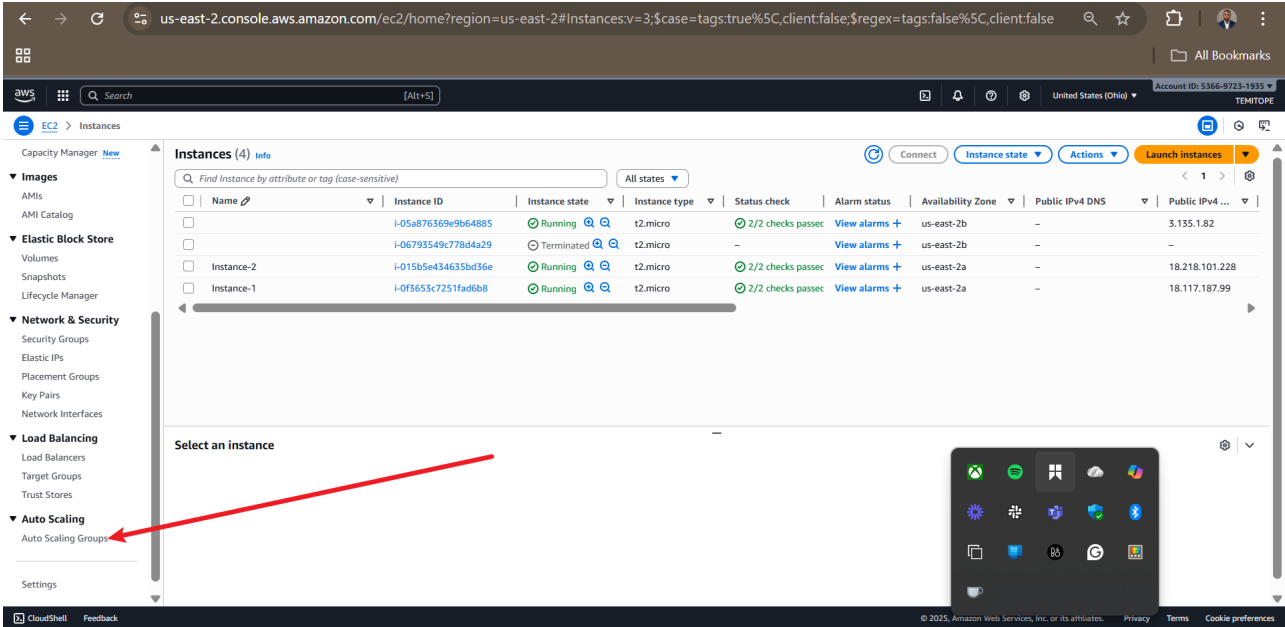
143. Install stress on new auto scaling instance.



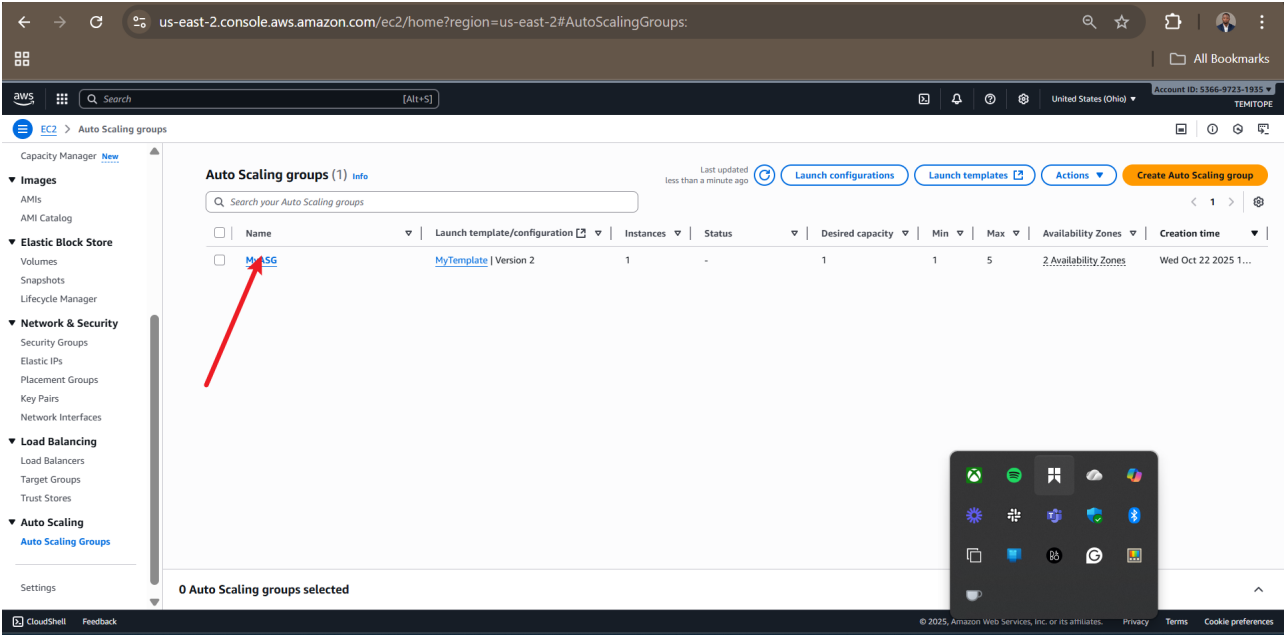
144. New auto scaling instance now stressed too.



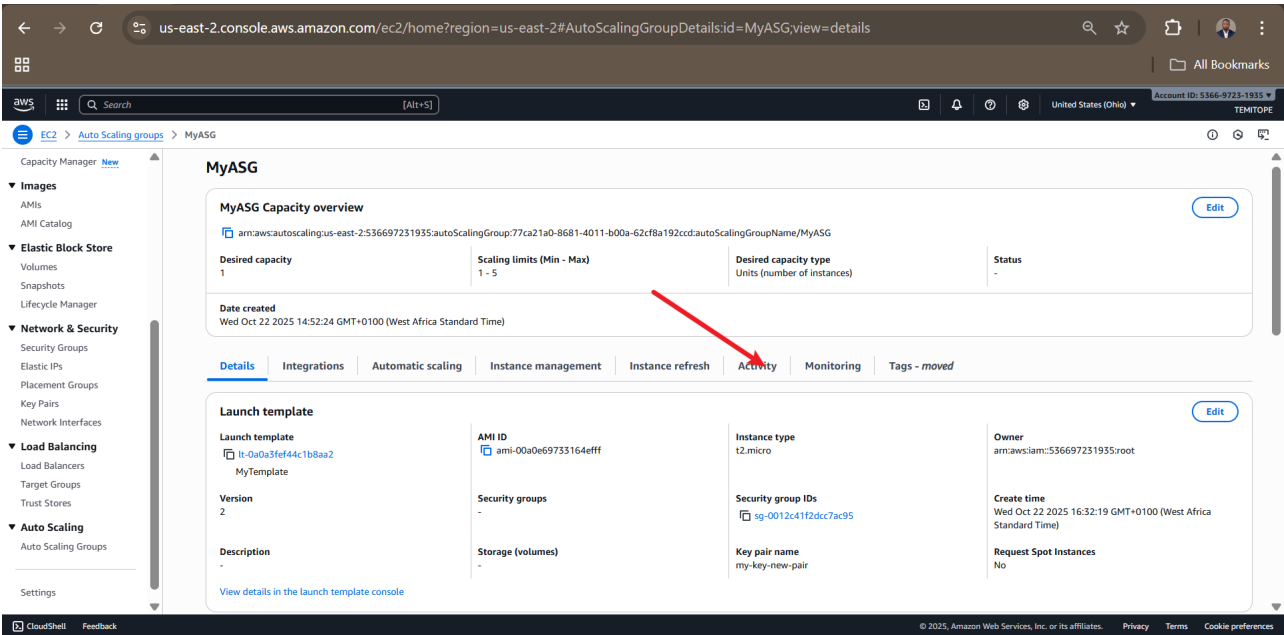
145. Navigate to EC2 instances page and under Auto Scaling click on ASG.



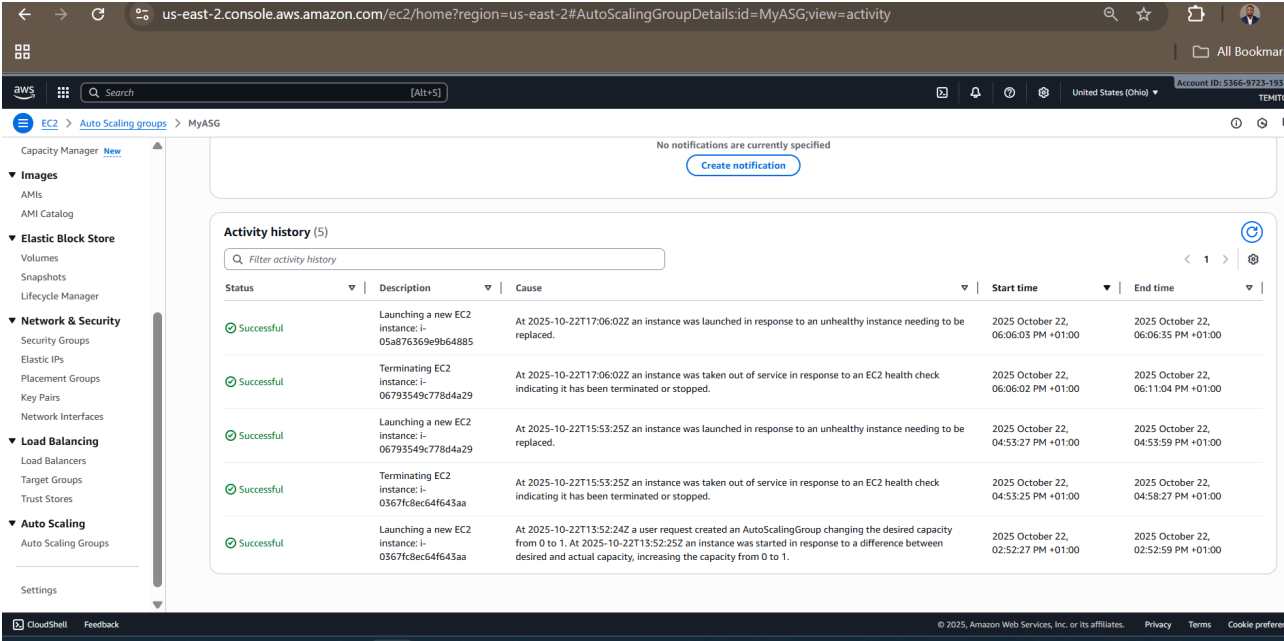
146. Click on the created auto scaling group.



147. On ASG dashboard click on activity.



148. Observe that there are now more instances created due to the surge in CPU consumption.



149. After a reload I now have instance with ID i-05a876369e9b64885.
![After a reload I now have instance with ID i-05a876369e9b64885]
(./img/149.after_a_reload_i_now_have_instace_with_id_ i-05a876369e9b64885.png)

150. After a reload I now have instance 2 with ID i-0f3653c7251fad6b8.
![After a reload I now have instance 2 with ID i-0f3653c7251fad6b8]
(./img/150.after_a_reload_i_now_have_instance_2_with_id_ i-0f3653c7251fad6b8.png)

151. After a reload I now have instance 5 with ID i-0c52afc9edb60330f.
![After a reload I now have instance 5 with ID i-0c52afc9edb60330f]
(./img/150.after_a_reload_i_now_have_instance_5_with_id_ i-0c52afc9edb60330f.png)

152. After a reload I now have instance 3 with ID i-015b5e434635bd36e.
![After a reload I now have instance 3 with ID i-015b5e434635bd36e]
(./img/151.after_a_reload_i_now_have_instance_3_with_id_ i-015b5e434635bd36e.png)

153. After a reload I now have instance 4 with ID i-015b5e434635bd36e.



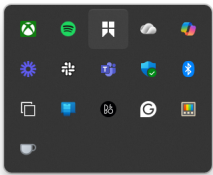
Welcome to Instance-1

Instance ID: i-05a876369e9b64885

Availability Zone: us-east-2b

Current CPU Load (1 min): 400%

Served by: i-05a876369e9b64885



154. Observe that all registered targets on target group are healthy.

us-east-2.console.aws.amazon.com/ec2/home?region=us-east-2#TargetGroup:targetGroupArn=arn:aws:elasticloadbalancing:us-east-2:53669723...

EC2 > Target groups > MyTargetGroup

Capacity Manager

▼ Images

- AMIs
- AMI Catalog

▼ Elastic Block Store

- Volumes
- Snapshots
- Lifecycle Manager

▼ Network & Security

- Security Groups
- Elastic IPs
- Placement Groups
- Key Pairs
- Network Interfaces

▼ Load Balancing

- Load Balancers
- Target Groups
- Trust Stores

▼ Auto Scaling

- Auto Scaling Groups

Settings

5
Total targets

5
Healthy

0
Unhealthy

0
Unused

0
Initial

0
Draining

0
Anomalous

Distribution of targets by Availability Zone (AZ)

Select values in this table to see corresponding Filters applied to the Registered targets table below.

Targets

Monitoring

Health checks

Attributes

Tags

Registered targets (5)

Anomaly mitigation: Not applicable

Deregister

Register targets

Filter targets

	Instance ID	Name	Port	Zone	Health status	Health status details	Administrative o...	Override details	Launch...	Anomaly
☐	i-0f4edec5d4acc85		80	us-east-2b (us...	Healthy	-	No override	No override is curre...	October 2...	Normal
☐	i-0c52afc9edb60330f		80	us-east-2a (us...	Healthy	-	No override	No override is curre...	October 2...	Normal
☐	i-05a876369e9b64885		80	us-east-2b (us...	Healthy	-	No override	No override is curre...	October 2...	Normal
☐	i-0f3653c7251fad6b8	Instance-1	80	us-east-2a (us...	Healthy	-	No override	No override is curre...	October 2...	Normal
☐	i-015b5e434635bd36e	Instance-2	80	us-east-2a (us...	Healthy	-	No override	No override is curre...	October 2...	Normal